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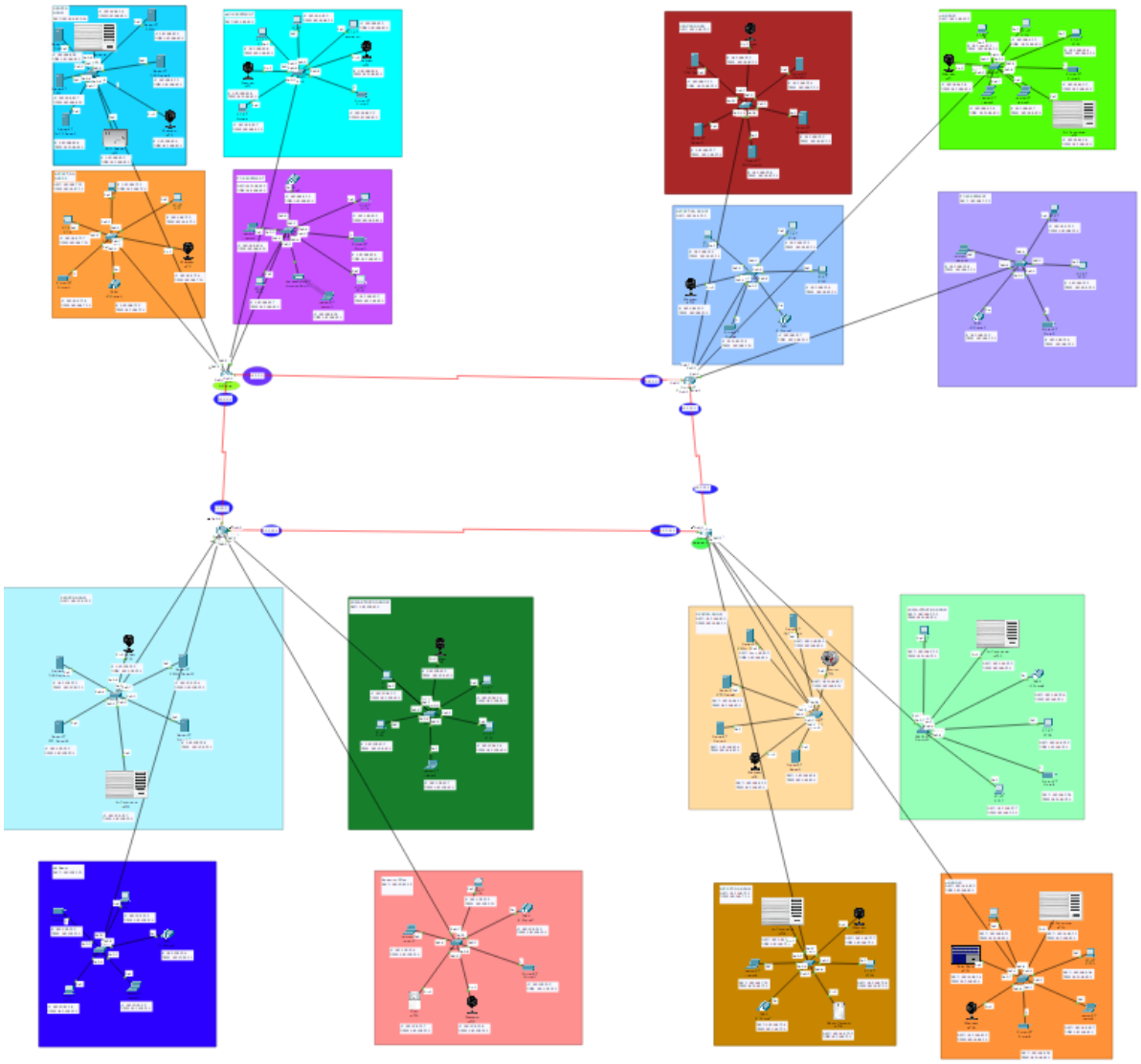
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1. Introduction

The purpose of this task is to create a reliable network design for the increasing number of startups called MACSATech. It has four branches based in KL, Penang, Johor, and Kelantan. We have created the design for this project using the most appropriate topology. We were convinced that the design would provide sufficient security and reliability to our users. Additionally, each staff member was trained to ensure that they were practical in the office. This task also includes justification of selected IP opinion schemes and network configuration methods for reliable and effective connection coordination. In the topology, the intermediary interconnections were established using the appropriate router.

## 2. TOPOLOGY



The topology of the network is a ring topology composed of four routers (R1, R2, R3, R4). The routers are interconnected to two adjacent routers and these make a closed loop. Such a structure provides basic redundancy and data flow even when one link goes dead.

Device Configuration Routers:

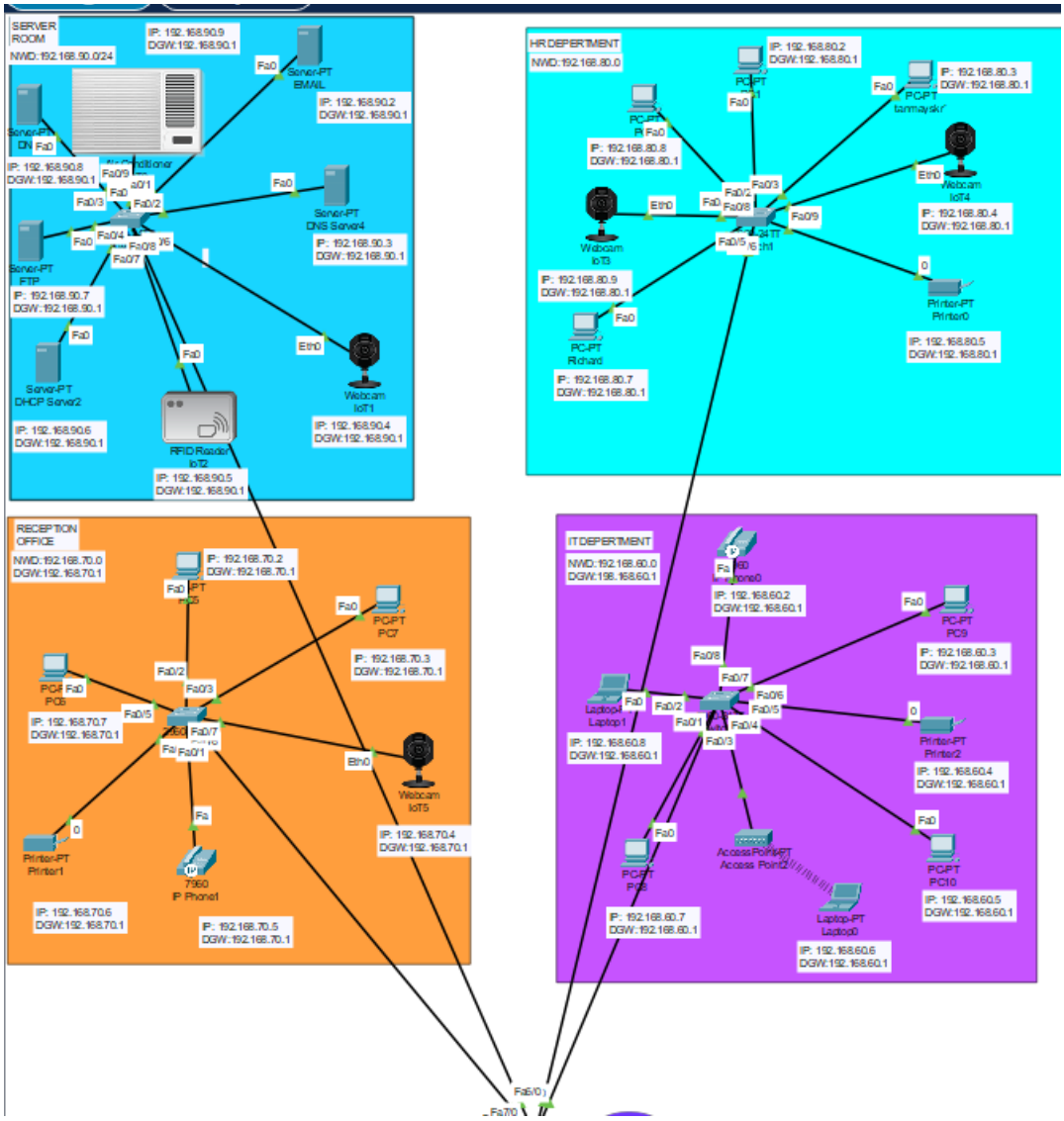
A router is set to have two FastEthernet interface. Switches: These are used in connections to each router and they are used to provide services to local PCs. Switch ports are switched and tagged to VLAN 1. End Devices: Local access is given to PCs with IP addresses and attached to switches.

**Routing Protocol :**

The network is implemented with Static Routing, and all the routes have been input by the network administrator. Static Routing means that data packets take a predetermined path to get to their destination. Unlike dynamic routing protocols such as OSPF, static routing does not adjust and recalculate routes when a link fails. Static routing is much simpler to set up, is more secure, and is better suited for small or stable networks that do not frequently change paths or routes.

## 3. KL-BRANCH : TANMAY SARKAR EMON (TP092959)

### *3.1 TOPOLOGY DESIGN FOR KL BRANCH*

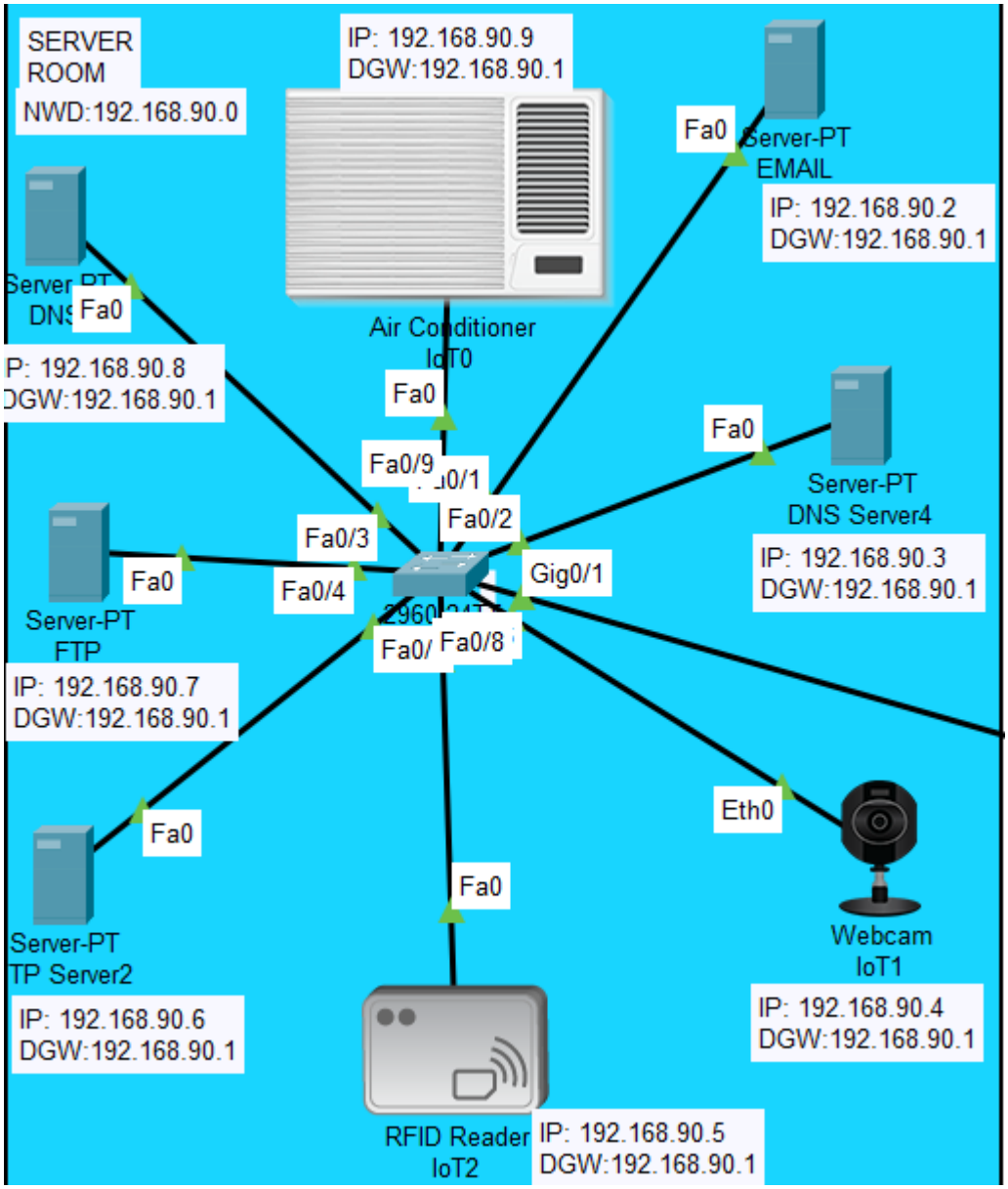


The picture above shows the topology for the KUALA LUMPUR Branch. Here we can see that the network is implemented using “Star Topology”. The branch has 4 separate rooms:

1. Server Room
2. Reception Office
3. HR Department
4. IT Department

Each of these departments is connected to the other through the Switch. And all the switches are connected to a router to establish a connection with other branches. In each room, there is one switch, and all the other network devices are connected to it.

### 3.1.2 Justification for SERVER ROOM




Server Room Description

This is a design of a server room in which a Star Topology is applied, in which all devices are linked to a main Cisco 2960 Switch. This makes the network simple to administer, dependable, and the failure of any individual device will not impact the entire network. The devices included are:

Server-PT DNS (192.168.90.8): Offers Domain Name System (DNS). It converts domain names into IP addresses, which simplifies the process of communication among users.

Server-PT EMAIL (192.168.90.2): Maintains email services at this branch. This server allows all users to send and receive emails making use of it.

Server-PT DNS Server4 (192.168.90.3): Redundancy Works as a backup DNS server to provide high availability. In case the main DNS goes offline, this server will be able to keep on with the DNS resolution.

Server-PT FTP (192.168.90.7): Applied to File Transfer Protocol (FTP).Shares also important files on the network.

Server-PT TP Server2 (192.168.90.6): To reduce the load and enhance performance, a secondary server is provided. It is also applicable as the fallback in the event of main server failures.

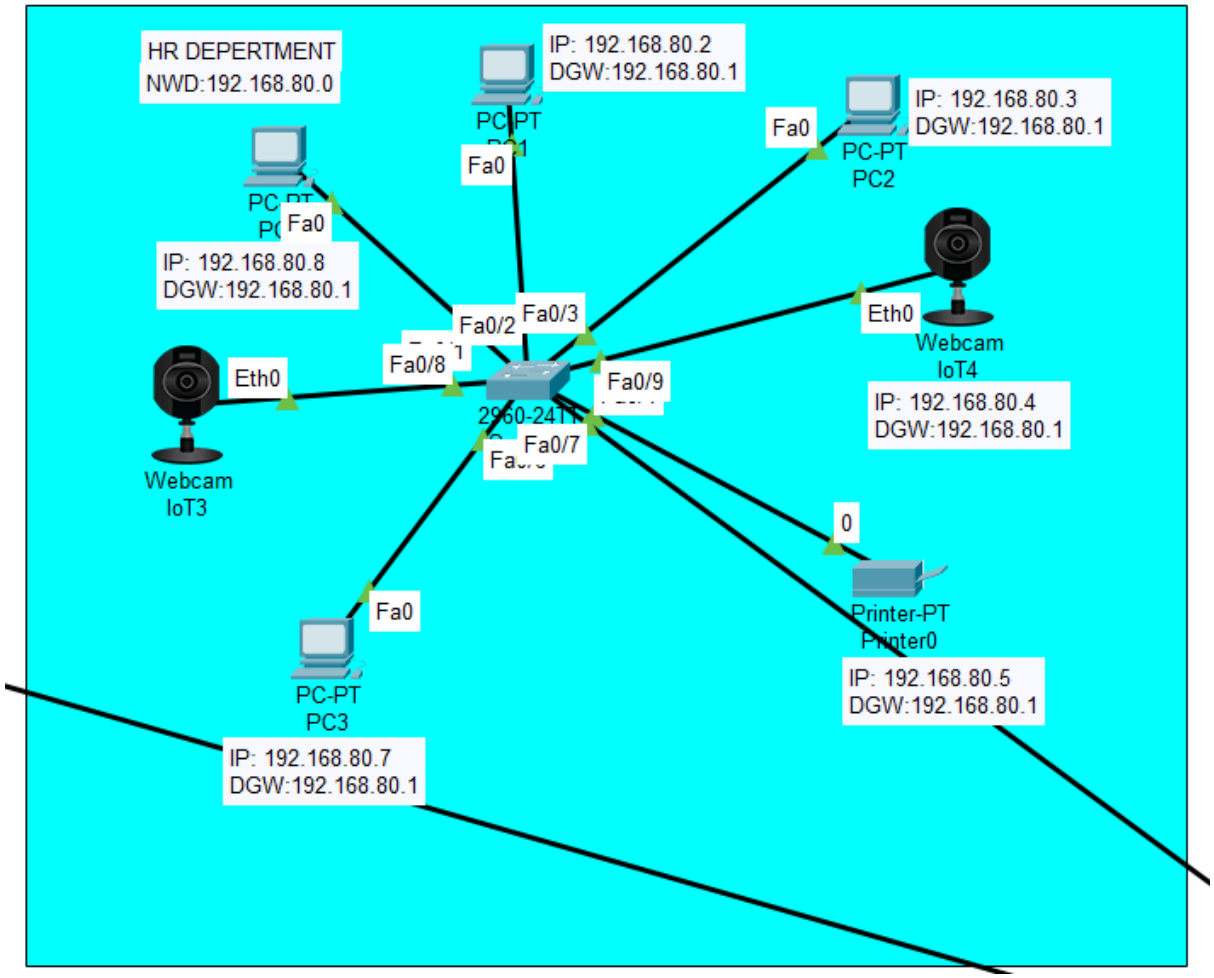
Air Conditioner IoT (192.168.90.9): Regulates the temperature of the server room. Avoids overheating of servers by ensuring that there is a cool atmosphere.

RFID Reader IoT (192.168.90.5): Security access control. Only authorized individuals who have ID cards are allowed to access the server room.

Webcam IoT (192.168.90.4): Gives 24/7 coverage to the server room. Assists in keeping track of operations and safety against any unauthorized entry.

Cisco 2960 Switch: The key equipment in such a Star Topology. Brings together all the servers and IoT devices and gives a communication flow.

3. Justification for HR Department:




This network architecture is a Local Area Network (LAN) design of the HR Department that is built in Cisco Packet Tracer. The network has a Cisco 2960-24TT switch to connect various devices that are set up in the 192.168.80.0/24 subnet.

Device Configuration

The default gateway is identical to all the devices: 192.168.80.1.

Devices include:

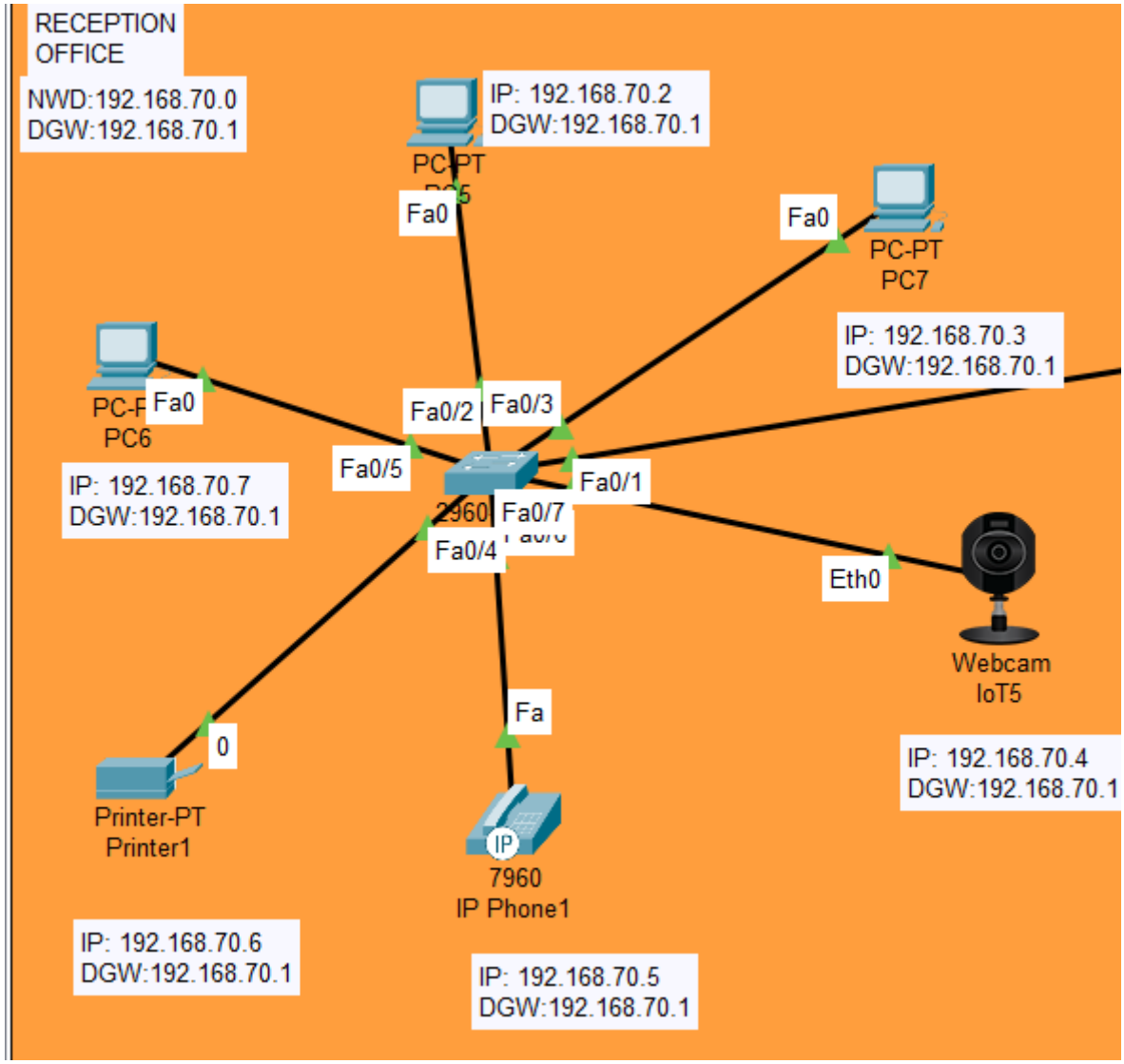
4 PCs (PC0-PC3)

2 Webcams (IoT3 & IoT4)

1 Printer

All the devices receive a fixed IP address and are connected to the switch through Fast Ethernet (Fa0/x) ports.

*4 Justification for Reception Office:*




This topology is an example of a Local Area Network (LAN) architecture of the Reception Office, created in Cisco Packet Tracer. It has a centralized switch (Cisco 2960) which links a range of devices such as PCs, a printer, an IP phone and a webcam- all set up in the 192.168.70.0/24 subnet.

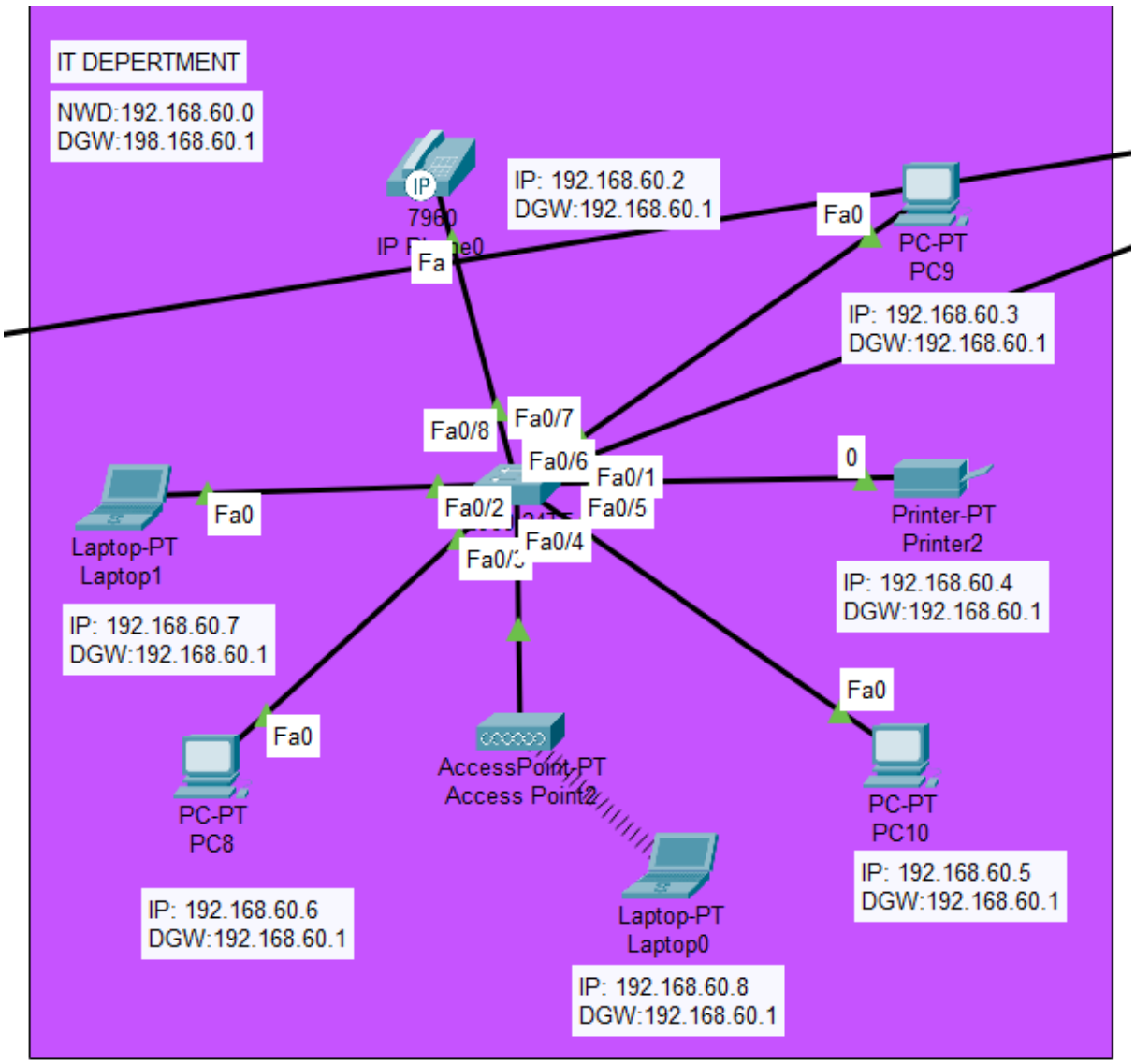
Device Configuration

- All devices use the same default gateway: 192.168.70.1
  Devices include:
- 3 PCs (PC5, PC6, PC7)
- 1 Printer
- 1 IP Phone

1. • 1 Webcam PC: For the manager to keep up with his duties in the branch.
2. IP phone: For communicating with other branch managers.
   Air conditioner: For maintaining the temperature.

• Each device is assigned a static IP address and connected to the switch via Fast Ethernet (Fa0/x) or Ethernet (Eth0) ports.

*5 Justification for IT Department:*




This topology is an example of a Local Area Network (LAN) configuration of the IT Department developed in Cisco Packet Tracer. It has a central router (IP 2960) that links all kinds of devices such as laptops, PCs, a printer, an access point, and a webcam- all configured in the 192.168.60.0/24 subnet.

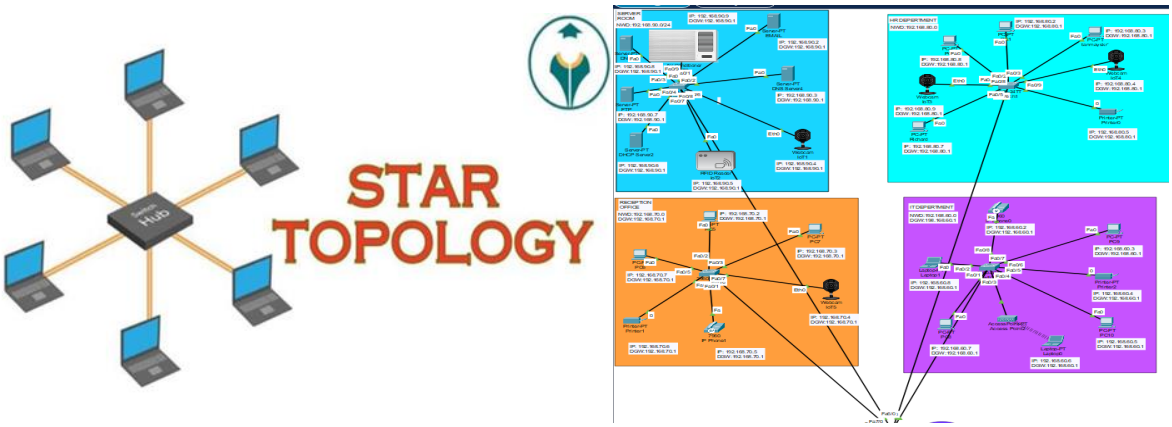
Device Configuration

- All devices use the same default gateway: 192.168.60.1
- Devices include:
  - 2 Laptops (Laptop01, Laptop02)
  - 3 PCs (PC8, PC9, PC10)
  - 1 Printer
  - 1 Access Point

- Each device is assigned a static IP address and connected to the router via Fast Ethernet (Fa0/x) ports.

*3.2 Justification for topology – KL Branch*

**Star Topology**




Both positive and negative sides:

Star topology is relatively simpler to install, and when one of the devices in the topology fails, it does not impact the rest of the other devices in the network. Because every device is linked to the switch, it enhances efficiency in communication. Any new device can be added to the network easily, and the chances of collision are very minimal. It has almost minimal chances of interception of data.

It has some drawbacks. One of them is that it is under the complete control of the central switch. In case of failure of the hub or switch, the entire network will be downed. It is normally an expensive installation as it requires many cables. This can even lower the performance of the performance when the central switch is of low capacity.

**MESH TOPOLOGY:**

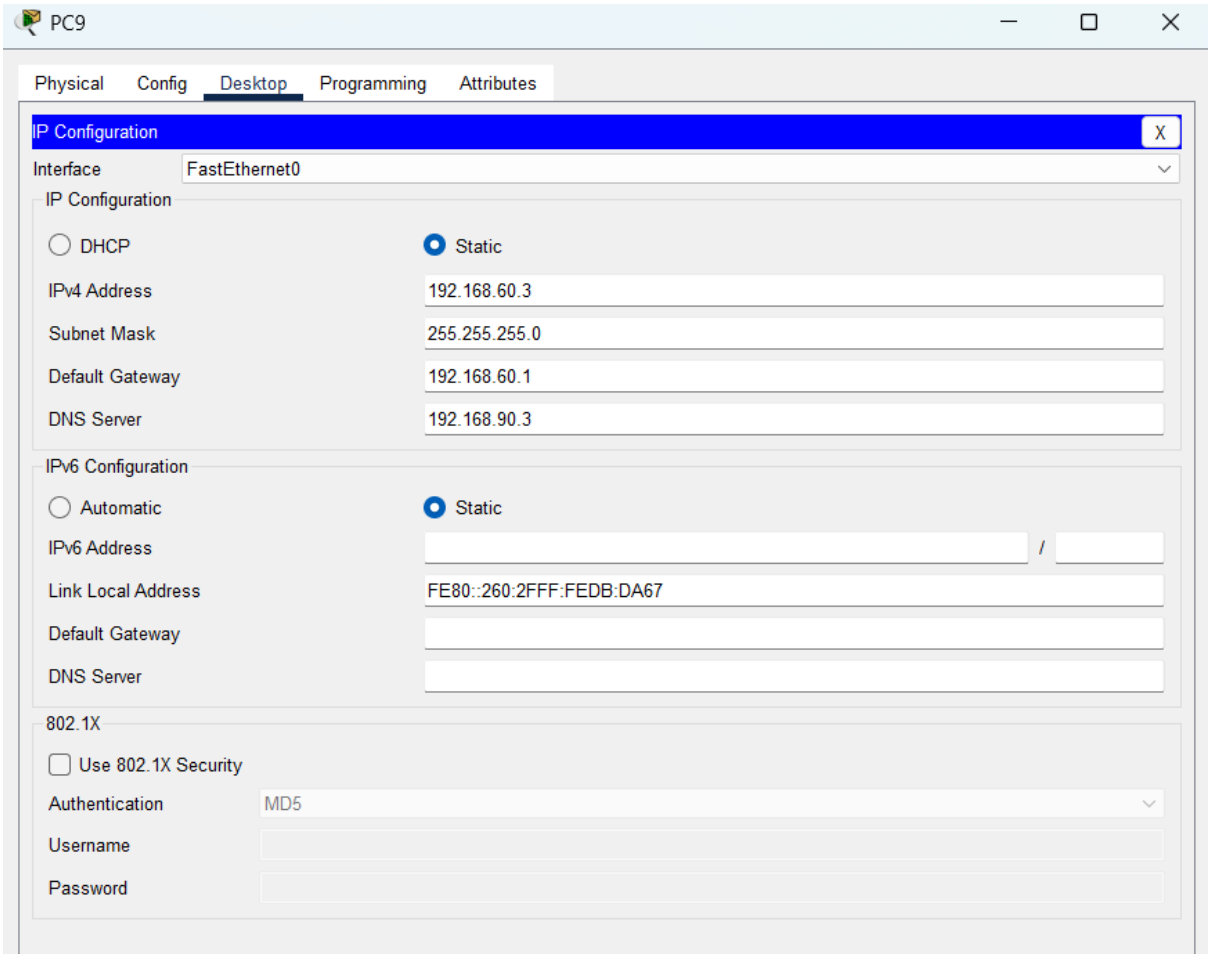
In the case of the KL Branch, I chose Star Topology over Mesh since it is the most economical, easy to maintain, and fits in the small to medium-sized office branches. Mesh would add both cost and complexity, which is unnecessary.

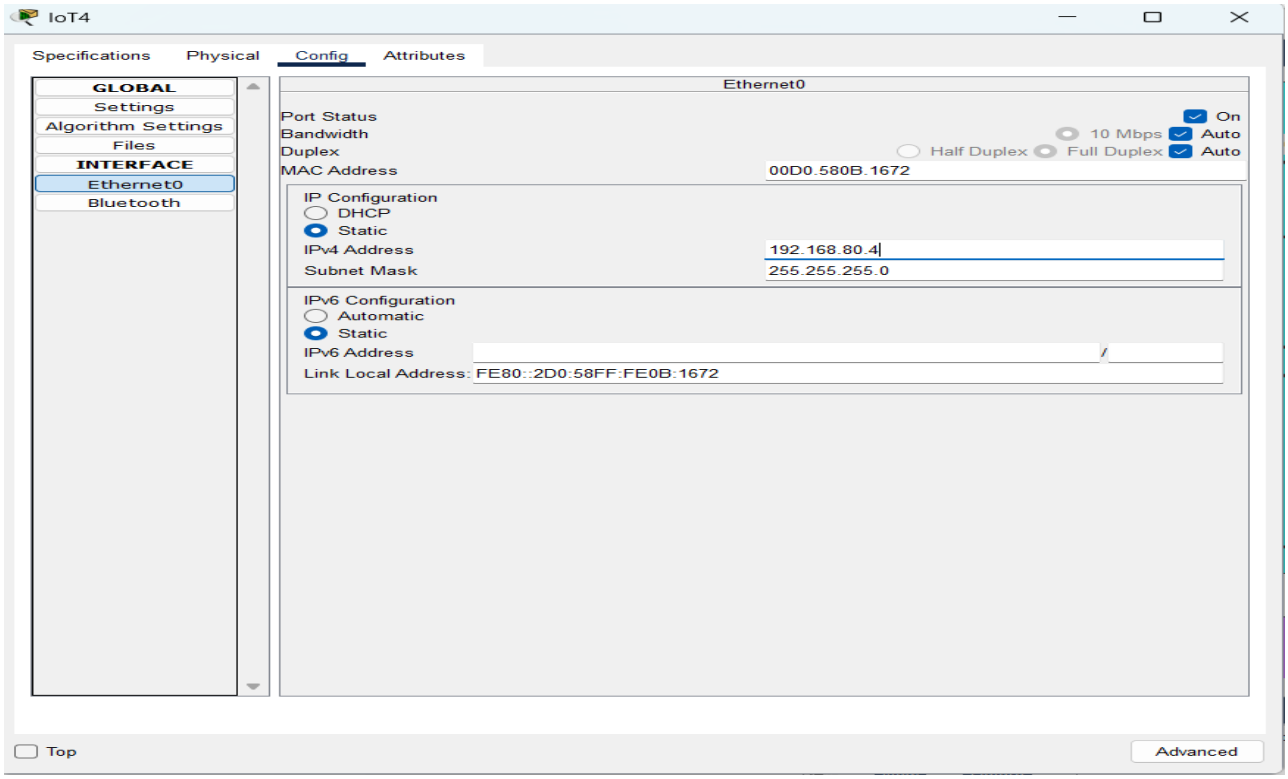
3.3 IP Addressing Scheme KL Branch:

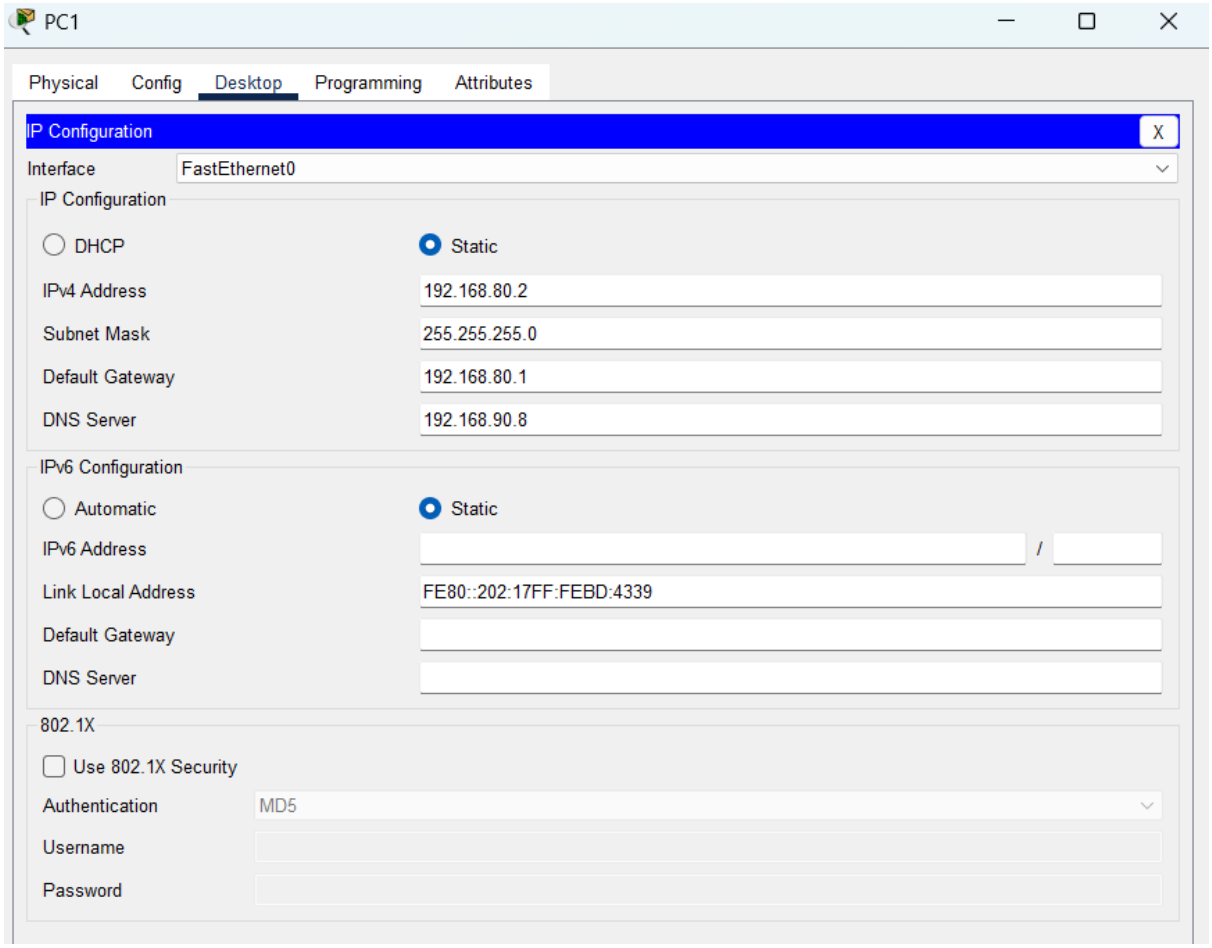
| Department | Device / Host | IP Address | Default Gateway |
|---|---|---|---|
| **Server Room** (192.168.90.0/24) | DNS Server | 192.168.90.8 | 192.168.90.1 |
| | Email Server | 192.168.90.2 | 192.168.90.1 |
| | Backup DNS Server | 192.168.90.3 | 192.168.90.1 |
| | FTP Server | 192.168.90.7 | 192.168.90.1 |
| | TP Server2 | 192.168.90.6 | 192.168.90.1 |
| | IoT Air Conditioner | 192.168.90.9 | 192.168.90.1 |
| | IoT RFID Reader | 192.168.90.5 | 192.168.90.1 |
| | IoT Webcam | 192.168.90.4 | 192.168.90.1 |
| | **Router Interface** | **192.168.90.1** | - |
| **Reception Office** (192.168.70.0/24) | PC0 | 192.168.70.2 | 192.168.70.1 |
| | Printer IoT | 192.168.70.5 | 192.168.70.1 |
| | Phone IoT | 192.168.70.6 | 192.168.70.1 |
| | IoT Switch Control | 192.168.70.7 | 192.168.70.1 |
| | **Router Interface** | **192.168.70.1** | - |
| **HR Department** (192.168.80.0/24) | PC0 | 192.168.80.2 | 192.168.80.1 |
| | PC1 | 192.168.80.3 | 192.168.80.1 |
| | PC2 | 192.168.80.4 | 192.168.80.1 |
| | Printer IoT | 192.168.80.5 | 192.168.80.1 |
| | Webcam IoT | 192.168.80.6 | 192.168.80.1 |
| | **Router Interface** | **192.168.80.1** | - |
| **IT Department** (192.168.60.0/24) | Laptop0 | 192.168.60.2 | 192.168.60.1 |
| **IT Department** (192.168.60.0/24) | PC0 | 192.168.60.3 | 192.168.60.1 |

| | | | |
|---|---|---|---|
| • | • Printer IoT | • 192.168.60.4 | • 192.168.60.1 |
| • | • Access Point IoT | • 192.168.60.5 | • 192.168.60.1 |
| • | • Laptop1 | • 192.168.60.6 | • 192.168.60.1 |
| • | • **Router Interface** | • **192.168.60.1** | • - |

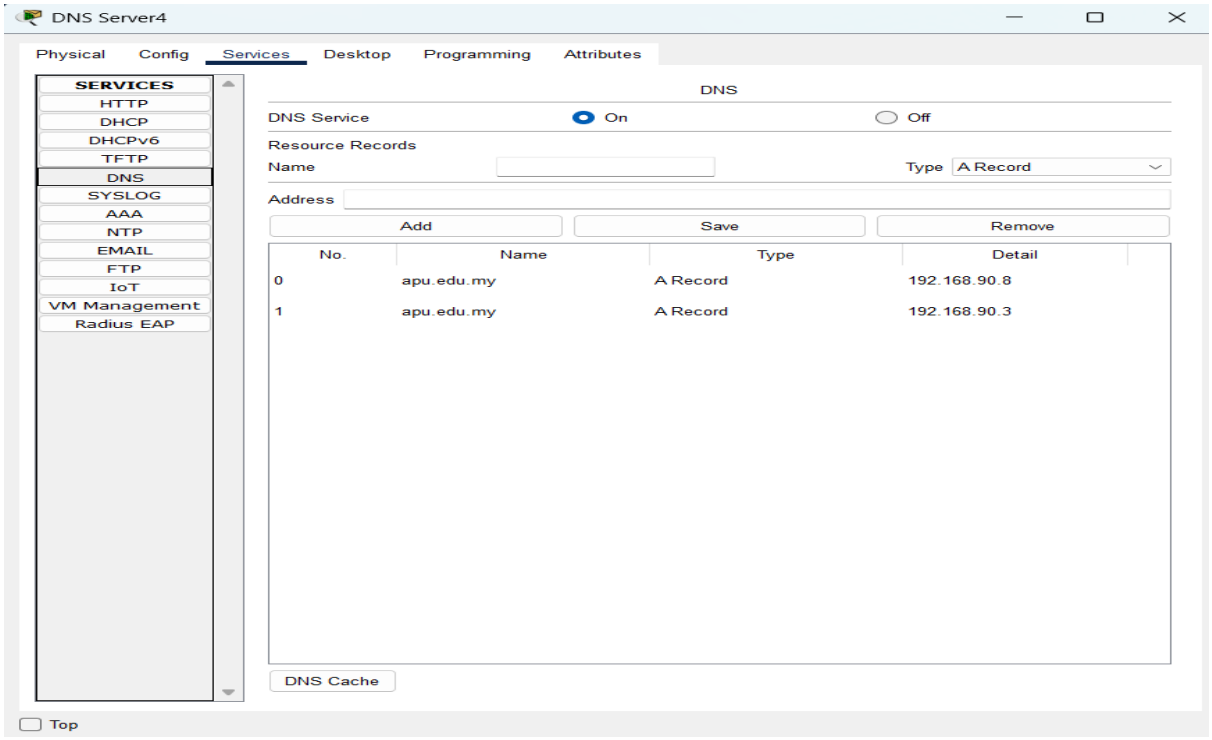
## 3.4 End devices for KL Branch:

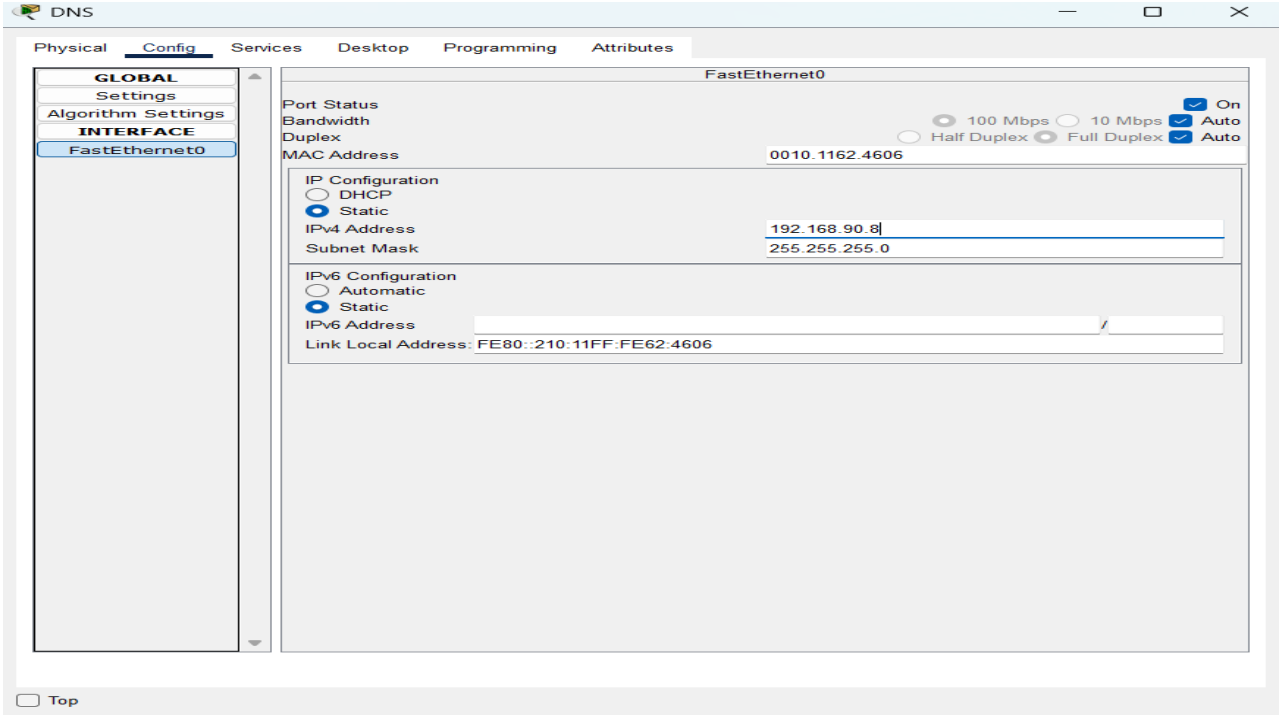



IoT4
Specifications
Physical
Config
Attributes
GLOBAL
Settings
Algorithm Settings
Files
INTERFACE
Ethernet0
Bluetooth
Ethernet0
Port Status
On
Bandwidth
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
00D0.580B.1672
IP Configuration
DHCP
Static
IPv4 Address
192.168.80.4
Subnet Mask
255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::2D0:58FF:FE0B:1672
Top
Advanced

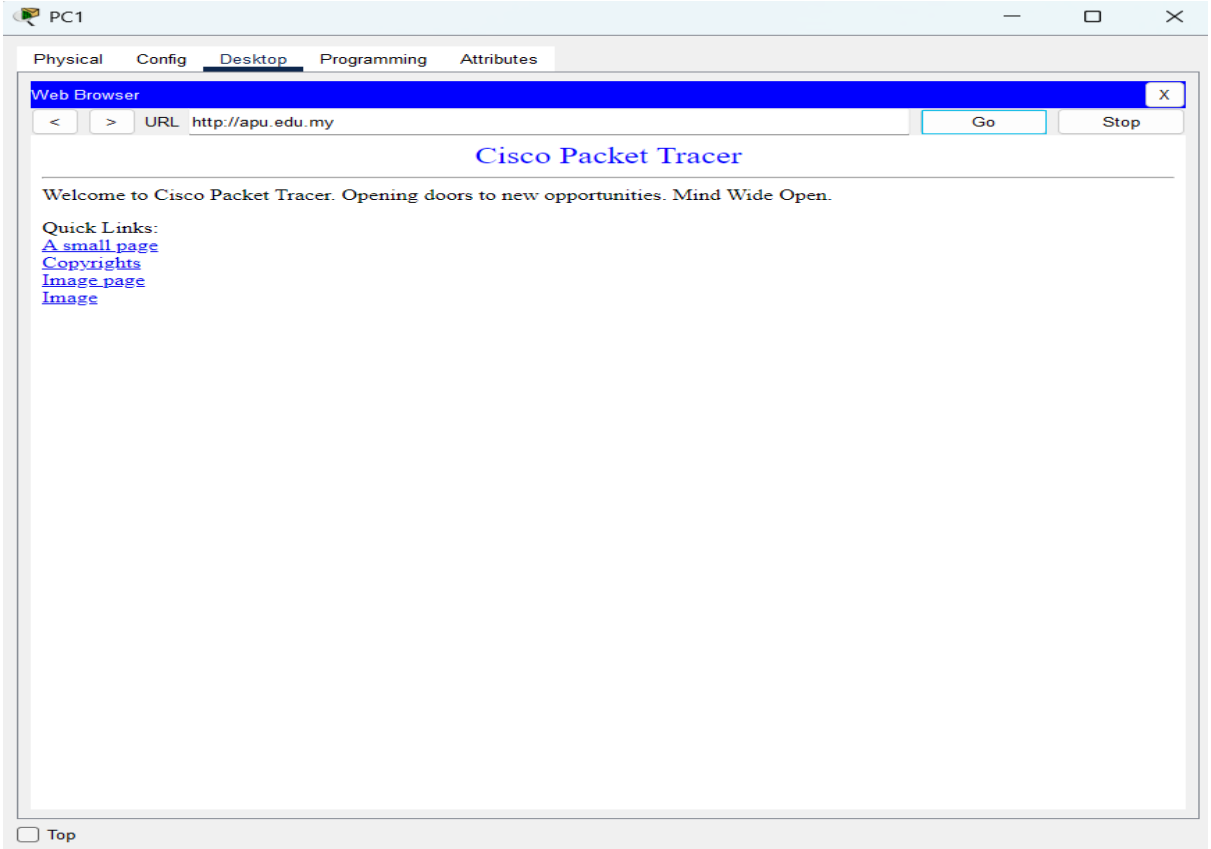

PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.80.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.80.1
DNS Server
192.168.90.8
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::202:17FF:FEBD:4339
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

## SERVERS:

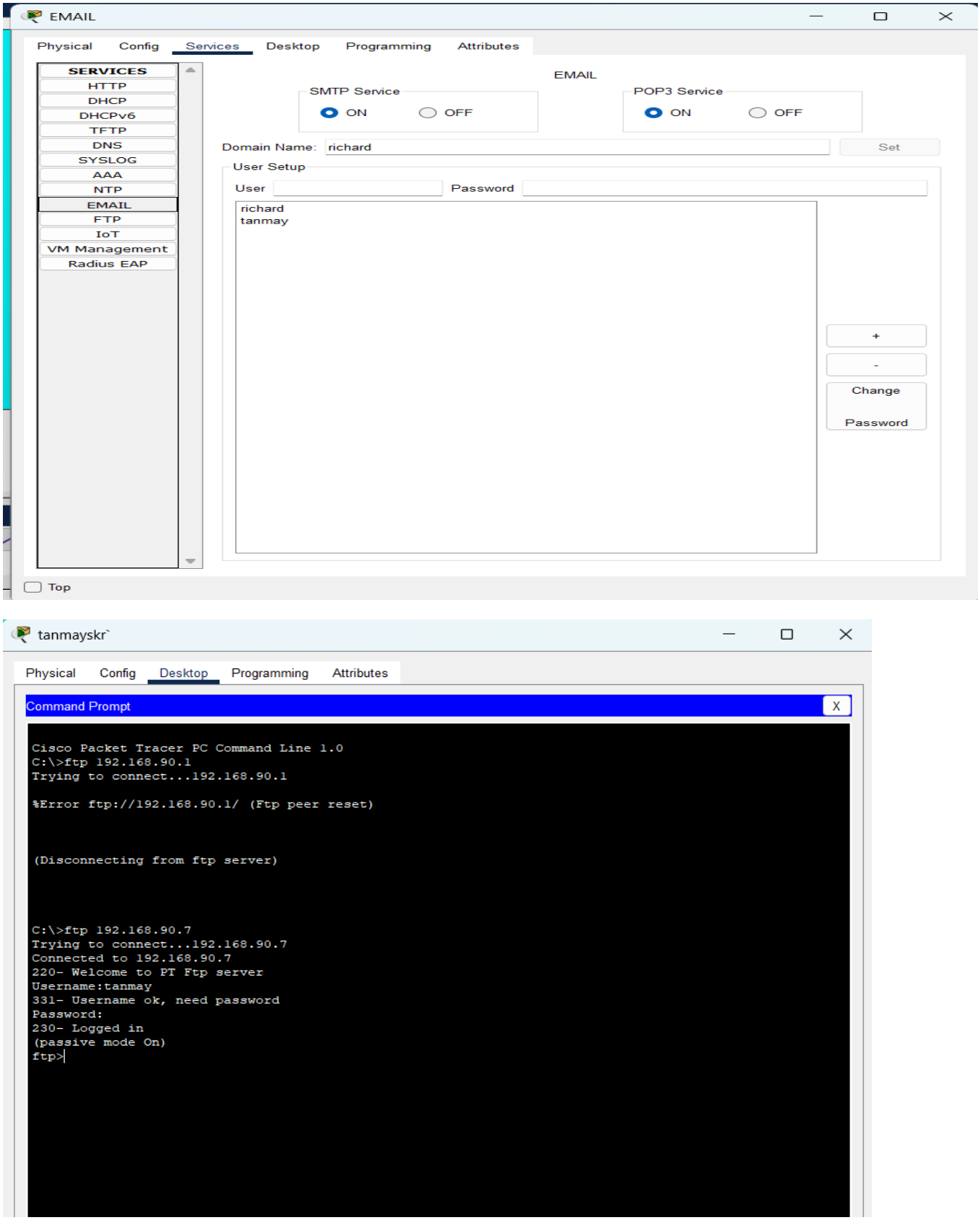

DNS Server4
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
DNS
DNS Service
On
Off
Resource Records
Name
Type
A Record
Address
Add
Save
Remove
No.
Name
Type
Detail
0
apu.edu.my
A Record
192.168.90.8
1
apu.edu.my
A Record
192.168.90.3
DNS Cache
Top




## ALL SERVERS ARE WORKING:




Picture: (http://apu.edu.my ).This is the URL name for our DNS server.



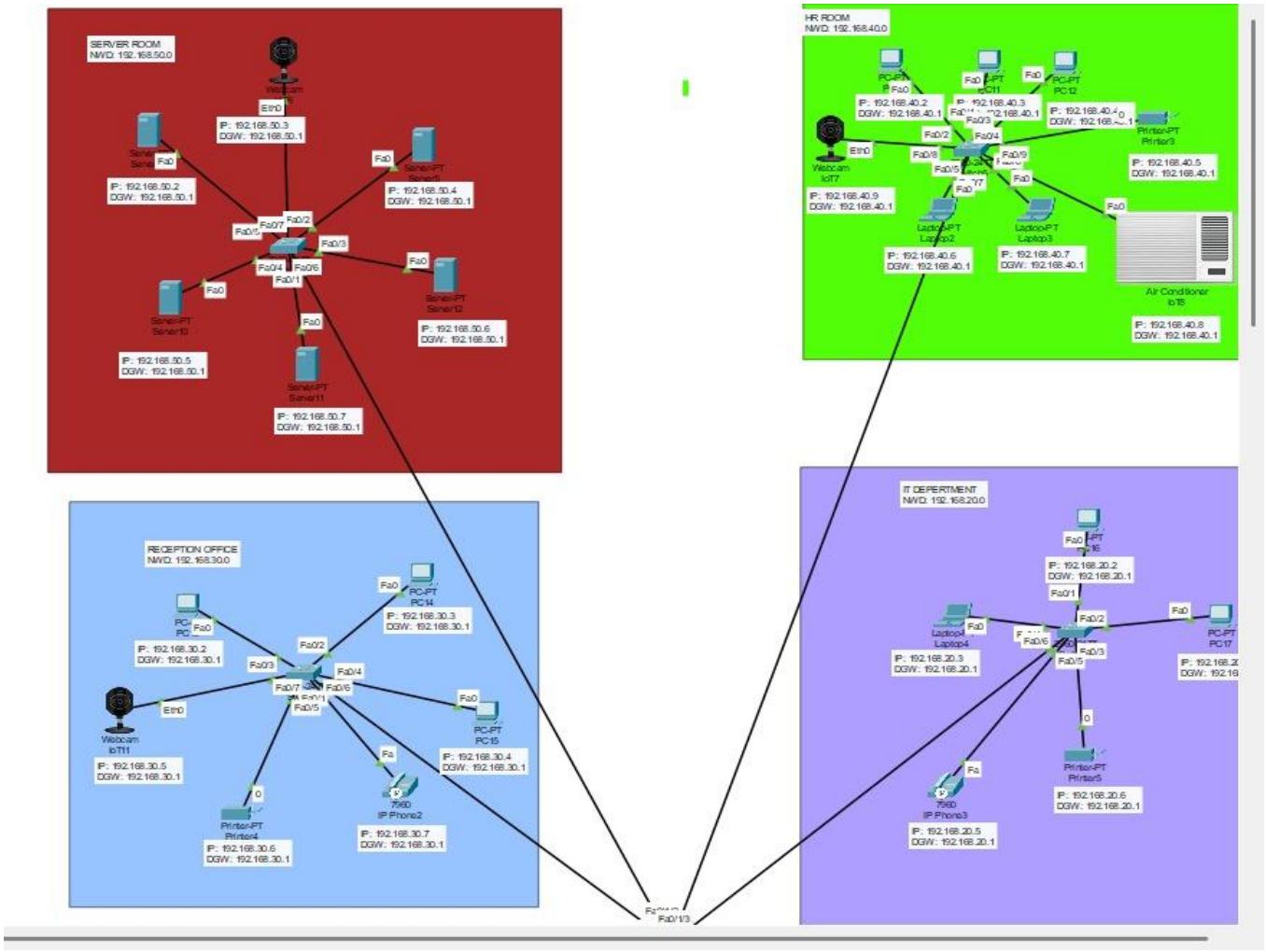

Picture: FTP SERVER WORKING

2.4 VLAN CONFIGURATION

1A. In case of Server Room, I add the following commands in order to configure the switch:

a. enable
b. configure terminal
c. interface vlan 1
d. ip address 192.168.90.10 255.255.255.0
e. no shutdown
f. ip default-gateway 192.168.90.1
g. exit
h. copy run start
i. wr

# PENANG BRANCH: MIRAJUL ALAM ( TP091208)

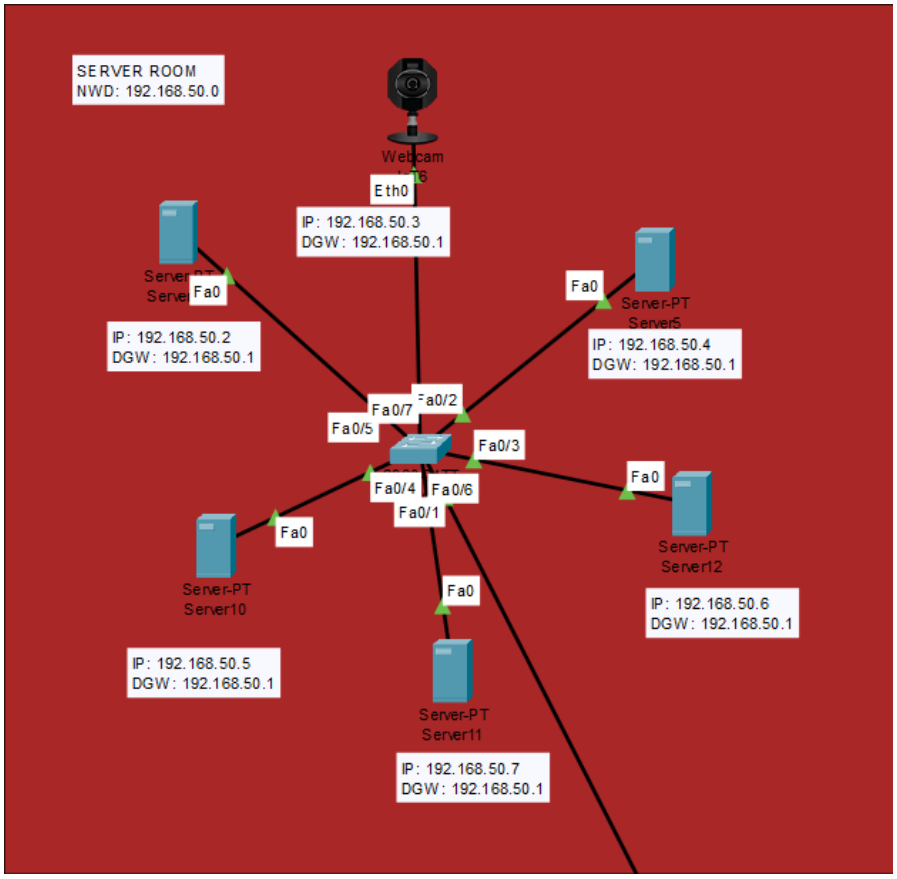



The picture above shows the topology for the **PENANG Branch**. Here we can see that the network is implemented using "Star Topology". The branch has 4 separate rooms:

1. Server Room
2. Reception Office
3. HR Department
4. IT Department

That way the Switch is basically interconnecting the departments to each other. And the switches are all connected to router in order to access the other branches in the network. Each room has a single switch and all the rest of the network equipment gets hardwired into it.

**JUSTIFICATION FOR SERVER ROOM:**




Then I have been designing a layout of a server room in which we would employ a star topology. Effectually all the devices are connected to this central Cisco 2960 switch. This is why the network can be easily managed, and it is also reliable and in case one equipment fails, the network does not collapse but continues with the other machines. Here's what's in it:

**Central Device**

The topology is based on a Switch.

It networked all servers and webcam by its FastEthernet interfaces **(Fa0/1 - Fa0/7).**

Local communication is maintained in the subnet by the switch.

The associated devices are able to communicate with one another without using an external network.

---

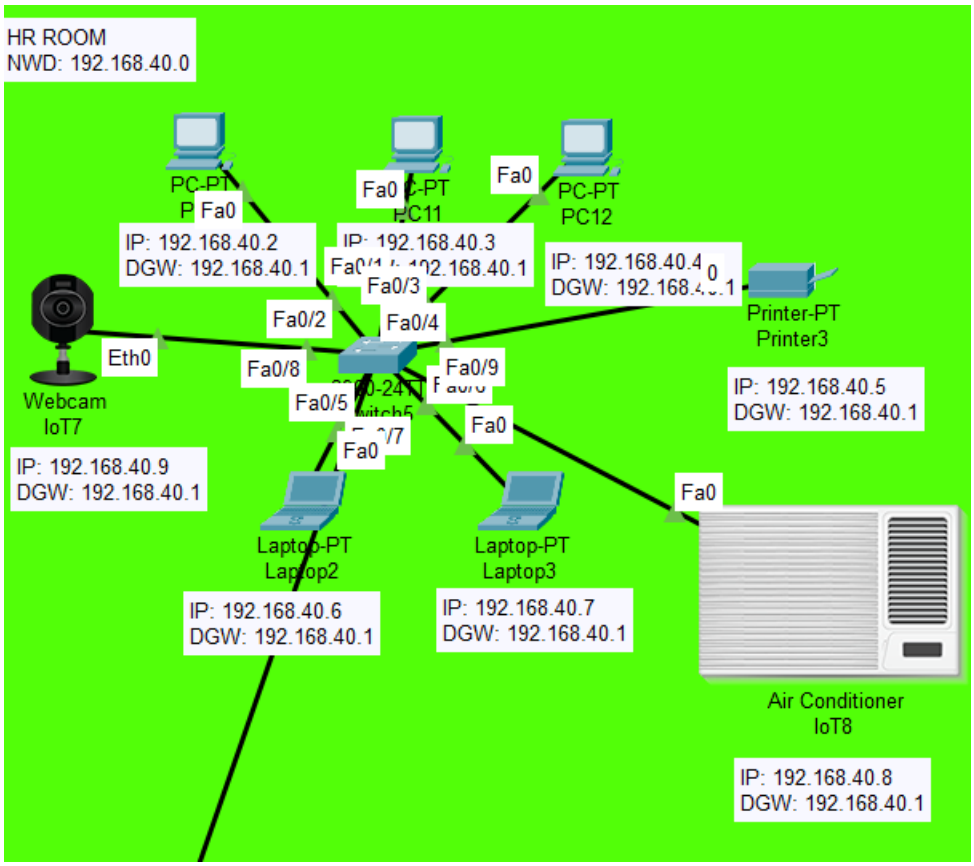
**3. Servers and Devices**

The network consists of **five servers** and **one webcam**, each configured with a **unique IP address** in the **192.168.50.0/24** range. All devices share the same **default gateway (DGW: 192.168.50.1)** for potential communication outside this local subnet.

- **DNS Server**

    - IP: 192.168.50.2
    - DGW: 192.168.50.1
    - Looks up domain names with both internal and external queries.
- **Webcam**
    - IP: 192.168.50.3
    - DGW: 192.168.50.1
    - Connected via **Eth0** interface.
    - Gives the server room security and surveillance features.
- **Email Server (Server5)**
    - IP: 192.168.50.4
    - DGW: 192.168.50.1
    - Manages company or in-house email as a communication media.
- **Web Server (Server10)**
    - IP: 192.168.50.5
    - DGW: 192.168.50.1
    - Hosts websites, applications and internal portals.
- **FTP Server (Server12)**
    - IP: 192.168.50.6
    - DGW: 192.168.50.1
    - Offers file transfer services to upload and download files.
- **Server11 (Generic Server)**
    - IP: 192.168.50.7
    - DGW: 192.168.50.1
    - Performs extra functions as a backup server, storage server, or some other specific function.

**JUSTIFICATION FOR THE HR ROOM:**




**Central Device**

The switch is the central device of the HR Room network, serving as the central point of connection of all the devices.

Its **FastEthernet ports (Fa0/1 - Fa0/9)** are used to connect PCs, laptops, and IoT devices.

This layout enables easy internal routing without having to have external routing.

---

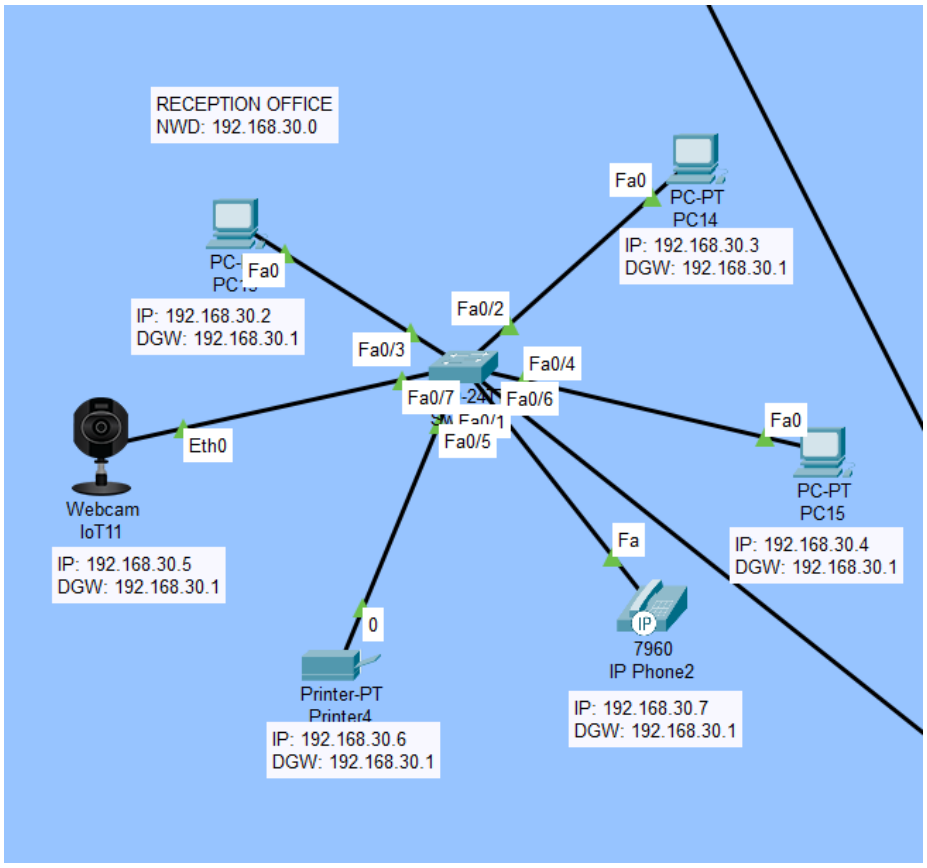
**2. Connected Devices**

The HR Room network includes a mix of **PCs, laptops, a printer, and IoT devices**. Each device has a **unique IP address** within the subnet and uses the **default gateway (DGW: 192.168.40.1)** for external communication.

- **PCs (Workstations)**
  - **PC10** – IP: 192.168.40.2
  - **PC11** – IP: 192.168.40.3
  - **PC12** – IP: 192.168.40.4

    - They are conventional office computer employees.
- **Printer (Printer3)**
    - IP: 192.168.40.5
    - HR staff share printers, which have the ability to print on network.
- **Laptops**
    - **Laptop2** – IP: 192.168.40.6
    - **Laptop3** – IP: 192.168.40.7
    - Employee workstations which can be carried around.
- **IoT Devices**
    - **Webcam (IoT7)** – IP: 192.168.40.9
        - Video surveils and monitors the HR room.
    - **Air Conditioner (IoT8)** – IP: 192.168.40.8
        - An intelligent that can be networked and controlled AC system.

**JUSTIFICATION FOR THE RECEPTION ROOM:**




The image shows a network diagram for a **reception office** with a network address (NWD) of 192.168.30.0. The network is centered around a switch, which connects multiple devices.

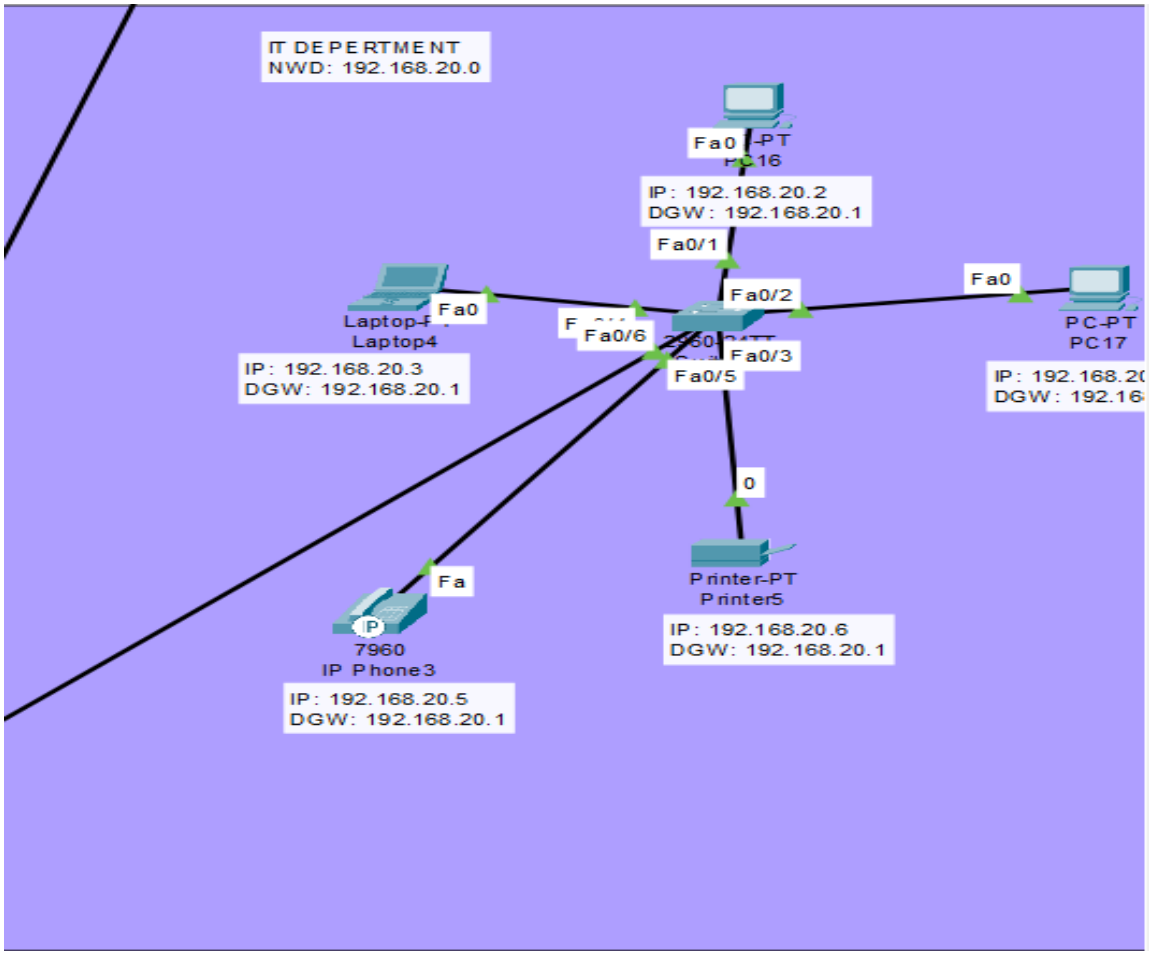
Here's a breakdown of the devices and their connections:

- **Switch:** This is the central device, and it has multiple ports labeled Fa0/1 through Fa0/7 and Fa0/1 through Fa0/6. Several devices connect to this switch.
- **PC-PT PC13:** A computer connected to the switch via port Fa0/3. It has the IP address 192.168.30.2 and a default gateway (DGW) of 192.168.30.1.
- **Webcam IoT11:** A webcam connected to the switch via port Fa0/7. Its IP address is 192.168.30.5, and its default gateway is 192.168.30.1.
- **Printer-PT Printer4:** A printer connected to the switch via an unspecified port. It has the IP address 192.168.30.6 and a default gateway of 192.168.30.1.
- **PC-PT PC14:** Another computer connected to the switch via port Fa0/2. It has the IP address 192.168.30.3 and a default gateway of 192.168.30.1.

- **PC-PT PC15:** A third computer connected to the switch via an unspecified port. It has the IP address 192.168.30.4 and a default gateway of 192.168.30.1.
- **IP Phone2:** An IP phone connected to the switch via port Fa0/6. Its IP address is 192.168.30.7 and its default gateway is 192.168.30.1.

They all are set to use a static IP address and all devices have the same default gateway which is 192.168.30.1. This implies that 192.168.30.1 is the router or the gateway which links this local network with the other networks like the internet. The connections are depicted as solid lines, which depict the physical connections (probably Ethernet cables) between the devices and switch.

**JUSTIFICATION FOR THE IT Department:**



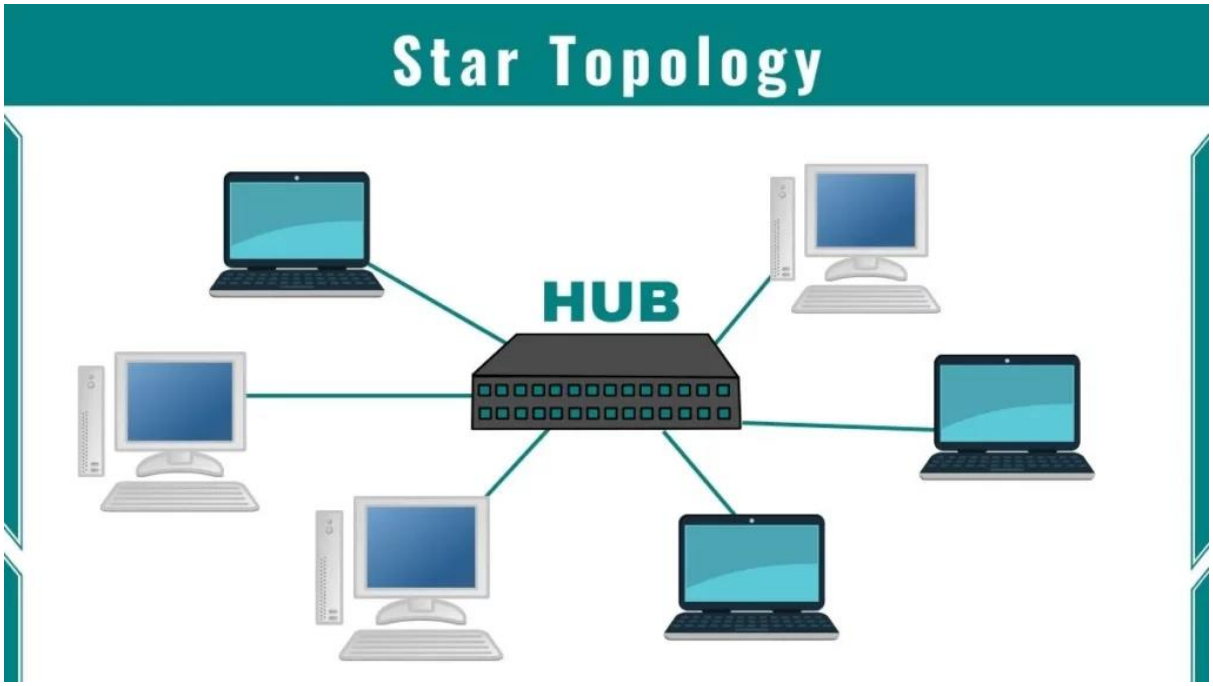

The image displays a network diagram for an **IT Department** with a network address (NWD) of 192.168.20.0. The central device is a **switch** that connects various devices within the department.

**Devices and Connections**

- **Switch:** This acts as the central hub, facilitating communication between all connected devices. It has multiple Fast Ethernet ports, indicated by Fa0/1, Fa0/2, Fa0/3, Fa0/5, and Fa0/6.
- **PC-PT PC16:** A desktop computer with an IP address of 192.168.20.2 and a default gateway (DGW) of 192.168.20.1. It's connected to the switch via port Fa0/1.
- **PC-PT PC17:** Another desktop computer with an IP address of 192.168.20.4 and a DGW of 192.168.20.1. It's connected to the switch via an unspecified Fast Ethernet port.
- **Laptop4:** A laptop with an IP address of 192.168.20.3 and a DGW of 192.168.20.1. It's connected to the switch via an unspecified Fast Ethernet port.
- **Printer-PT Printer5:** A printer with an IP address of 192.168.20.6 and a DGW of 192.168.20.1. It's connected to the switch via port Fa0/5.
- **IP Phone3:** An IP phone with an IP address of 192.168.20.5 and a DGW of 192.168.20.1. It's connected to the switch via an unspecified Fast Ethernet port.

Justification for topology – PENANG Branch

STAR TOPOLOGY




**Star Topology Description**

A Star Topology is a network design where all devices (computers, servers, printers or IoT devices) are linked to a central device which is typically a switch or hub. The devices in this structure are connected to the central device by a point to point connection and the central device is in charge of communication between the devices.

When one device fails in a star network, the other devices are not affected because all the devices are in contact with each other through the central hub. Nevertheless, the central device is essential to the performance and integrity of the network- failure of the hub or switch will bring the whole network to a halt.

**Advantages of Star Topology**

Easy to install and manage.

Failure of one does not impact on others.

Devices can be added and removed easily without disrupting the network.

Traffic management in a centralized manner.

Star Topology has drawbacks.

Has more cabling requirements than other topologies.

When the central networking equipment (hub/switch) fails, the entire network will not work.

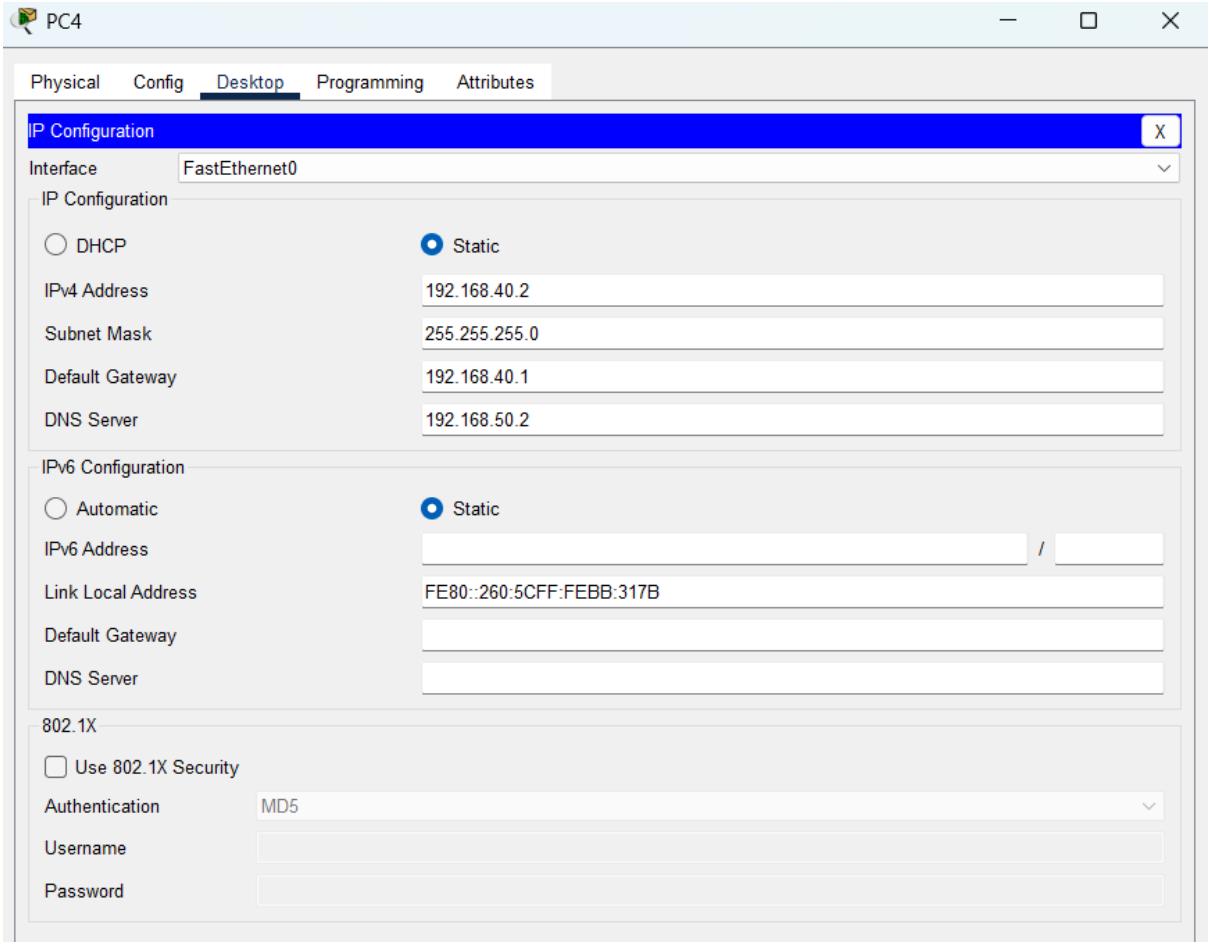
IP Addressing Scheme PENANG Branch:

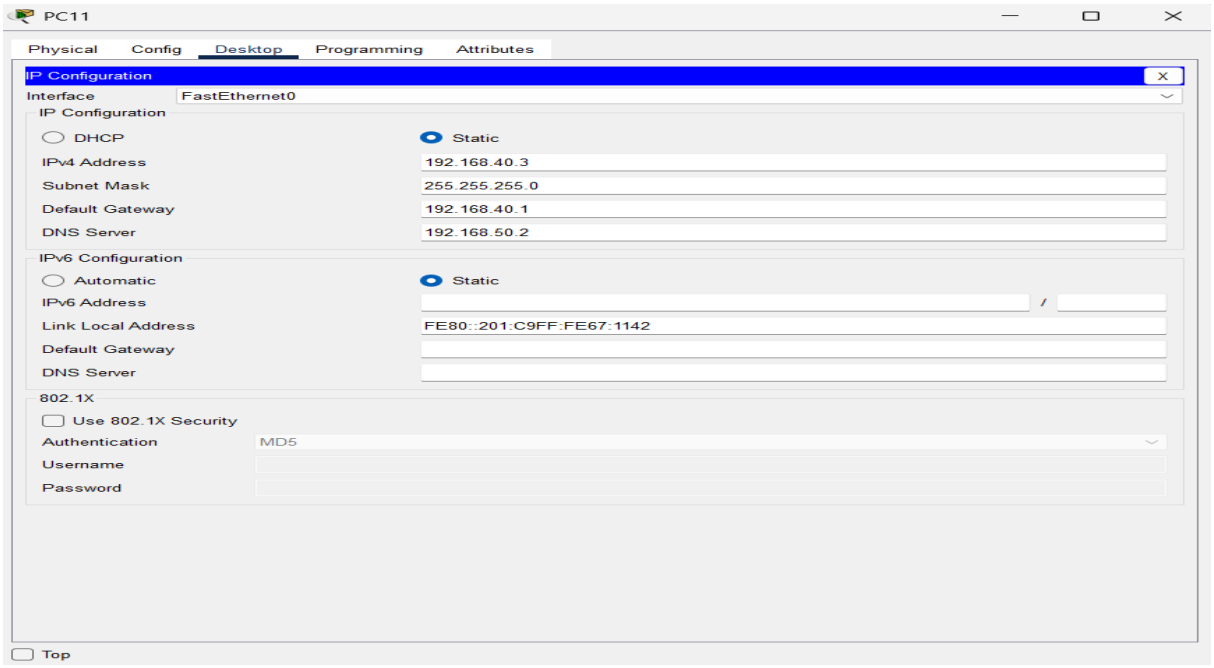
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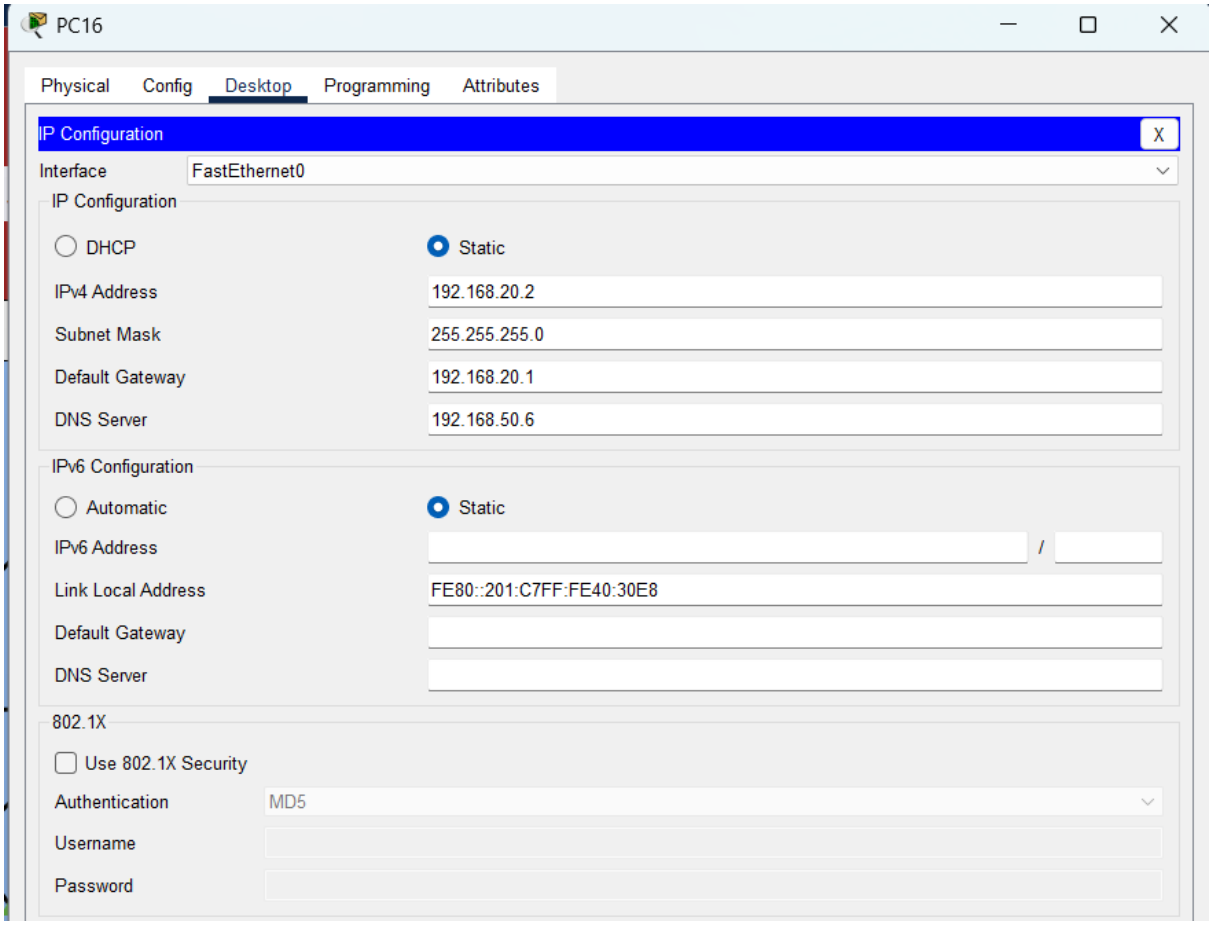
| Department | Device | IP Address | Default Gateway |
|---|---|---|---|
| **Server Room (192.168.50.0/24)** | DNS Server | 192.168.50.5 | 192.168.50.1 |
| | Web Server | 192.168.50.6 | 192.168.50.1 |
| | FTP Server | 192.168.50.7 | 192.168.50.1 |
| | Other Servers | 192.168.50.8 | 192.168.50.1 |
| **Reception Office (192.168.30.0/24)** | PC1 | 192.168.30.2 | 192.168.30.1 |
| | PC2 | 192.168.30.3 | 192.168.30.1 |
| | PC3 | 192.168.30.4 | 192.168.30.1 |
| | Printer | 192.168.30.6 | 192.168.30.1 |
| | IP Phone | 192.168.30.7 | 192.168.30.1 |
| | Webcam | 192.168.30.5 | 192.168.30.1 |
| **HR Room (192.168.40.0/24)** | PC1 | 192.168.40.2 | 192.168.40.1 |
| | PC2 | 192.168.40.3 | 192.168.40.1 |
| | PC3 | 192.168.40.4 | 192.168.40.1 |
| | Printer | 192.168.40.5 | 192.168.40.1 |
| | Laptop1 | 192.168.40.6 | 192.168.40.1 |
| | Laptop2 | 192.168.40.7 | 192.168.40.1 |
| | Webcam | 192.168.40.9 | 192.168.40.1 |
| | Air Conditioner | 192.168.40.8 | 192.168.40.1 |

| Department | Device | IP Address | Default Gateway |
|---|---|---|---|
| **IT Department (192.168.20.0/24)** | PC1 | 192.168.20.2 | 192.168.20.1 |
| | PC2 | 192.168.20.3 | 192.168.20.1 |
| | Laptop1 | 192.168.20.4 | 192.168.20.1 |
| | Laptop2 | 192.168.20.5 | 192.168.20.1 |
| | Printer | 192.168.20.6 | 192.168.20.1 |
| | IP Phone | 192.168.20.7 | 192.168.20.1 |

**End devices for PENANG Branch**: we also added the DNS server for all branches.

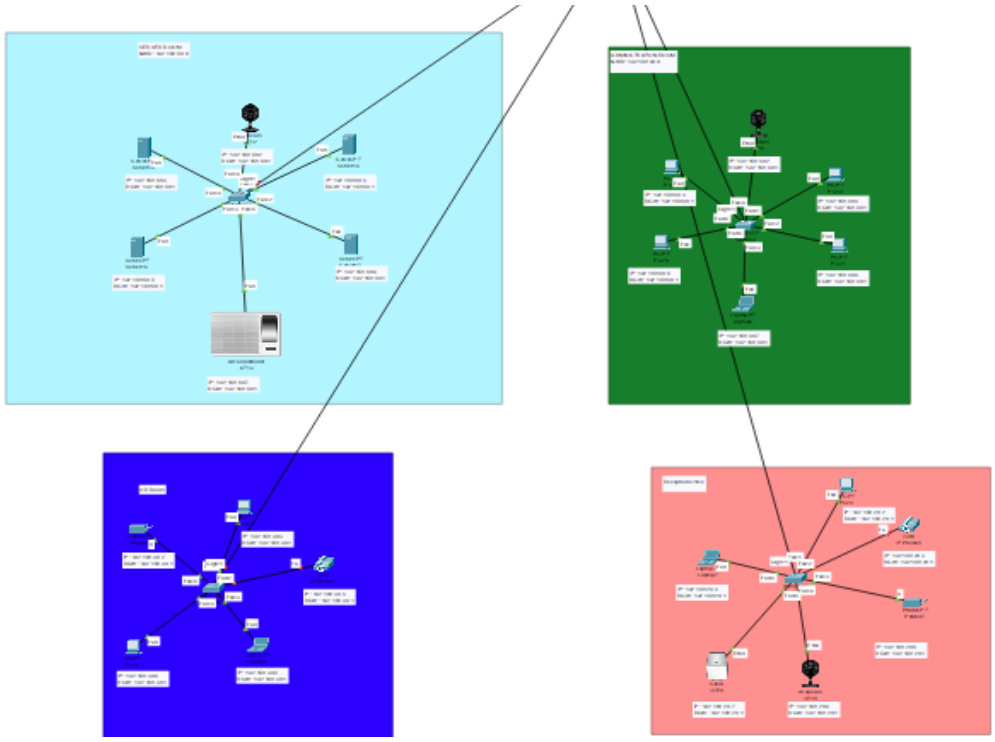



PC11
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.40.3
Subnet Mask
255.255.255.0
Default Gateway
192.168.40.1
DNS Server
192.168.50.2
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::201:C9FF:FE67:1142
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top


PC16
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.20.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.20.1
DNS Server
192.168.50.6
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::201:C7FF:FE40:30E8
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

# JOHOR BRANCH-

# Name: K.M. ROUNAK UDDIN AHMED (TP090616)

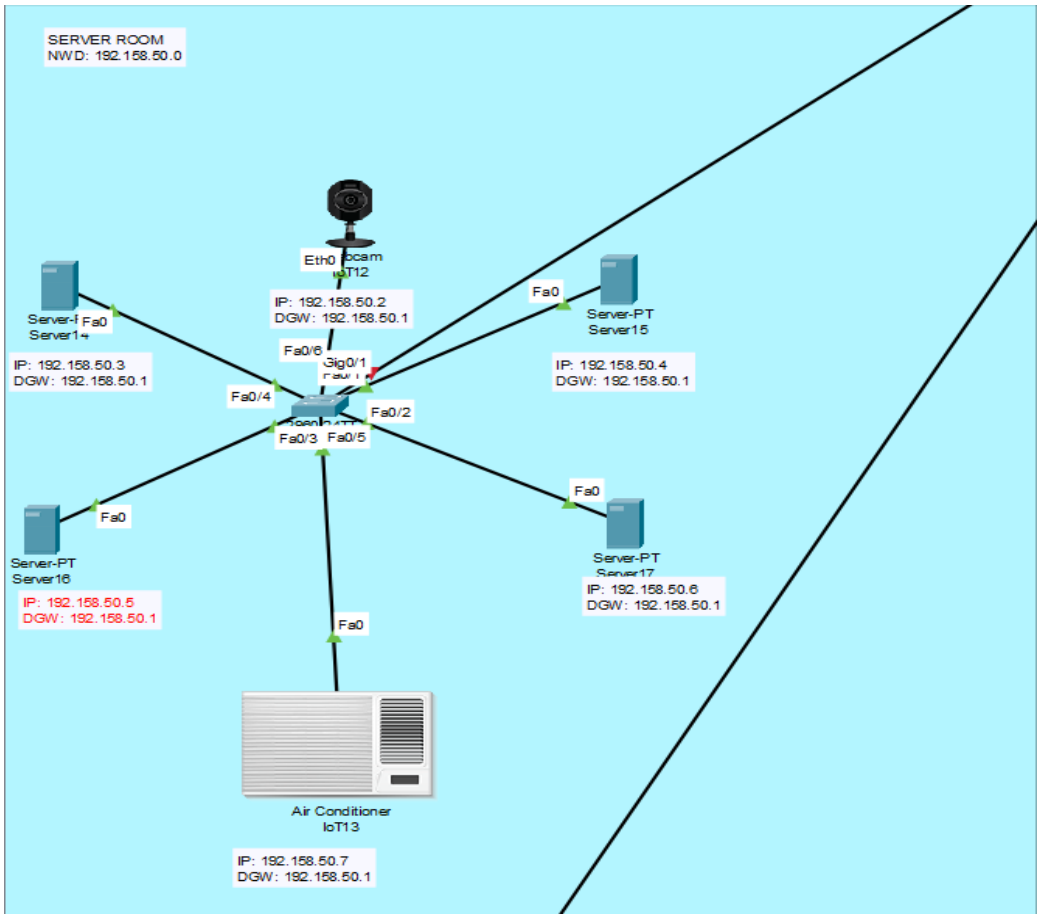


The picture above shows the topology for the PENANG Branch. Here we can see that the network is implemented using “Star Topology”. The branch has 4 separate rooms:

1. Server Room
2. Reception Office
3. HR Department
4. Administration Office

In that manner the Switch is fundamentally connecting the departments to one another. And the switches in turn are linked to router to reach the other branches in the network. A single switch is put in each room and all the rest of the network equipment is hardwired into it.

## JUSTIFICATION FOR SERVER ROOM:




### Description

This server room layout exhibits a star network design in which all the devices lead to one central switch which is further connected to the router. The addressing is organised into an IP scheme structure in the 192.158.50.0 network which makes it easy to manage and communicate. The installation also incorporates IoT devices (CCTV camera and air conditioner) into the server environment, a contemporary infrastructure kit, where IT and smart systems become one in order to provide security and environmental control..

### Network Components

1. **Router (Central Gateway)**
    - Interface: **Gig0/1** connected to the switch
    - Default Gateway for all devices: **192.158.50.1**
    - Acts as the main communication hub for all devices.
2. **Switch (Central Connection Point)**
    - Linked to the router and distributes connections to server, IoT devices and cameras.Servers

- **Server14**
  - IP: 192.158.50.3
  - DGW: 192.158.50.1
- **Server15**
  - IP: 192.158.50.4
  - DGW: 192.158.50.1
- **Server16**
  - IP: 192.158.50.5
  - DGW: 192.158.50.1
- **Server17**
  - IP: 192.158.50.6
  - DGW: 192.158.50.1

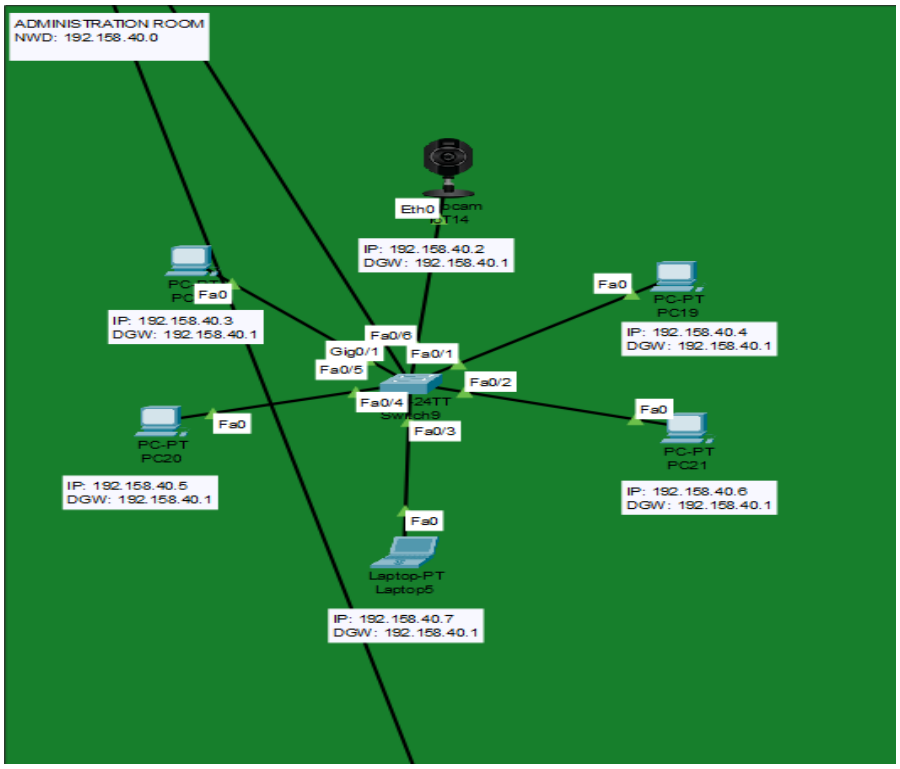
The uses of these servers are probably storage, applications or database management in the server room.

**3.IoT Devices**

- **IP Camera (IoT12)**
  - IP: 192.158.50.2
  - DGW: 192.158.50.1
- **Air Conditioner (IoT13)**
  - IP: 192.158.50.7
  - DGW: 192.158.50.1

These intelligent devices have connections to the network, which are used to monitor (camera) and manage (AC) remotely.

**Justification for Administration Room:**




**Network Diagram Description: Administration Room**

This network diagram represents the layout and connections of devices within an Administration Room, identified by the network address 192.158.40.0.

**Network Components:**

Router (NWD: 192.158.40.0)

This is a point through which the network is presented, and it enables communication between the different devices and even other networks.

Switch (24TT Switch9)

A central device to which various devices are connected using Fast Ethernet (Fa) ports.

The switch has multiple ports labeled Fa0/1 through Fa0/6 and a Gigabit port Gig0/1.

Devices Connected to the Switch:

**PCs**

PC18 connected via port Fa0/5

IP: 192.158.40.3

Gateway (DGW): 192.158.40.1

PC19 connected via port Fa0/2

IP: 192.158.40.4

Gateway: 192.158.40.1

PC20 connected via port Fa0/0

IP: 192.158.40.5

Gateway: 192.158.40.1

PC21 connected via port Fa0/3

IP: 192.158.40.6

Gateway: 192.158.40.1

Laptop

Laptop5 connected via port Fa0

IP: 192.158.40.7

Gateway: 192.158.40.1

IP Camera (pcam)

Connected via Ethernet 0 (Eth0)

IP: 192.158.40.2

Gateway: 192.158.40.1

**Connections:**

All devices use the default gateway IP 192.158.40.1, which is likely the router's IP.

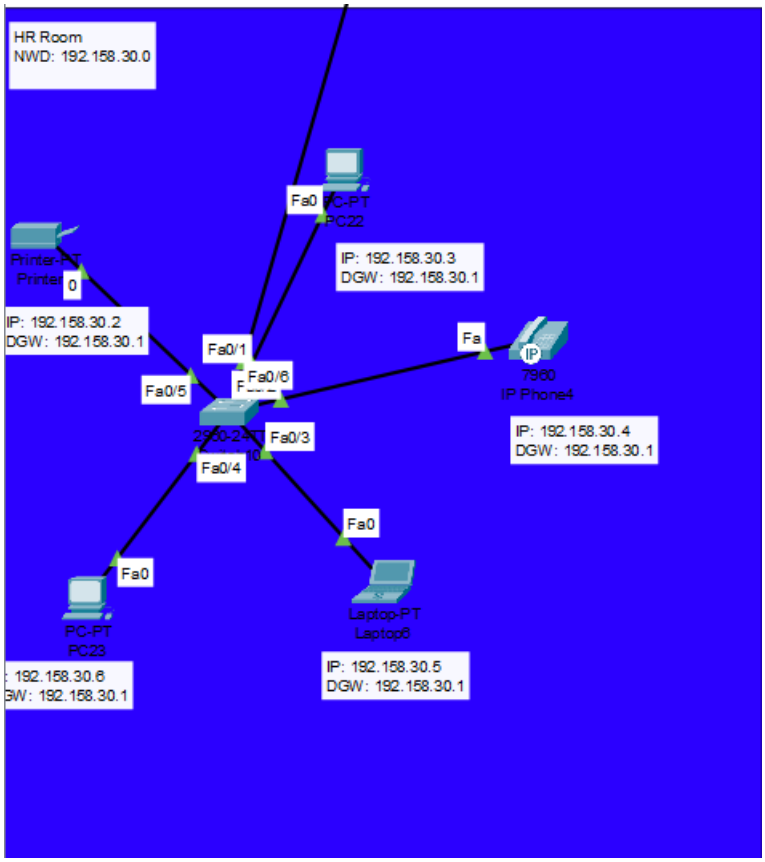
The switch serves as the point in which all the PCs, laptop, and the IP camera are connected, through their respective Fast Ethernet ports.

The IP camera is linked through Ethernet 0 port (Eth0), which is different than the PCs and the laptop that are connected through Fast Ethernet (Fa0).

The router is connected to the switch through the port Fa0/1.

This configuration is recommended as a typical local area network (LAN) setup in an administration room with the network of several devices connected by a switch, and a router is the gateway.

**Justification for HR Room:**




This network diagram depicts the layout and interconnection of devices within the HR Room, with the network address 192.158.30.0.

**Network Components:**

Switch (2960-1)

Central networking device that connects all other devices in the HR room.

It has multiple Fast Ethernet (Fa) ports: Fa0/2, Fa0/3, Fa0/4, Fa0/5, and a Gigabit port Gig0/1.

**Devices Connected to the Switch:**

**PCs:**

PC22 connected via port Fa0/2

IP: 192.158.30.3

Default Gateway (DGW): 192.158.30.1

PC23 connected via port Fa0/4

IP: 192.158.30.6

Gateway: 192.158.30.1

Laptop:

Laptop0 connected via port Fa0

IP: 192.158.30.5

Gateway: 192.158.30.1

Printer:

Connected via port Fa0/5

IP: 192.158.30.2

Gateway: 192.158.30.1

IP Phone (7960):

Connected via port Fa0/3

IP: 192.158.30.4

Gateway: 192.158.30.1

**Connections:**

All devices share the default gateway IP 192.158.30.1, likely the router's IP address.

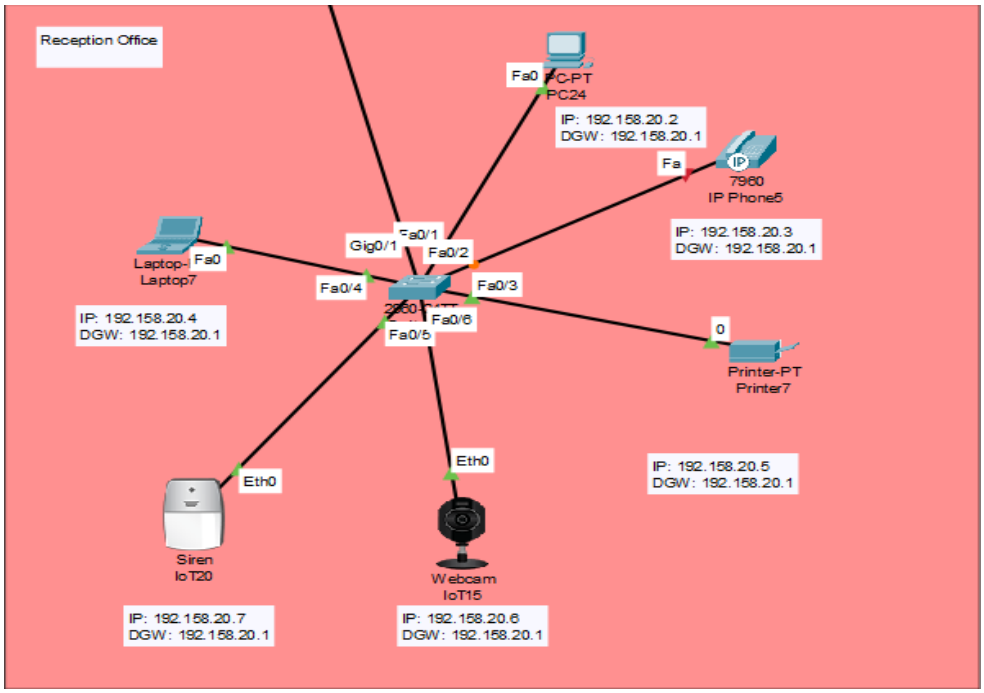
The switch will be the central node of the network, which will connect PCs, laptop, printer, and IP phone.

The printer and IP phone have special dedicated connections on special Fast Ethernet ports.

The PCs and the laptop are also directly linked with Fast Ethernet interfaces.

Gig0/1 is a Gigabit port, which links the switch to the rest of the network infrastructure or router.

**Justification for RECEPTION OFFICE:**




This network diagram depicts the layout and interconnection of devices within the HR Room, with the network address 192.158.30.0.

**Network Components:**

Switch (2960-1)

Central networking device that connects all other devices in the HR room.

It has multiple Fast Ethernet (Fa) ports: Fa0/2, Fa0/3, Fa0/4, Fa0/5, and a Gigabit port Gig0/1.

Devices Connected to the Switch:

**PCs:**

PC22 connected via port Fa0/2

IP: 192.158.30.3

Default Gateway (DGW): 192.158.30.1

PC23 connected via port Fa0/4

IP: 192.158.30.6

Gateway: 192.158.30.1

**Laptop:**

Laptop0 connected via port Fa

IP: 192.158.30.5

Gateway: 192.158.30.1

**Printer:**

Connected via port Fa0/5

IP: 192.158.30.2

Gateway: 192.158.30.1

**IP Phone (7960):**

Connected via port Fa0/

IP: 192.158.30.4

Gateway: 192.158.30.1

**Connections:**

All devices share the default gateway IP 192.158.30.1, likely the router's IP address.

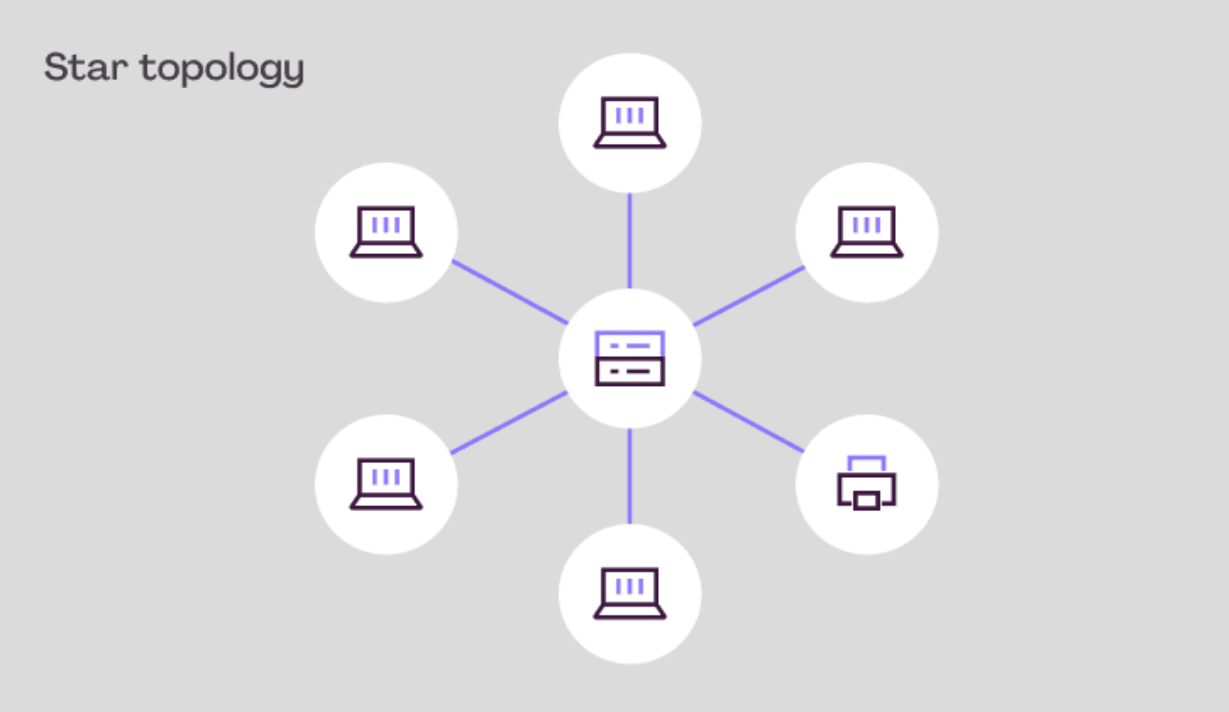
The switch will be the main point of the network, which will connect PCs, laptop, printer, and IP phone.

The printer and IP phone have special connections on certain Fast Ethernet port.

The laptop and the PCs are also directly linked together with Fast Ethernet interfaces.

The Gigabit port Gig0/1 links the switch to the remainder of the network infrastructure/ router.

**Star Topology Description**




A Star Topology This is a type of network architecture in which all devices (computers, servers, printers or IoT devices) are connected to a central device that is usually a switch or hub. The devices in this structure are linked to the central device through a point to point connection and the central device is in control of communication between the devices.

Failure of one device in a star network does not impact the other devices since all the devices are connected to the central hub. However, the main device is vital to the functioning and integrity of the network- the breakdown of the hub or switch will crippled the entire network.

**Advantages of Star Topology**

Easy to install and manage.

Failure will not affect other people.

The addition and removal of devices can be easily carried out without interfering with the network.

Managing traffic centrally.

Star Topology has drawbacks.

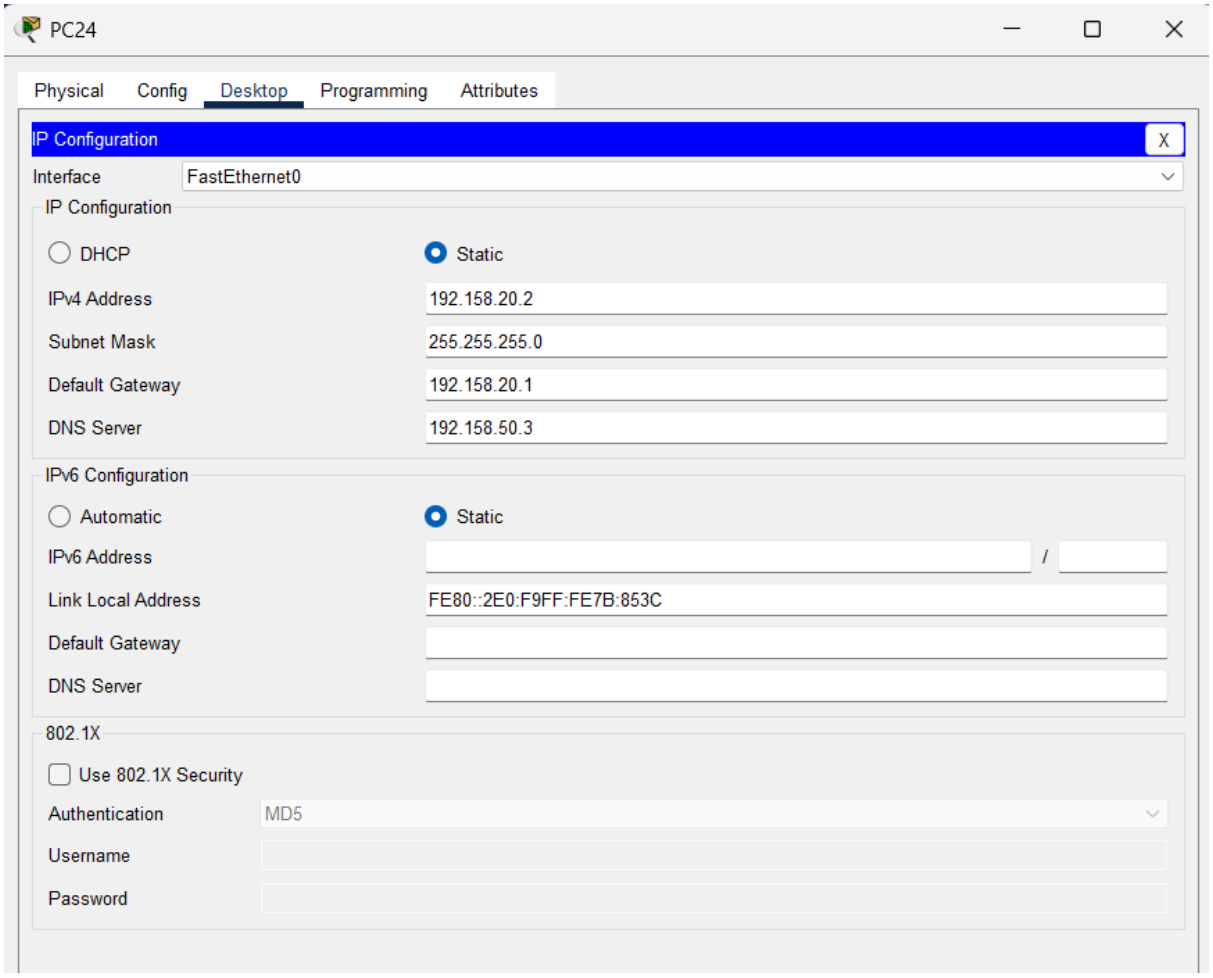
More cabling than other topologies.

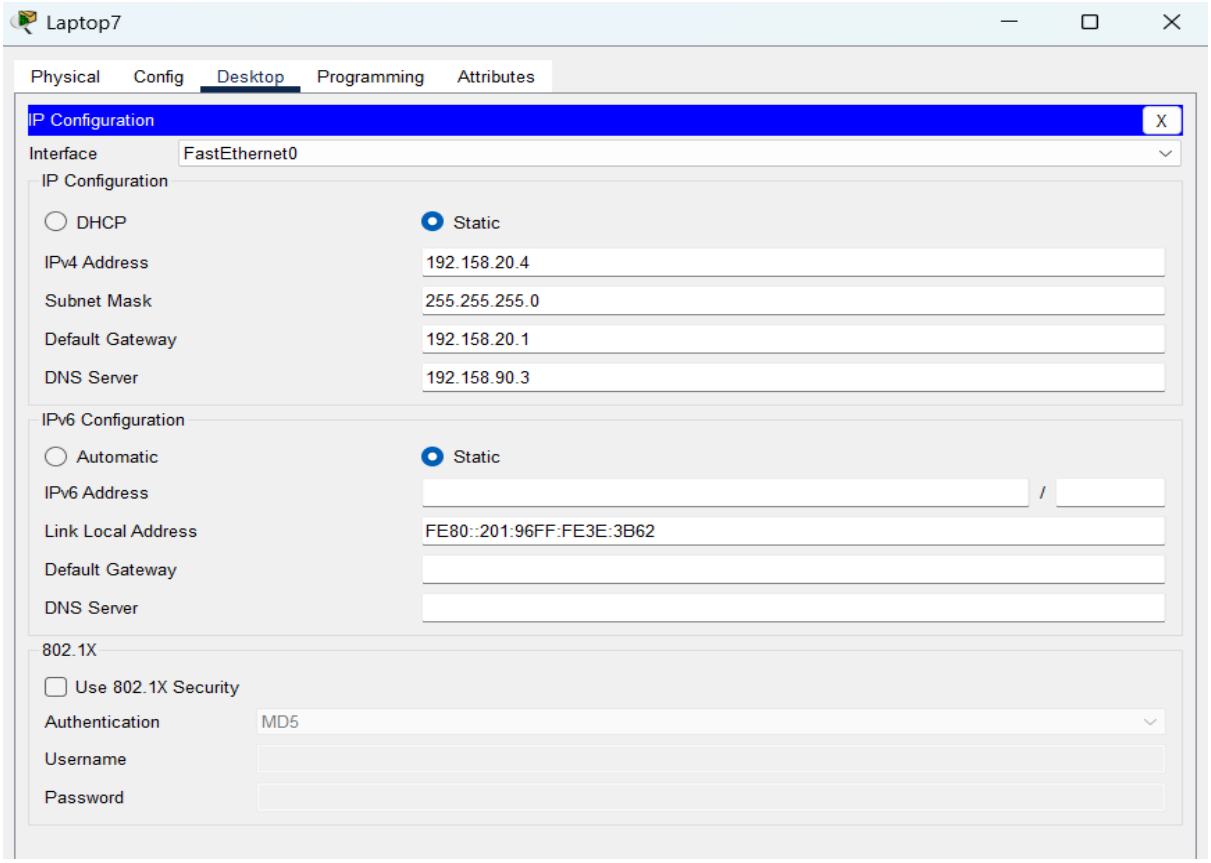
The central networking equipment (hub/switch) will fail and the whole network will stop functioning.

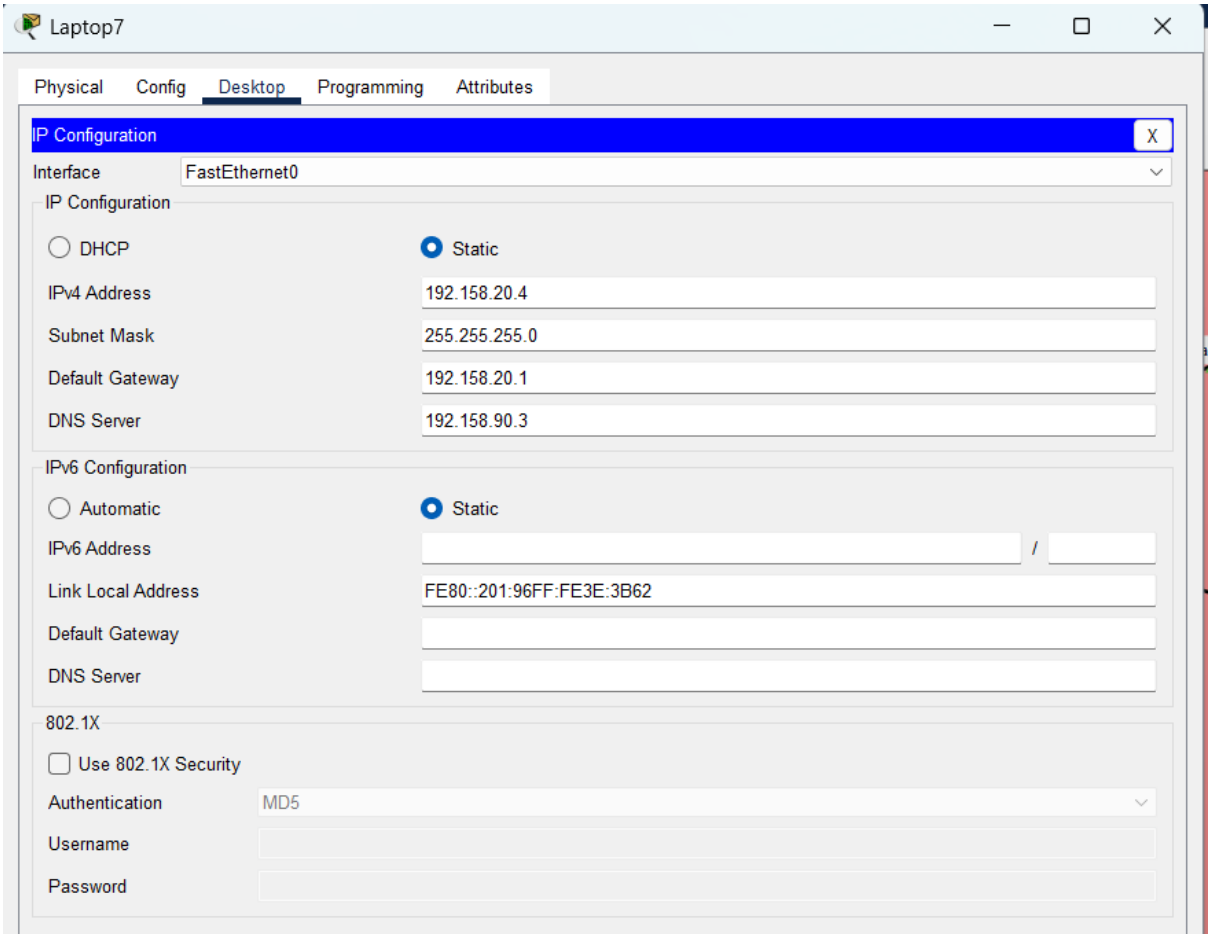
**IP Addressing Schema Table**

| Zone (Color) | Device/Role | IP Address | Subnet Mask | Default Gateway |
|---|---|---|---|---|
| Light Blue (Server Zone) | Router Interface | 192.168.10.1 | 255.255.255.0 | |
| Server | 192.168.10.2 | 255.255.255.0 | 192.168.10.1 | |
| PC1 | 192.168.10.10 | 255.255.255.0 | 192.168.10.1 | |
| PC2 | 192.168.10.11 | 255.255.255.0 | 192.168.10.1 | |
| Printer | 192.168.10.30 | 255.255.255.0 | 192.168.10.1 | |
| Green (Office LAN 1) | Router Interface | 192.168.20.1 | 255.255.255.0 | |
| PC1 | 192.168.20.10 | 255.255.255.0 | 192.168.20.1 | |
| PC2 | 192.168.20.11 | 255.255.255.0 | 192.168.20.1 | |
| Printer | 192.168.20.100 | 255.255.255.0 | 192.168.20.1 | |
| \| Dark Blue (Lab / Office LAN 2) | Router Interface | 192.168.30.1 | 255.255.255.0 | |
| PC1 | 192.168.30.10 | 255.255.255.0 | 192.168.30.1 | |
| PC2 | 192.168.30.11 | 255.255.255.0 | 192.168.30.1 | |
| PC3 | 192.168.30.12 | 255.255.255.0 | 192.168.30.1 | |
| Printer | 192.168.30.100 | 255.255.255.0 | 192.168.30.1 | |
| Pink (Admin LAN) | Router Interface | 192.168.40.1 | 255.255.255.0 | |
| PC1 | 192.168.40.10 | 255.255.255.0 | 192.168.40.1 | |
| PC2 | 192.168.40.11 | 255.255.255.0 | 192.168.40.1 | |
| Printer | 192.168.40.100 | 255.255.255.0 | 192.168.40.1 | |

**End devices for JOHOR Branch**:




Laptop7
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.158.20.4
Subnet Mask
255.255.255.0
Default Gateway
192.158.20.1
DNS Server
192.158.90.3
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::201:96FF:FE3E:3B62
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

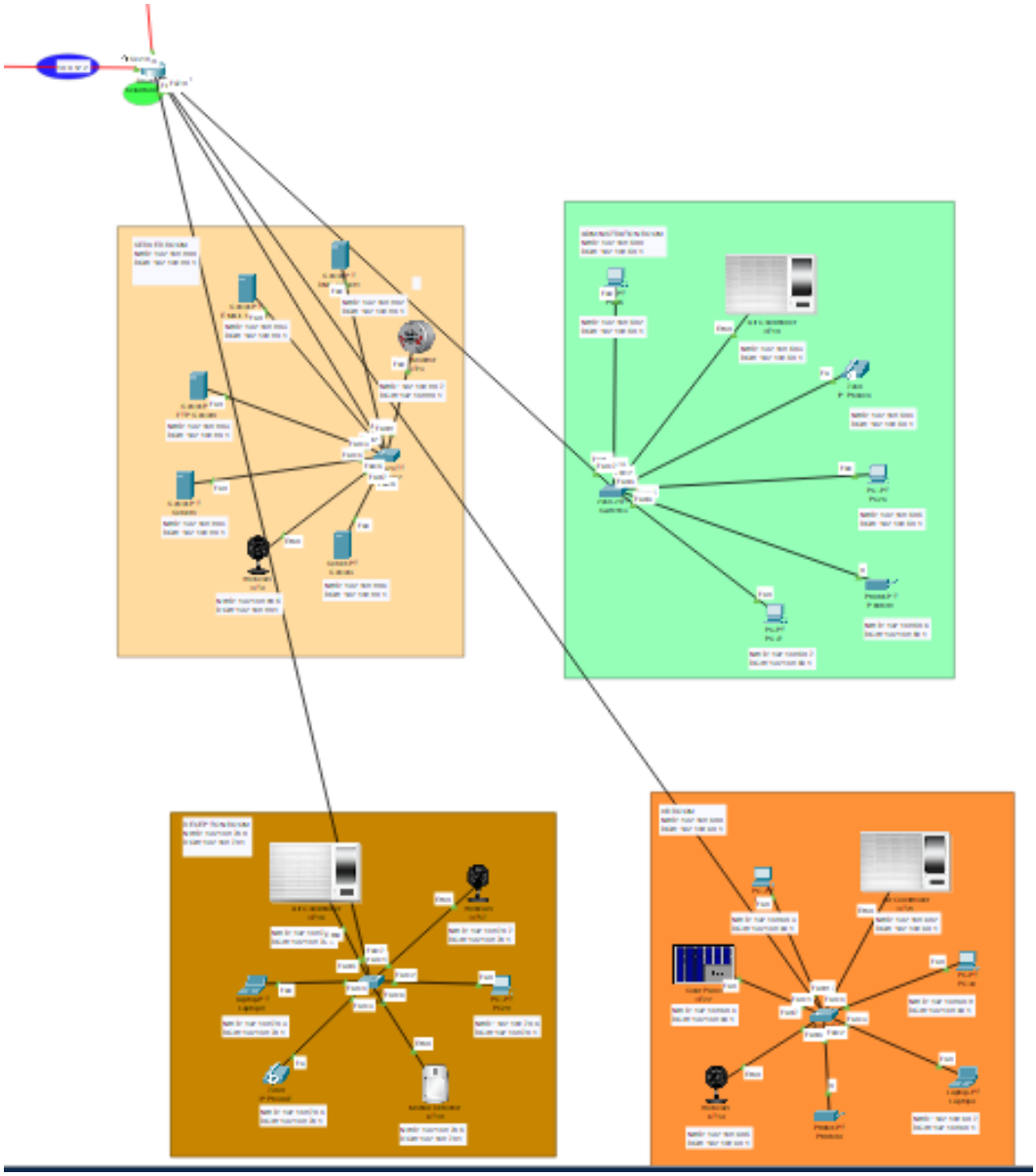

Laptop7
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.158.20.4
Subnet Mask
255.255.255.0
Default Gateway
192.158.20.1
DNS Server
192.158.90.3
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::201:96FF:FE3E:3B62
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

# KELANTAN BRANCH: MD SHAIKH JOBYER (TP092052)

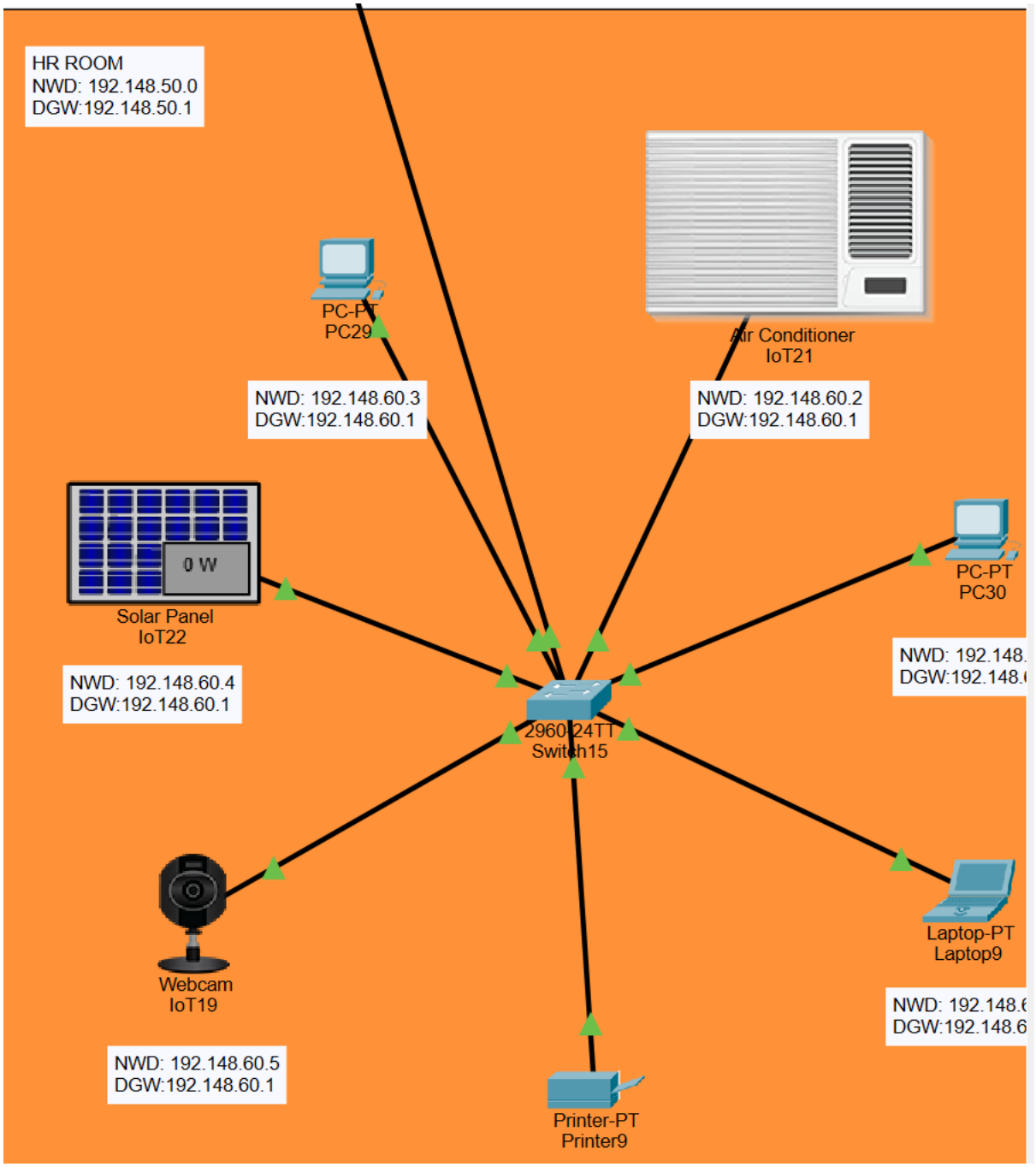
The picture below shows the topology for the KELANTAN Branch. The network is implemented using “Star Topology”. The branch has 4 separate rooms:

1. Hr room
2. Reciption room
3. Server room
4. Administration Room

In this topology the switch interconnects devices within each room, and the room switches connect to the router. Each room uses its own /24 subnet with a room gateway of x.x.x.1 for clear segmentation and easy troubleshooting.



# JUSTIFICATION FOR THE HR ROOM:




The image shows the hr room network with a network address (NWD) of 192.148.60.0/24. All devices connect in a star to a 2960 switch and use the default gateway (DGW) 192.148.60.1.

**Devices and Connections**

• Air Conditioner IoT21: IP 192.148.60.2; DGW 192.148.60.1.

• PC-PT PC29: IP 192.148.60.3; DGW 192.148.60.1.

• Solar Panel IoT22: IP 192.148.60.4; DGW 192.148.60.1.

• Webcam IoT19: IP 192.148.60.5; DGW 192.148.60.1.

• Laptop-PT Laptop9: IP 192.148.60.6; DGW 192.148.60.1.

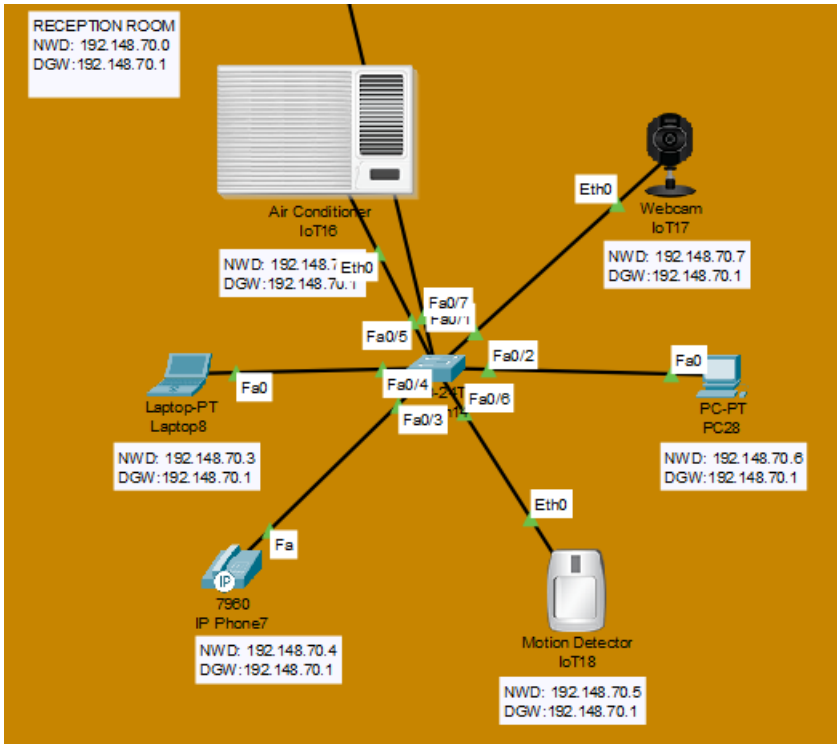
• Printer-PT Printer9: IP 192.148.60.7; DGW 192.148.60.1.

• PC-PT PC30: IP 192.148.60.8; DGW 192.148.60.1.

**Notes:**

• Room switch aggregates office endpoints and green-energy IoT (solar panel telemetry).

## JUSTIFICATION FOR THE RECEPTION ROOM:




The image shows the reception room network with a network address (NWD) of 192.148.70.0/24. All devices connect in a star to a 2960 switch and use the default gateway (DGW) 192.148.70.1.

**Devices and Connections**

• Air Conditioner IoT16: IP 192.148.70.2; DGW 192.148.70.1.

• Laptop-PT Laptop8: IP 192.148.70.3; DGW 192.148.70.1.

• IP Phone7 (7960): IP 192.148.70.4; DGW 192.148.70.1.

• Motion Detector IoT18: IP 192.148.70.5; DGW 192.148.70.1.

• PC-PT PC28: IP 192.148.70.6; DGW 192.148.70.1.

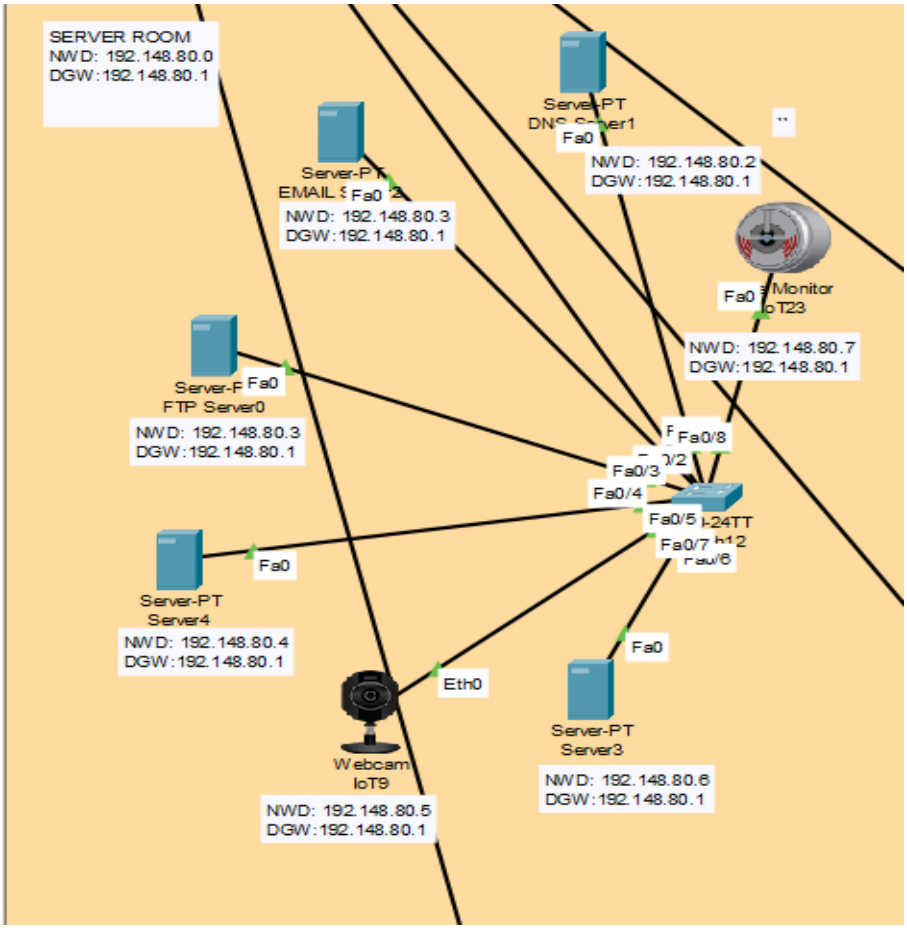
• Webcam IoT17: IP 192.148.70.7; DGW 192.148.70.1.

**Notes:**

• Switch 2960-24TT concentrates all endpoints; router link provides branch connectivity.

• IoT/security devices (webcam, motion detector) are statically addressed for monitoring.

## JUSTIFICATION FOR THE SERVER ROOM:



The image shows the server room network with a network address (NWD) of 192.148.80.0/24. All devices connect in a star to a 2960 switch and use the default gateway (DGW) 192.148.80.1.

**Devices and Connections**

• Server-PT Server1: IP 192.148.80.2; DGW 192.148.80.1.

• Server-PT Server2: IP 192.148.80.3; DGW 192.148.80.1.

• Server-PT Server4: IP 192.148.80.4; DGW 192.148.80.1.

• Webcam IoT9: IP 192.148.80.5; DGW 192.148.80.1.

• Server-PT Server3: IP 192.148.80.6; DGW 192.148.80.1.

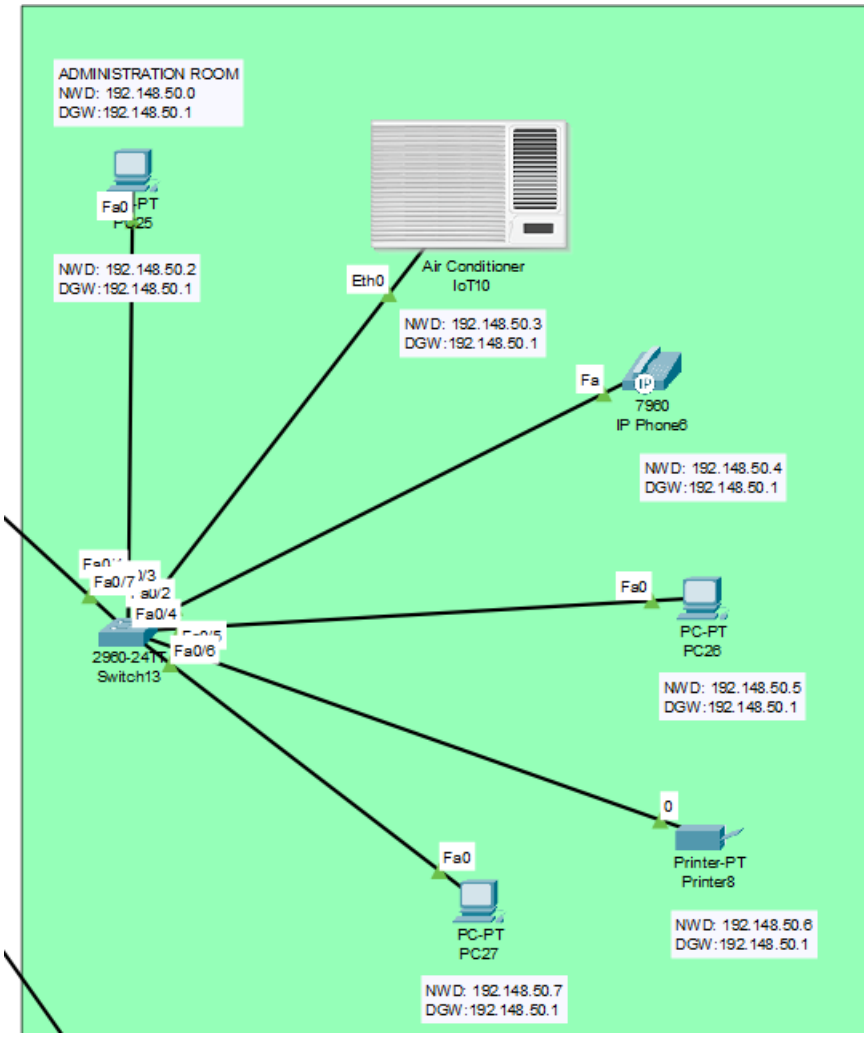
• Fire Monitor IoT23: IP 192.148.80.7; DGW 192.148.80.1.

**Notes:**

• Centralized services (web/DNS/file) are placed here to support other rooms.

• Static addressing used for deterministic access to servers and CCTV.

# JUSTIFICATION FOR THE ADMINISTRATION ROOM:




The image shows the administration room network with a network address (NWD) of 192.148.50.0/24. All devices connect in a star to a 2960 switch and use the default gateway (DGW) 192.148.50.1.

**Devices and Connections**

• PC-PT PC25: IP 192.148.50.2; DGW 192.148.50.1.

• Air Conditioner IoT10: IP 192.148.50.3; DGW 192.148.50.1.

• IP Phone 7960 (Phone6): IP 192.148.50.4; DGW 192.148.50.1.

• PC-PT PC26: IP 192.148.50.5; DGW 192.148.50.1.

• Printer-PT Printer8: IP 192.148.50.6; DGW 192.148.50.1.

• PC-PT PC27: IP 192.148.50.7; DGW 192.148.50.1.

**Notes:**

• Gig0/1 uplink connects the room switch to the branch router for inter-room communication.

• Static IPs keep devices reachable and consistent with the addressing plan.

## IP Addressing Scheme – KELANTAN Branch

| Department (Subnet) | Device | IP Address | Default Gateway |
|---|---|---|---|
| Server Room (192.148.80.0/24) | Server-PT Server1 | 192.148.80.2 | 192.148.80.1 |
| Server Room (192.148.80.0/24) | Server-PT Server2 | 192.148.80.3 | 192.148.80.1 |
| Server Room (192.148.80.0/24) | Server-PT Server4 | 192.148.80.4 | 192.148.80.1 |
| Server Room (192.148.80.0/24) | Webcam IoT9 | 192.148.80.5 | 192.148.80.1 |
| Server Room (192.148.80.0/24) | Server-PT Server3 | 192.148.80.6 | 192.148.80.1 |
| Server Room (192.148.80.0/24) | Fire Monitor IoT23 | 192.148.80.7 | 192.148.80.1 |
| Administration Room (192.148.50.0/24) | PC-PT PC25 | 192.148.50.2 | 192.148.50.1 |
| Administration Room (192.148.50.0/24) | Air Conditioner IoT10 | 192.148.50.3 | 192.148.50.1 |
| Administration Room (192.148.50.0/24) | IP Phone 7960 (Phone6) | 192.148.50.4 | 192.148.50.1 |
| Administration Room (192.148.50.0/24) | PC-PT PC26 | 192.148.50.5 | 192.148.50.1 |

| | | | |
|---|---|---|---|
| Administration Room (192.148.50.0/24) | Printer-PT Printer8 | 192.148.50.6 | 192.148.50.1 |
| Administration Room (192.148.50.0/24) | PC-PT PC27 | 192.148.50.7 | 192.148.50.1 |
| Reception Room (192.148.70.0/24) | Air Conditioner IoT16 | 192.148.70.2 | 192.148.70.1 |
| Reception Room (192.148.70.0/24) | Laptop-PT Laptop8 | 192.148.70.3 | 192.148.70.1 |
| Reception Room (192.148.70.0/24) | IP Phone7 (7960) | 192.148.70.4 | 192.148.70.1 |
| Reception Room (192.148.70.0/24) | Motion Detector IoT18 | 192.148.70.5 | 192.148.70.1 |
| Reception Room (192.148.70.0/24) | PC-PT PC28 | 192.148.70.6 | 192.148.70.1 |
| Reception Room (192.148.70.0/24) | Webcam IoT17 | 192.148.70.7 | 192.148.70.1 |
| HR Room (192.148.60.0/24) | Air Conditioner IoT21 | 192.148.60.2 | 192.148.60.1 |
| HR Room (192.148.60.0/24) | PC-PT PC29 | 192.148.60.3 | 192.148.60.1 |
| HR Room (192.148.60.0/24) | Solar Panel IoT22 | 192.148.60.4 | 192.148.60.1 |
| HR Room (192.148.60.0/24) | Webcam IoT19 | 192.148.60.5 | 192.148.60.1 |
| HR Room (192.148.60.0/24) | Laptop-PT Laptop9 | 192.148.60.6 | 192.148.60.1 |
| HR Room (192.148.60.0/24) | Printer-PT Printer9 | 192.148.60.7 | 192.148.60.1 |
| HR Room (192.148.60.0/24) | PC-PT PC30 | 192.148.60.8 | 192.148.60.1 |

## Conclusion

The Kelantan Branch uses a clean Star Topology with four segregated /24 subnets. Each room's switch provides a single point of connection and the router at DGW x.x.x.1 enables

inter-room reachability. Static addressing across servers, user devices, and IoT endpoints keeps management simple and aligns with the team's KL/Johor/Penang formats.

### 6.5 PING TESTS: Tanmay Sarker Emon (TP092959)

Ping tests are used to ensure the presence of the network topology, the ping test must have a success rate of at least 50 percent before the topology can be regarded as viable. After doing two or three ping attempts, the process is fruitful by being successful.

Pinging from Router to the Switches :

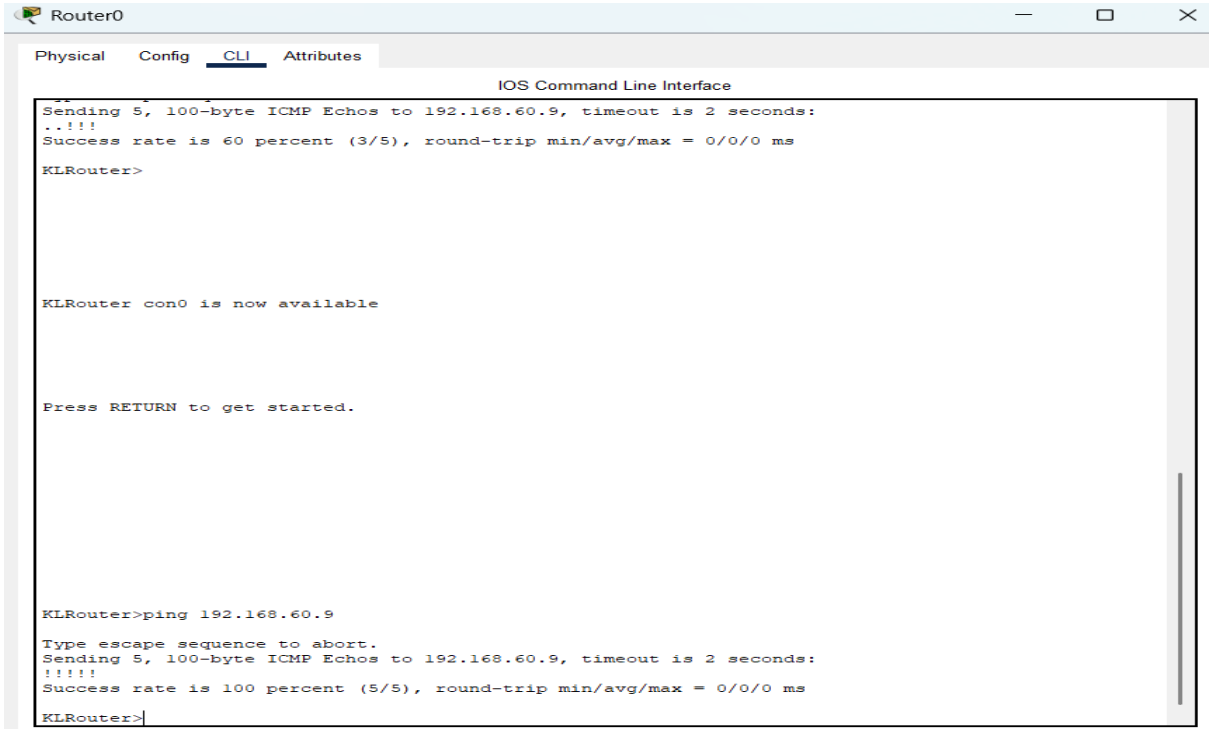



Figure 1: KL router to IT Department

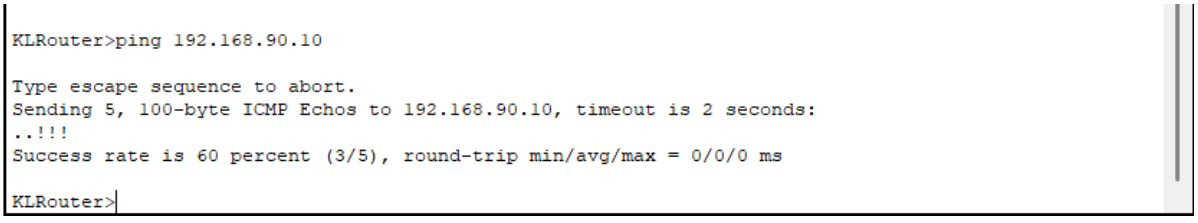



**Figure 2: KLrouter to Server Room**

ICMP echo requests (ping) sent between the KLRouter and two switches in different departments were made to check the existence of a network connection. The switches have different IP addresses and physical locations: - KLRouter – IP: (IT Department) Ping test was

successfully showing 100 percent (five out of five packets received), successfully passing the test, hence showing stable and reliable communication.

- K1Router – IP: (Server Room) Ping test had a 60 percent success rate (three out of five packets). The known packet loss can be due to the delay in ARP resolutions or the interface starting, or minor network traffic congestion. These findings point to the fact that Router0 is capable of connecting with both switches in various departments, which means proper IP addresses, interface settings, and physical links. The reduced compatibility of the Server Room switch indicates that they should either undertake more monitoring or testing.

*6.6 Pinging between two End Devices:Tanmay Sarkar Emon(TP092959)*

We can also ping between two different PCs (for eg.PC-10 of IT Department to PC-2 of HR Department ) as shown below:

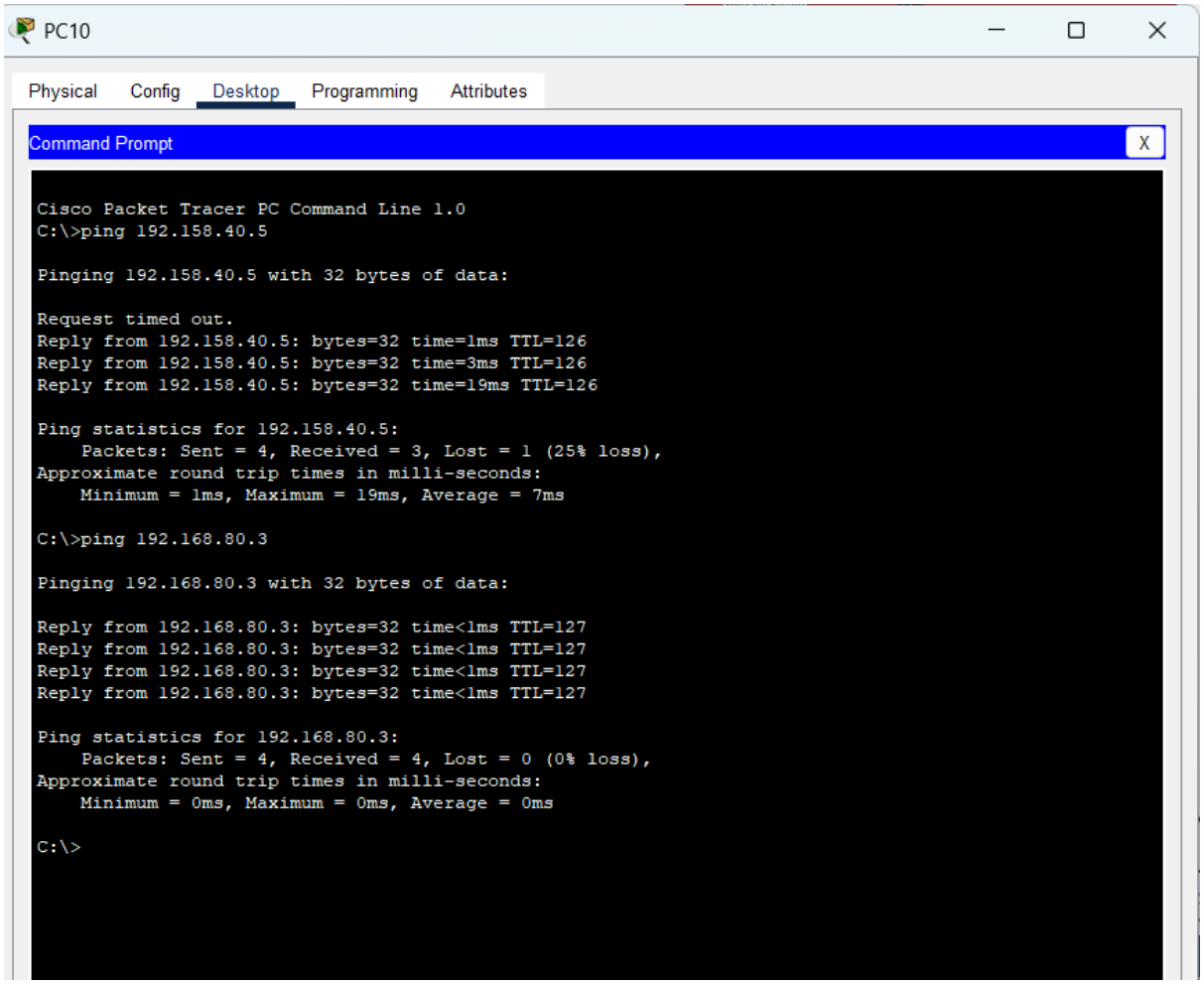



Figure 3: Here 192.168.80.3 is the HR dept pc 2

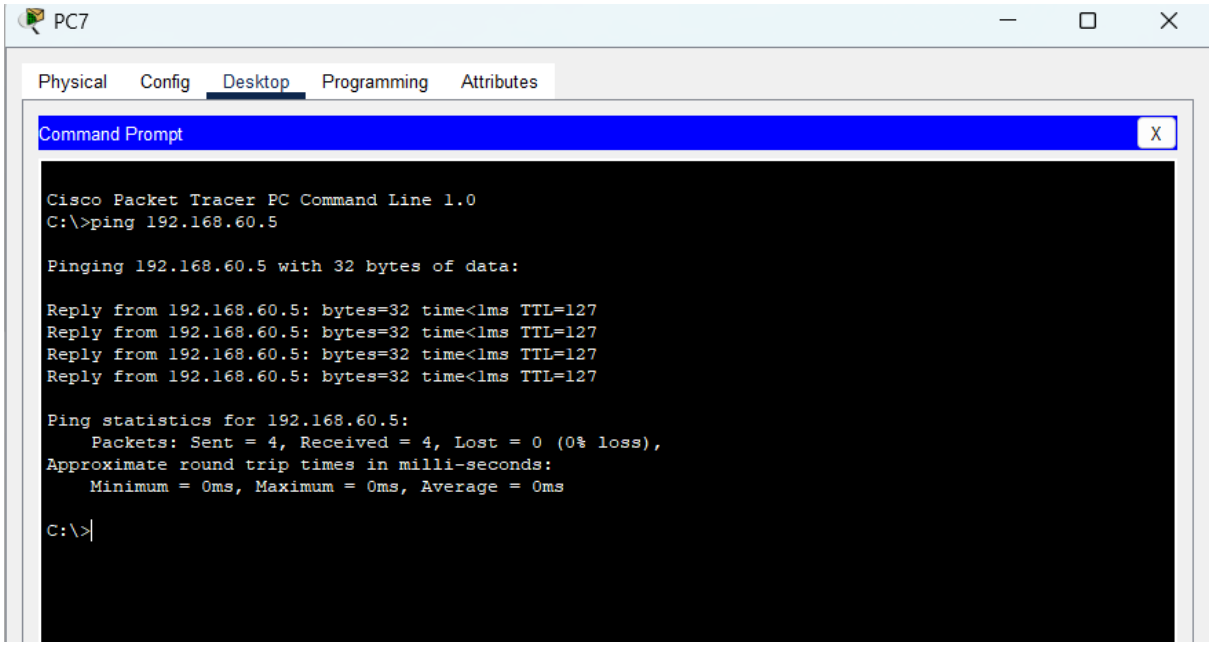



Figure 4: here is Pinging Reception Pc 7 to IT department Pc 10

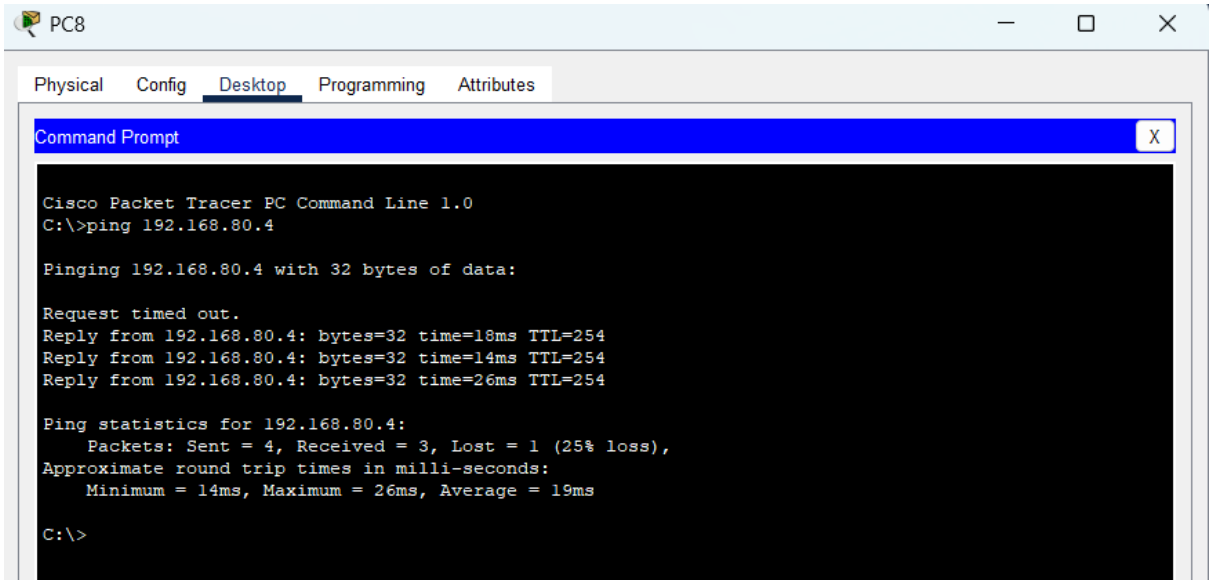



Figure 5: IT department to HR department (CCTV)

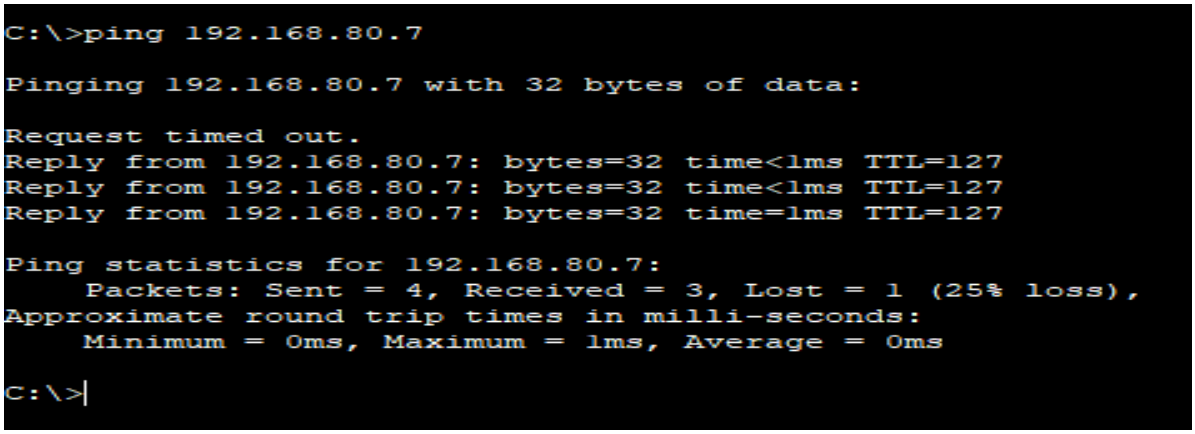
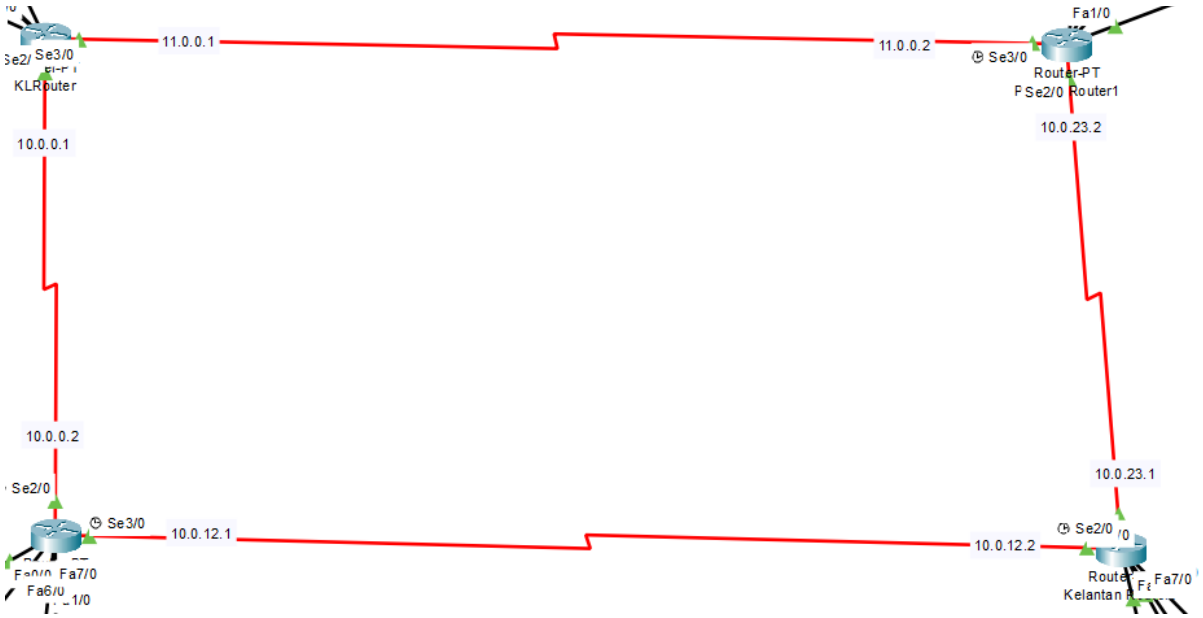


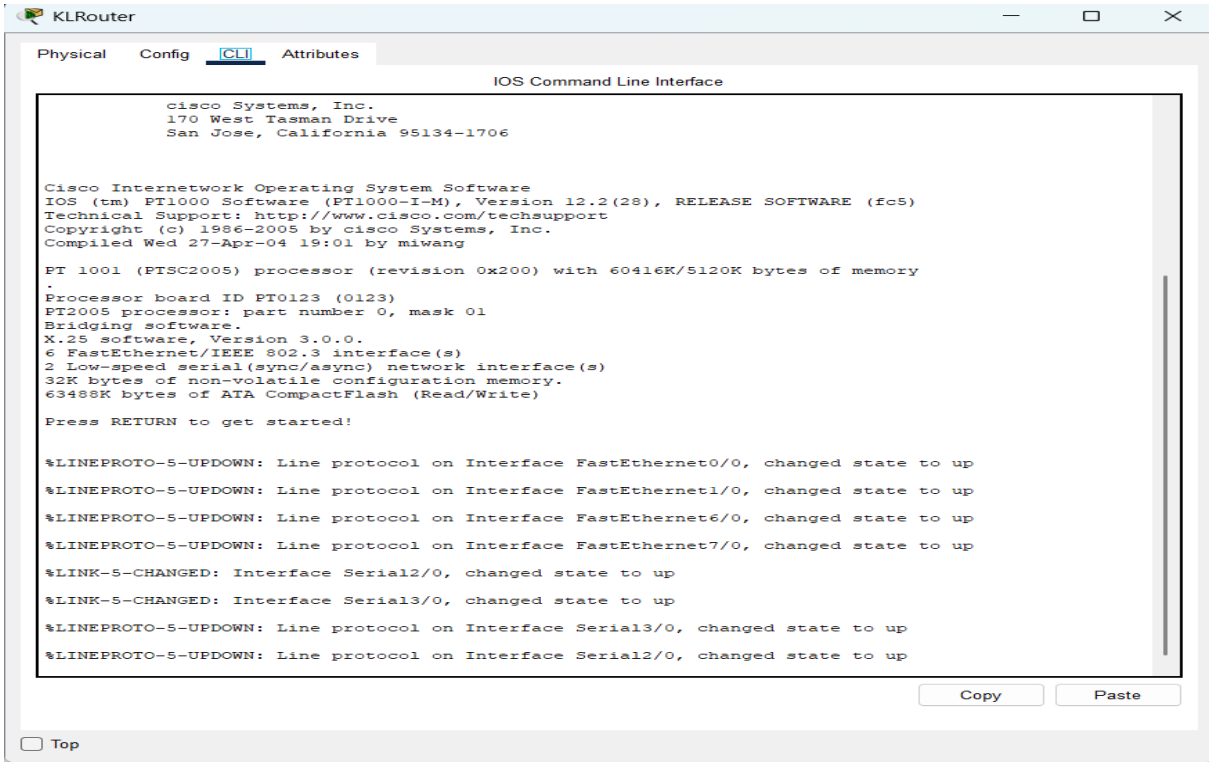
Figure 6: Reception pc5 to HR department Pc 3

## 7.BRANCH ROUTERS JUSTIFIED:




### 7.1 CLI Config. Tanmay Sarkar Emon (TP092959)

### 1.KL BRANCH:

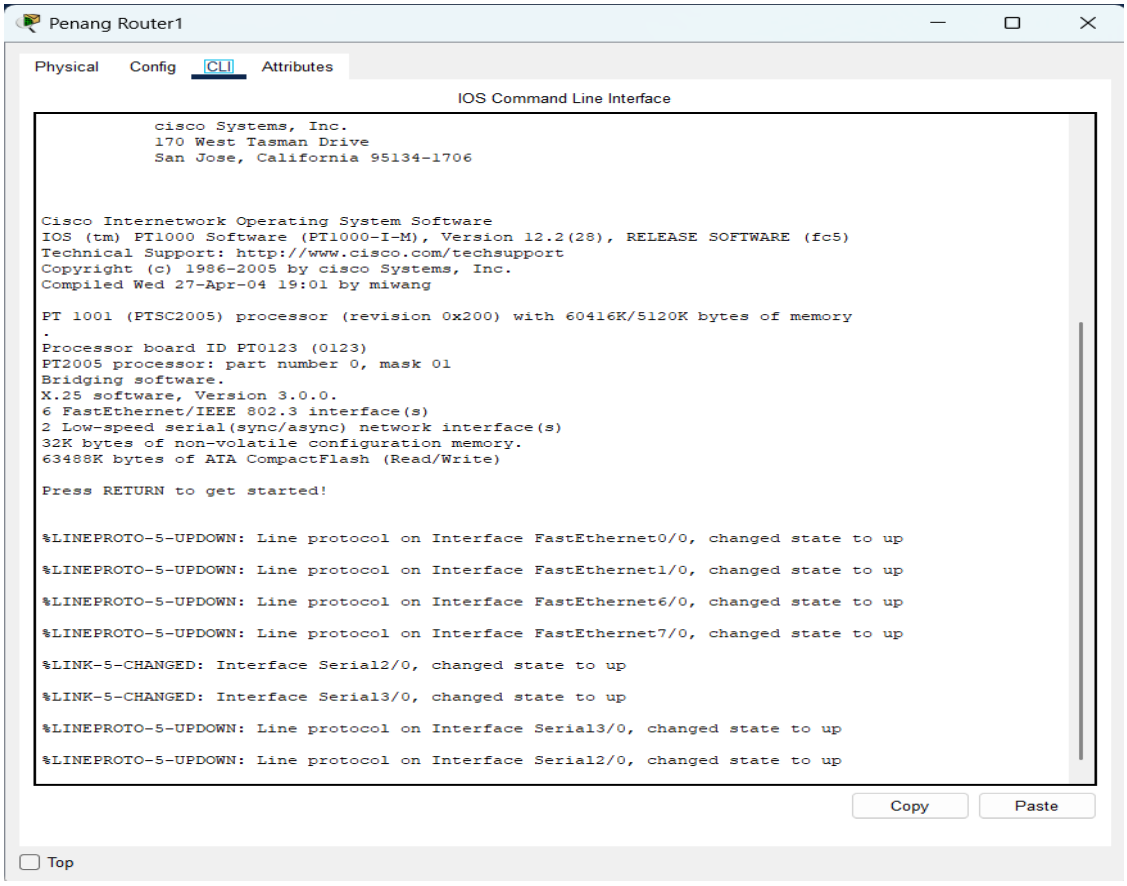


```
Device Name: Router0
Device Model: Router-PT
Hostname: KLRouter

Port              Link   IP Address        IPv6 Address                          MAC Address
FastEthernet0/0   Up     192.168.60.1/24   <not set>                             00E0.A3A3.AC5C
FastEthernet1/0   Up     192.168.70.1/24   <not set>                             0060.3E85.5ABC
Serial2/0         Up     10.0.0.1/30       <not set>                             <not set>
Serial3/0         Up     11.0.0.1/8        <not set>                             <not set>
FastEthernet4/0   Down   <not set>         <not set>                             00E0.F701.0D90
FastEthernet5/0   Down   <not set>         <not set>                             0000.0CD4.90A3
FastEthernet6/0   Up     192.168.80.1/24   <not set>                             0090.2B0C.ECB2
FastEthernet7/0   Up     192.168.90.1/24   <not set>                             0001.4250.1ACA

Physical Location: Intercity > Home City > Corporate Office > Wiring Closet > Rack > Router0
```

## PENANG BRANCH

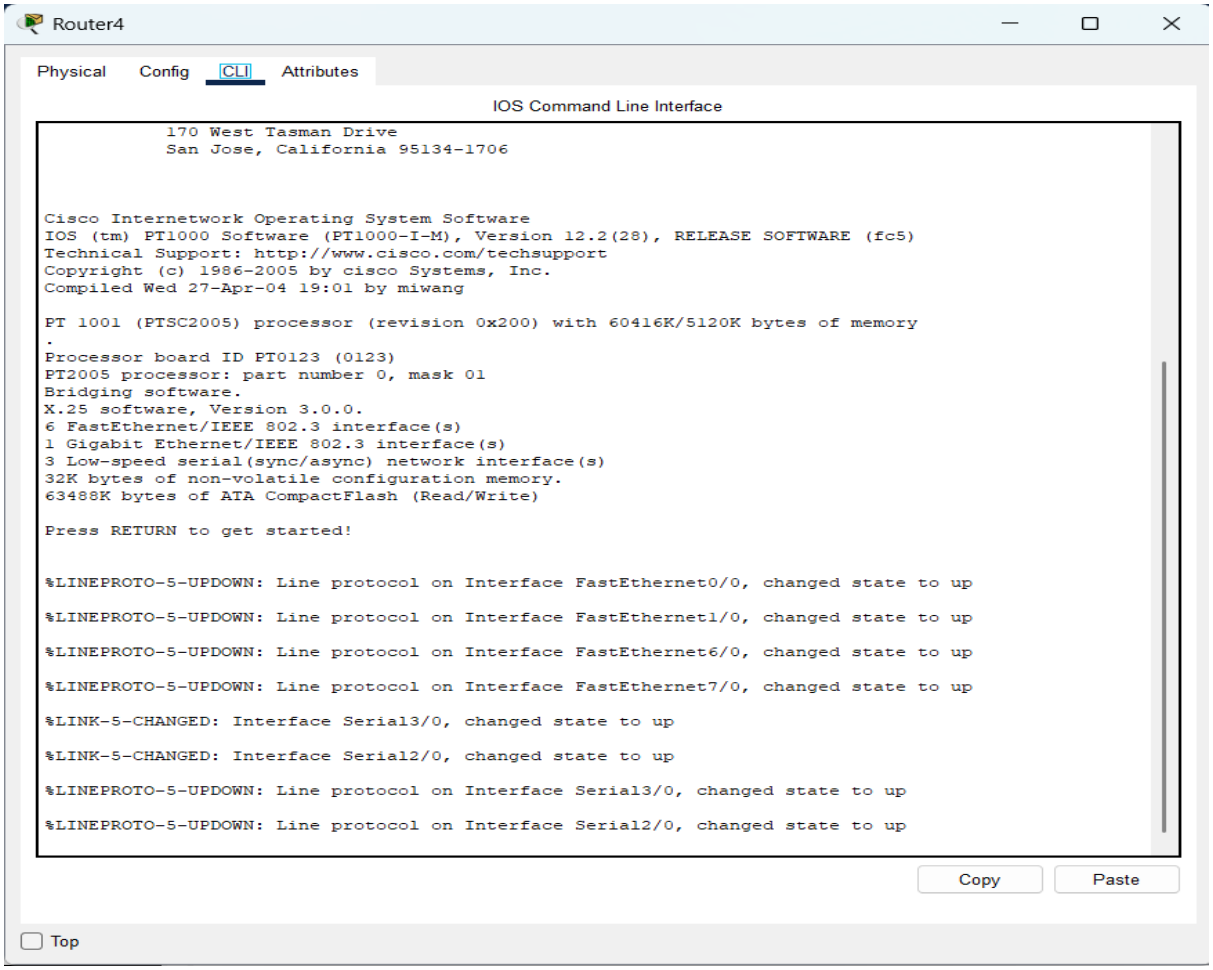



```
Device Name: Router1
Device Model: Router-PT
Hostname: Penang_Router

Port              Link   IP Address        IPv6 Address                          MAC Address
FastEthernet0/0   Up     192.168.30.1/24   <not set>                             0001.C78A.6487
FastEthernet1/0   Up     192.168.20.1/24   <not set>                             0005.5E20.5339
Serial2/0         Up     10.0.23.2/30      <not set>                             <not set>
Serial3/0         Up     11.0.0.2/8        <not set>                             <not set>
FastEthernet4/0   Down   <not set>         <not set>                             0001.637D.923C
FastEthernet5/0   Down   <not set>         <not set>                             00D0.97C2.27BA
FastEthernet6/0   Up     192.168.50.1/24   <not set>                             0030.F224.D68A
FastEthernet7/0   Up     192.168.40.1/24   <not set>                             0060.3E5A.0E11

Physical Location: Intercity > Home City > Corporate Office > Wiring Closet > Rack > Router1
```

### 7.1.1. *JOHOR BRANCH*

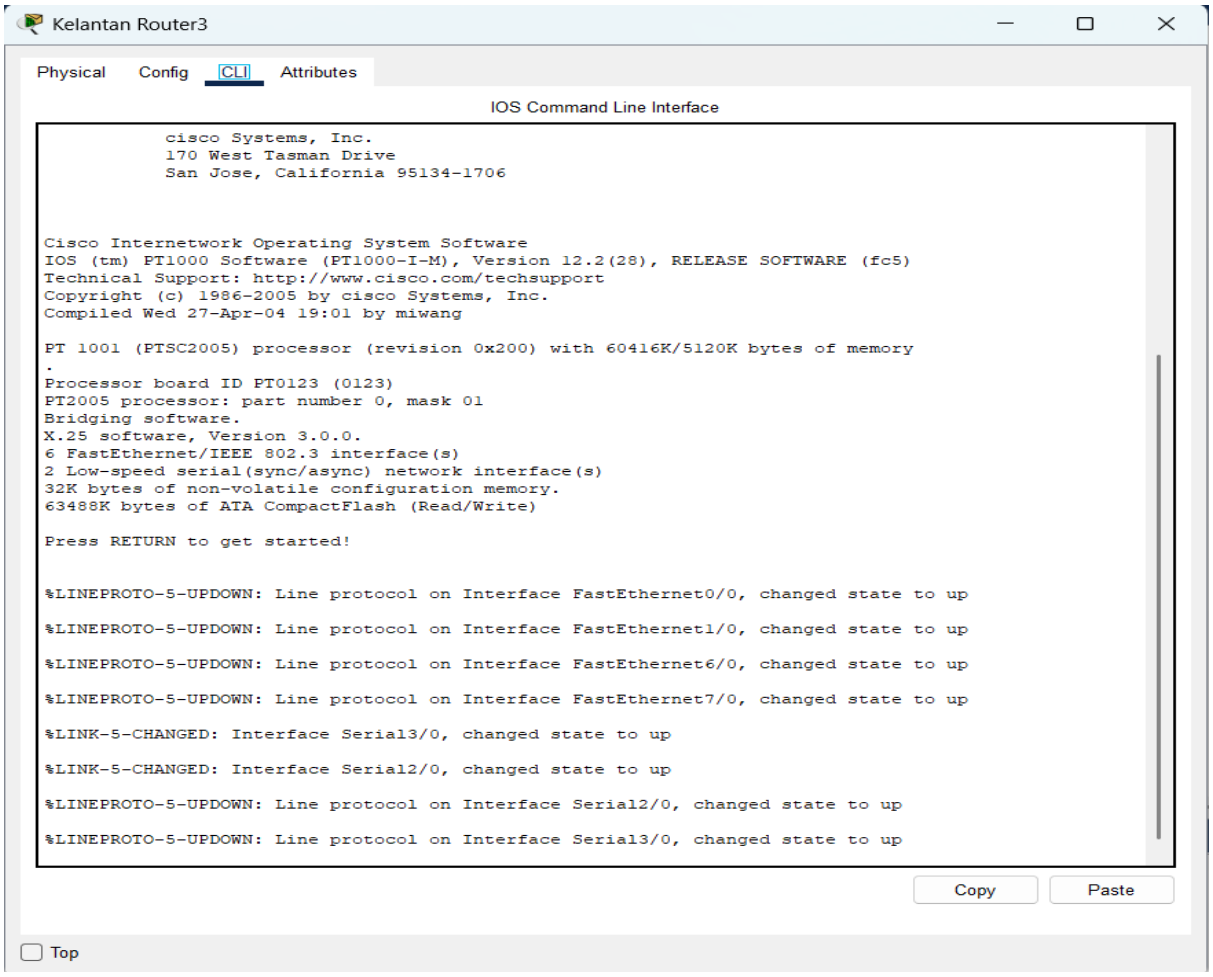



```
Device Name: Router4
Device Model: Router-PT
Hostname: Johor_Router

Port                Link   IP Address          IPv6 Address                          MAC Address
FastEthernet0/0     Up     192.158.50.1/24     <not set>                             000C.85EA.7B3C
FastEthernet1/0     Up     192.158.40.1/24     <not set>                             00E0.B08D.75CA
Serial2/0           Up     10.0.0.2/30         <not set>                             <not set>
Serial3/0           Up     10.0.12.1/30        <not set>                             <not set>
FastEthernet4/0     Down   <not set>           <not set>                             0002.4A86.B530
FastEthernet5/0     Down   <not set>           <not set>                             000C.850C.BDAA
FastEthernet6/0     Up     192.158.30.1/24     <not set>                             0050.0F80.A801
FastEthernet7/0     Up     192.158.20.1/24     <not set>                             00D0.FFC0.B694
Serial8/0           Down   <not set>           <not set>                             <not set>
GigabitEthernet9/0  Down   <not set>           <not set>                             00E0.8F32.A1BD

Physical Location: Intercity > Home City > Corporate Office > Wiring Closet > Rack > Router4
```

### 7.1.2. KELANTAN BRANCH




```
Device Name: Router3
Device Model: Router-PT
Hostname: KelantanRouter

Port              Link   IP Address         IPv6 Address                          MAC Address
FastEthernet0/0   Up     192.148.80.1/24    <not set>                             00D0.9704.31E9
FastEthernet1/0   Up     192.148.50.1/24    <not set>                             00D0.D3E0.1081
Serial2/0         Up     10.0.23.1/30       <not set>                             <not set>
Serial3/0         Up     10.0.12.2/30       <not set>                             <not set>
FastEthernet4/0   Down   <not set>          <not set>                             0007.EC5B.E437
FastEthernet5/0   Down   <not set>          <not set>                             0090.2B01.8412
FastEthernet6/0   Up     192.148.70.1/24    <not set>                             0090.0CCD.486D
FastEthernet7/0   Up     192.148.60.1/24    <not set>                             000B.BEE6.14C9

Physical Location: Intercity > Home City > Corporate Office > Wiring Closet > Rack > Router3
```

7.2 IP ROUTING:

---

**I've now rebuilt all four branch tables with the correct LAN subnets, serial links, and routing paths based on the topology: KL → Johor → Kelantan → Penang.**

---

**KL Branch – Router0**

**LAN Subnets**

| Room | Network/Subnet | Gateway |
|---|---|---|
| Room1 | 192.168.90.0/24 | 192.168.90.1 |
| Room2 | 192.168.80.0/24 | 192.168.80.1 |
| Room3 | 192.168.70.0/24 | 192.168.70.1 |
| Room4 | 192.168.60.0/24 | 192.168.60.1 |

**◈ Serial Link**

| Link | Subnet | Router0 IP | Peer Router IP |
|---|---|---|---|
| KL ↔ JOHOR | 10.0.0.0/30 | 10.0.0.1 | 10.0.0.2 |

**◈ Routing Table**

| Destination Branch | Destination NWID(s) | Mask | Next Hop |
|---|---|---|---|
| JOHOR | 192.158.50.0/24, etc. | 255.255.255.0 | 10.0.0.2 |
| KELANTAN | 192.148.60.0/24, etc. | 255.255.255.0 | 10.0.0.2 |
| PENANG | 192.168.20.0/24, 192.168.50.0/24 | 255.255.255.0 | 10.0.0.2 |

---

**JOHOR Branch – Router1**

**LAN Subnets**

| Room | Network/Subnet | Gateway |
|---|---|---|
| Room1 | 192.158.50.0/24 | 192.158.50.1 |
| Room2 | 192.158.40.0/24 | 192.158.40.1 |
| Room3 | 192.158.30.0/24 | 192.158.30.1 |

**Room Network/Subnet Gateway**

**Room4 192.158.20.0/24 192.158.20.1**

**◈ Serial Links**

| Link | Subnet | Router1 IP | Peer Router IP |
|---|---|---|---|
| JOHOR ↔ KL | 10.0.0.0/30 | 10.0.0.2 | 10.0.0.1 |
| JOHOR ↔ KELANTAN | 10.0.12.0/30 | 10.0.12.1 | 10.0.12.2 |

**◈ Routing Table**

| Destination Branch | Destination NWID(s) | Mask | Next Hop |
|---|---|---|---|
| KL | 192.168.90.0/24, etc. | 255.255.255.0 | 10.0.0.1 |
| KELANTAN | 192.148.60.0/24, etc. | 255.255.255.0 | 10.0.12.2 |
| PENANG | 192.168.20.0/24, 192.168.50.0/24 | 255.255.255.0 | 10.0.12.2 |

---

**KELANTAN Branch – Router2**

**◈ LAN Subnets**

**Room Network/Subnet Gateway**

**Room1 192.148.60.0/24 192.148.60.1**

**Room2 192.148.50.0/24 192.148.50.1**

**Room3 192.148.40.0/24 192.148.40.1**

**Room4 192.148.30.0/24 192.148.30.1**

**◈ Serial Links**

| Link | Subnet | Router2 IP | Peer Router IP |
|---|---|---|---|
| KELANTAN ↔ JOHOR | 10.0.12.0/30 | 10.0.12.2 | 10.0.12.1 |
| KELANTAN ↔ PENANG | 10.0.23.0/30 | 10.0.23.1 | 10.0.23.2 |

**◈ Routing Table**

| Destination Branch | Destination NWID(s) | Mask | Next Hop |
|---|---|---|---|
| JOHOR | 192.158.50.0/24, etc. | 255.255.255.0 | 10.0.12.1 |
| KL | 192.168.90.0/24, etc. | 255.255.255.0 | 10.0.12.1 |
| PENANG | 192.168.20.0/24, 192.168.50.0/24 | 255.255.255.0 | 10.0.23.2 |

---

**PENANG Branch – Router3**

◈ **LAN Subnets**

| Room | Network/Subnet | Gateway |
|---|---|---|
| Room1 | 192.168.20.0/24 | 192.168.20.1 |
| Room2 | 192.168.50.0/24 | 192.168.50.1 |
| Room3 | —192.168.30.0/24 | —192.168.30.1 |
| Room4 | —192.168.20.0/24 | —192.168.20.1 |

◈ **Serial Link**

| Link | Subnet | Router3 IP | Peer Router IP |
|---|---|---|---|
| PENANG ↔ KELANTAN | 10.0.23.0/30 | 10.0.23.2 | 10.0.23.1 |

◈ **Routing Table**

| Destination Branch | Destination NWID(s) | Mask | Next Hop |
|---|---|---|---|
| KELANTAN | 192.148.60.0/24, etc. | 255.255.255.0 | 10.0.23.1 |
| JOHOR | 192.158.50.0/24, etc. | 255.255.255.0 | 10.0.23.1 |
| KL | 192.168.90.0/24, etc. | 255.255.255.0 | 10.0.23.1 |

---

THE PING TEST STATUS: TANMAY SARKAR EMON

Here I can send a message from a device to another device in a different branch and as it is shown in the image below it successfully reached.

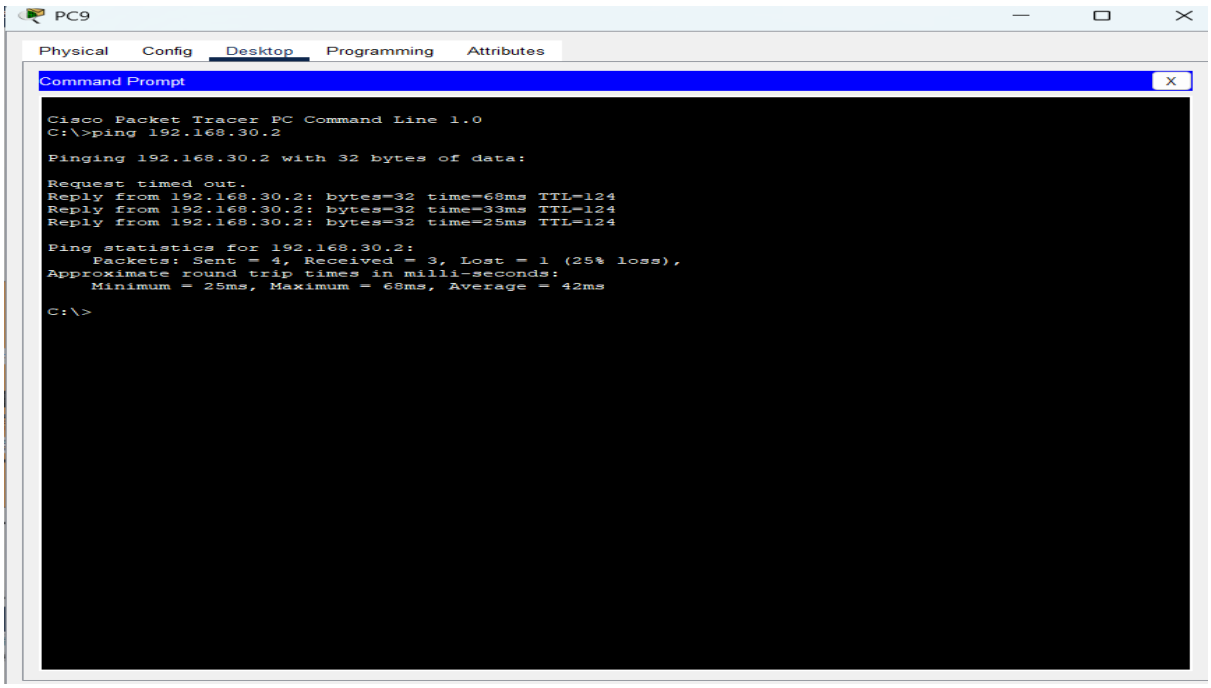
| Fire | Last Status | Source | Destination | Type | Color | Time(sec) | Periodic | Num | Edit | Delete |
|---|---|---|---|---|---|---|---|---|---|---|
| | Successful | PC19 | PC10 | ICMP | | 0.994 | N | 4 | (edit) | |
| | Successful | PC19 | PC25 | ICMP | | 1.002 | N | 5 | (edit) | |
| | Successful | tanm... | Server10 | ICMP | | 0.000 | N | 6 | (edit) | |

| Fire | Last Status | Source | Destination | Type | Color | Time(sec) | Periodic | Num | Edit | Delete |
|---|---|---|---|---|---|---|---|---|---|---|
| | Successful | IoT10 | IoT16 | ICMP | | 0.993 | N | 0 | (edit) | |
| | Successful | IoT7 | Laptop4 | ICMP | | 1.025 | N | 1 | (edit) | |
| | Successful | PC21 | PC10 | ICMP | | 0.994 | N | 2 | (edit) | |
| | Successful | tanm | PC9 | ICMP | | 1.302 | N | 3 | (edit) | |

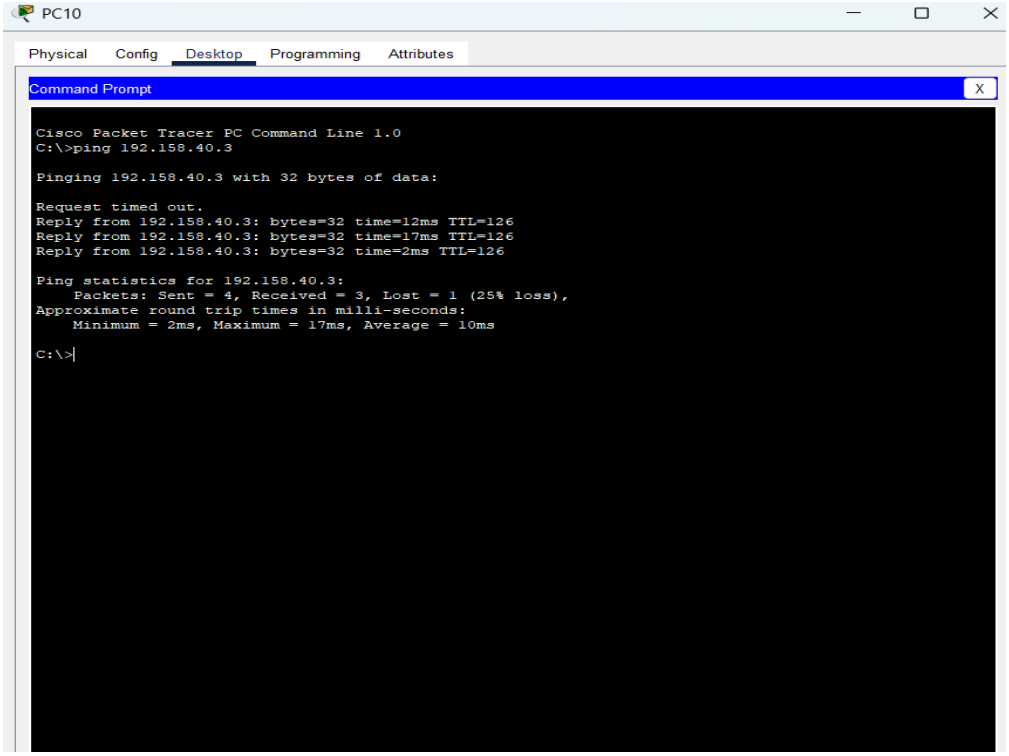
PING TEST. Tanmay Sarkar Emon (TP092959)

KL BRANCH.

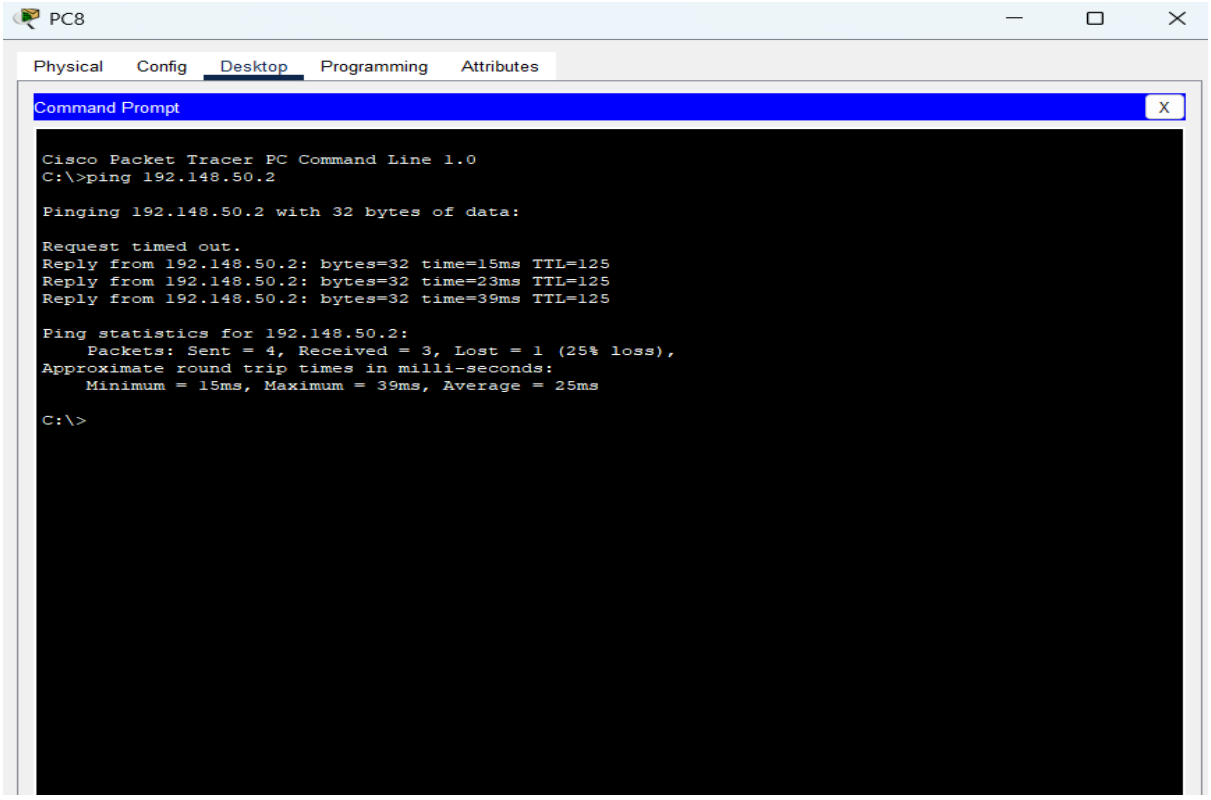
Kl to PENANG:




Kl to JOHOR:

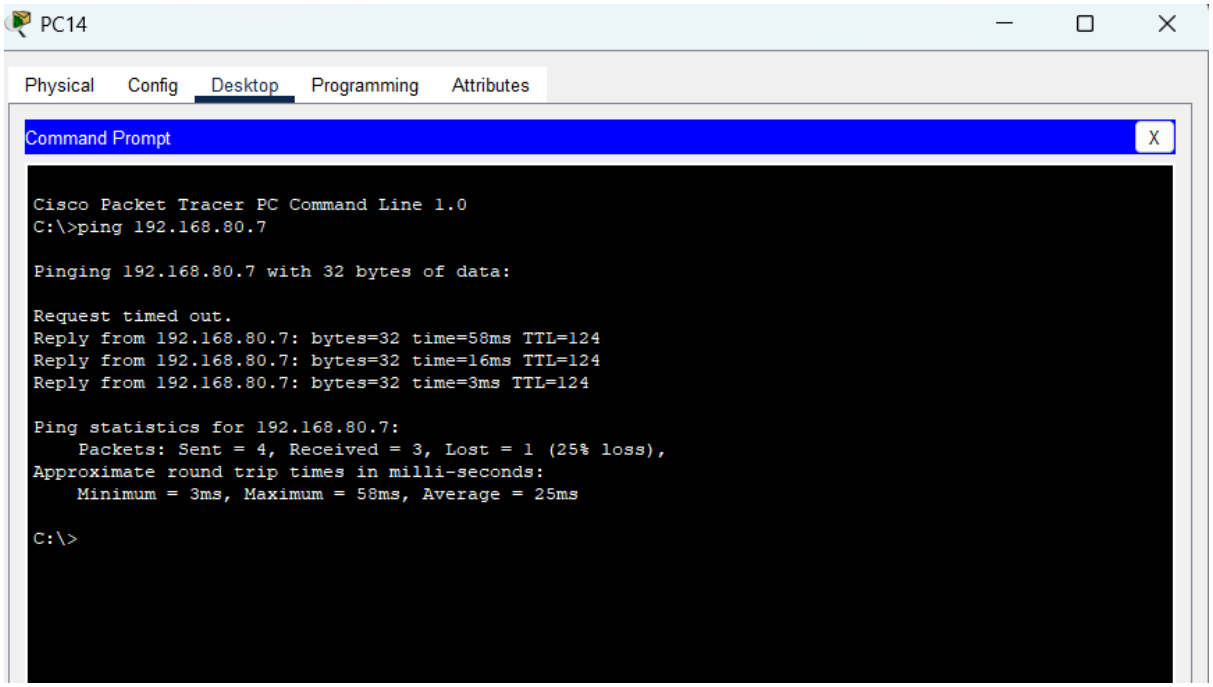



Kl to KELANTAN:

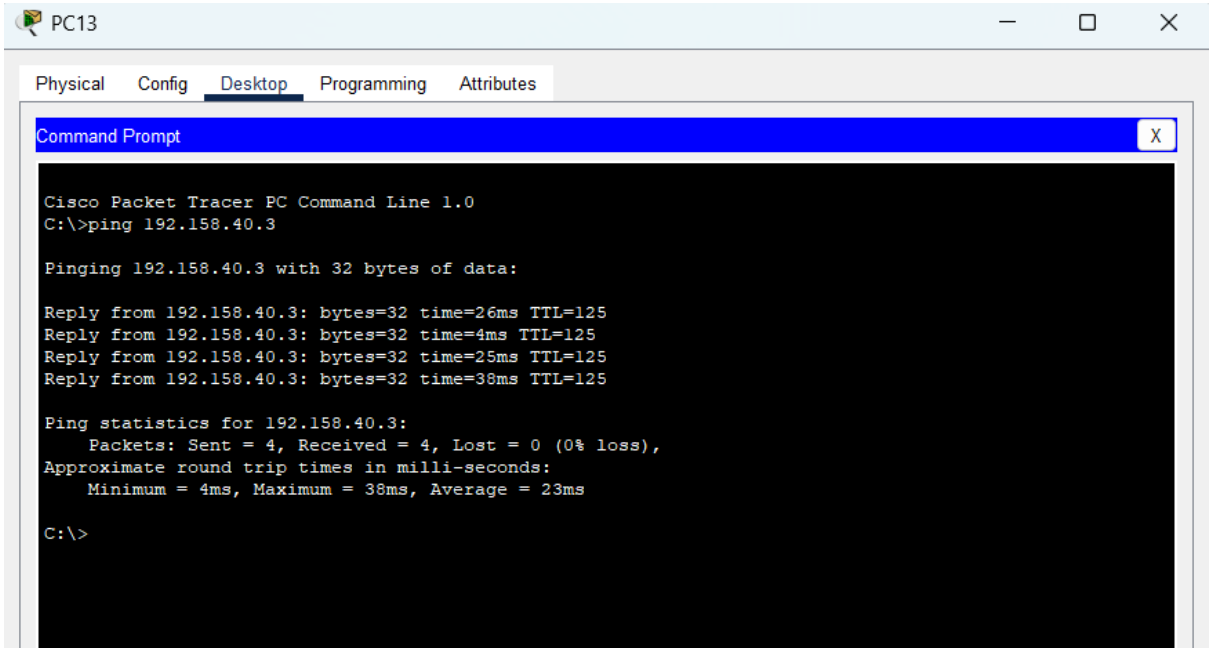



PENANG BRANCH.

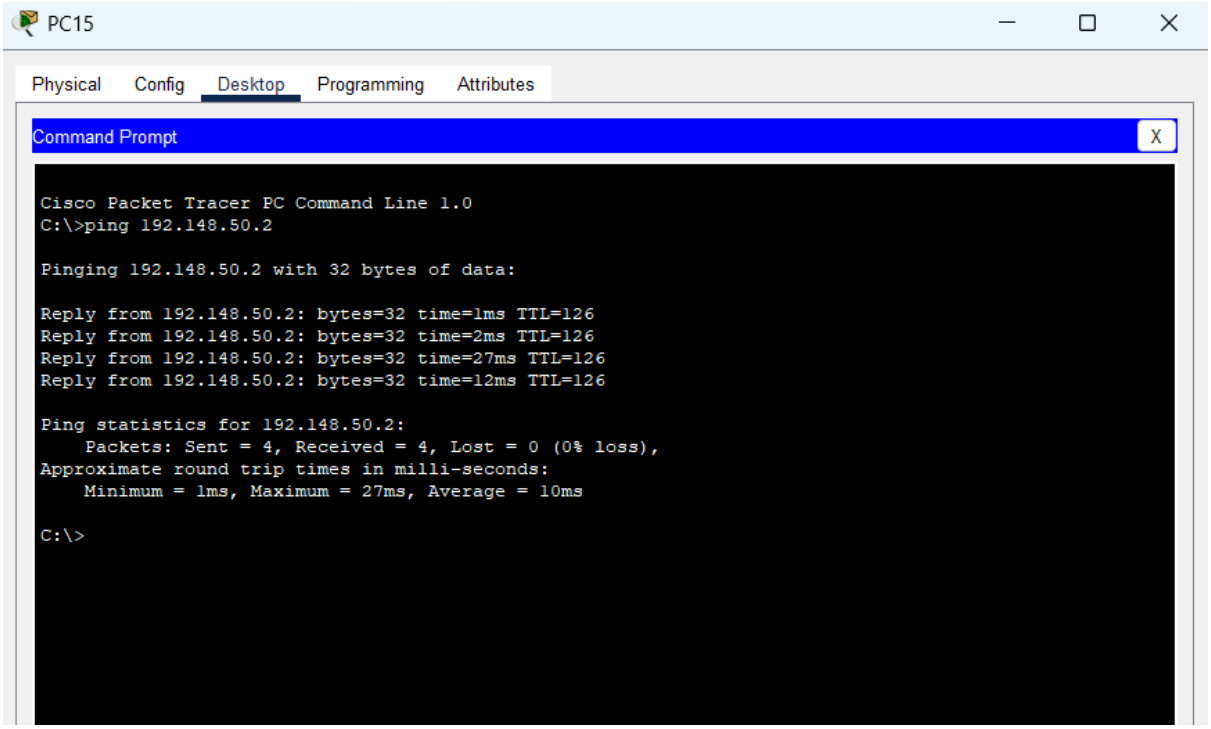
PENANG to KL:




PENANG to JOHOR:

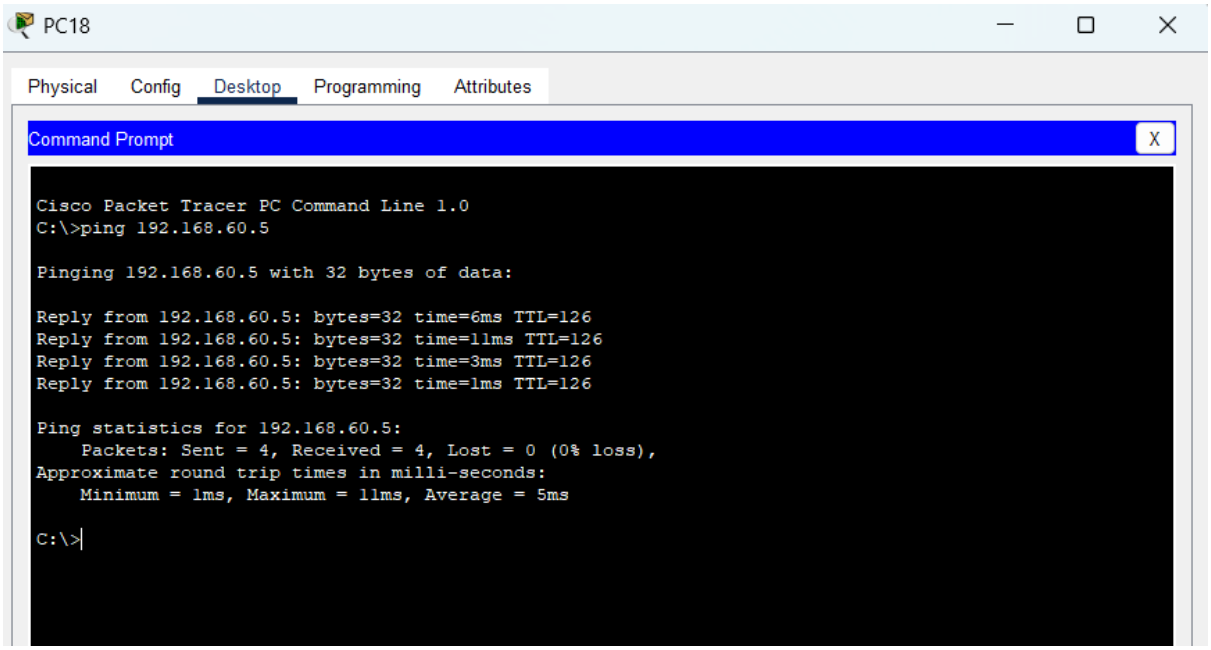



PENANG to
KELANTAN:

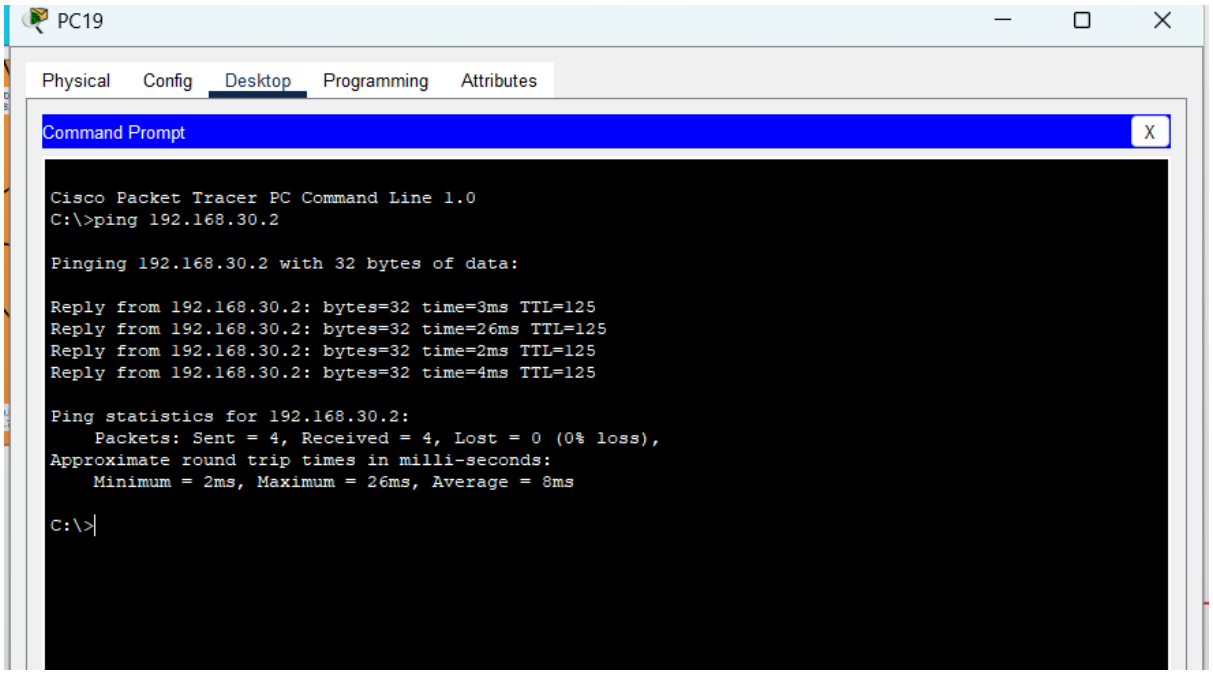



## *8.3 JOHOR BRANCH.*

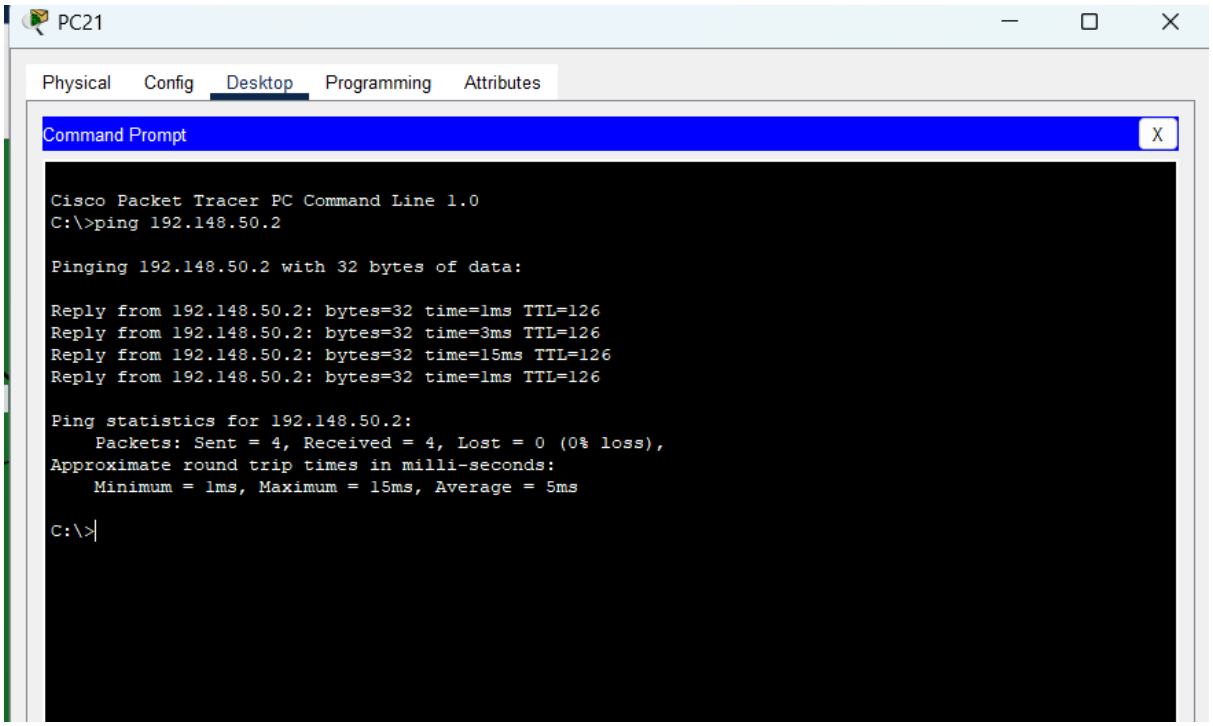
JOHOR to KL:



JOHOR to PENANG:



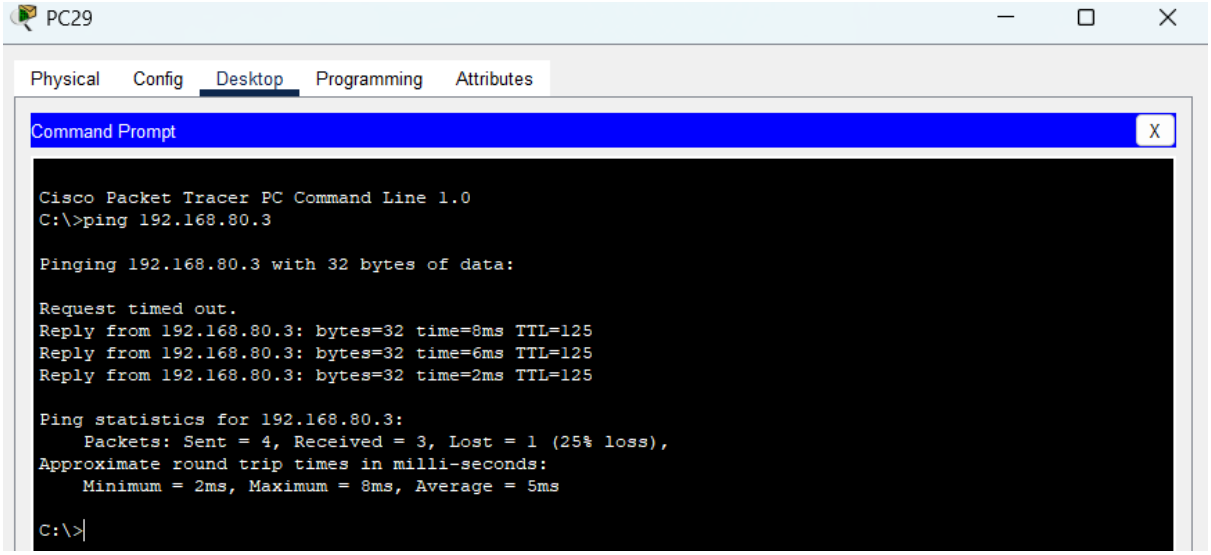

JOHOR to KELANTAN:



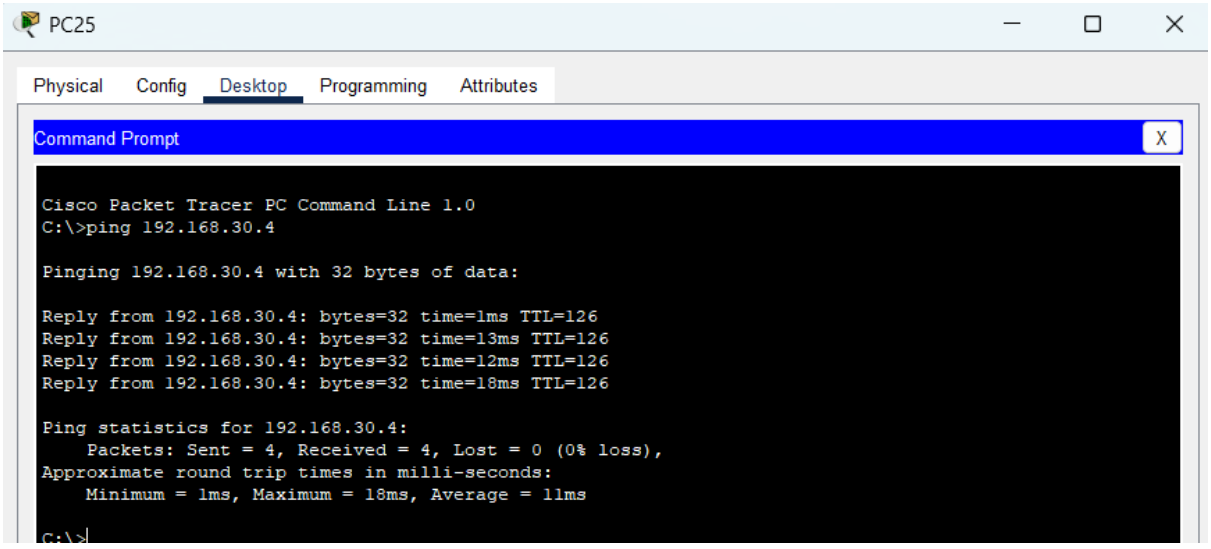
## 8.4 KELANTAN

### BRANCH.

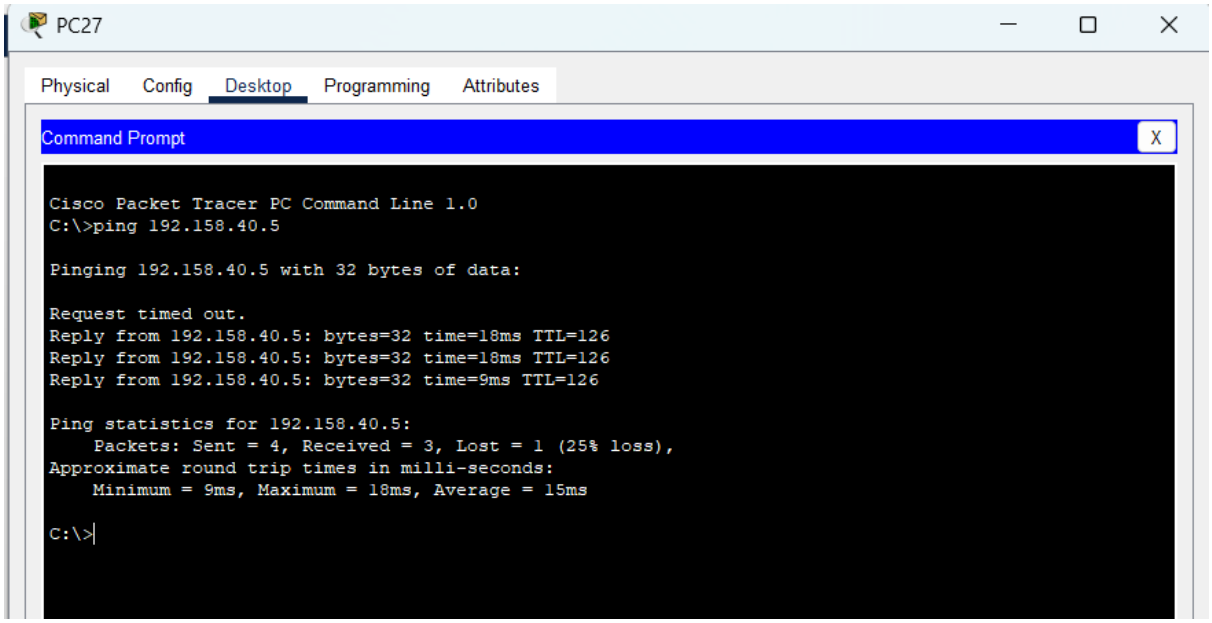
KELANTAN to KL:




KELANTAN to PENANG:



KELANTAN to JOHOR:




Conclusion :

The topology design of MACSATech follows a story approach whereby the outcomes of the design are applicable in practice and free of errors. Strictly defined IP addressing scheme ensures extreme security, scalability, efficient network utilization, and optimized performance all over the topology; each branch has got an assigned default gateway, a discrete subnet, and suitably set the router, switch configurations. Using the best topology to meet the desired requirements, we have come up with an appropriate solution, which is economically feasible to the organization.

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